



PROJECTIVE DETERMINACY

Obrad KASUM

Boban Veličković & Grigor Sargsyan

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Abstract

We prove the (game-theoretic) determinacy of projective sets of reals, under suitable large cardinal assumptions. In addition, we develop the necessary theory of tree-like iterations of the ultrapower construction.

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0 Introduction

This paper is my Master's Thesis (*mémoire de stage*). It was done in the academic year 2020/2021 at the University of Paris, as a part of the *Logique Mathématique et Fondements de l'Informatique* master's program. The work was jointly supervised by Boban Veličković (University of Paris) and Grigor Sargsyan (Rutgers University, USA). The content of this thesis was inspired by the lectures given by Sargsyan at the Baltic Set Theory Seminar, hosted by Rutgers in Spring 2021. The Seminar, and consequently the thesis as well, refer to Neeman [7] as their primary source (at least as far as the proof itself is concerned).

The outline of the paper is as follows. We want to prove that projective sets are determined; this is a result of Martin & Steel published in [5]. We show in the section 8 that homogeneously Suslin sets are determined, hence reducing the problem to showing that projective sets are homogeneously Suslin. This is done in the section 9, where we need iteration trees in order to show that a certain set is homogeneously Suslin. The iteration trees themselves are studied in the first seven sections, in the following conceptual order.

1. We briefly describe the notion of iterating ultrapowers in the simplest possible case.
2. We then introduce (short) extenders. They are families of measures which provide for a more sophisticated ultrapower construction.
3. We go on to demonstrate how one uses these extenders to capture large cardinal notions between a measurable and a superstrong cardinal. We also introduce the crucial notion of a Woodin cardinal, which is used to formulate the large cardinal assumption in our proof of the projective determinacy.
4. In this section, we prove technical propositions that provide a basic toolkit for approaching iteration trees.
5. Then we introduce the iteration trees themselves.
6. Here, we prove that every iteration tree on V of length ω has a cofinal branch whose direct limit is well-founded. This will be used for the final conclusion in the section 8.
7. In the proof of the projective determinacy, we will need to actually construct some non-linear iteration trees. This section describes a construction of the simplest such iteration tree. This construction will need to be technically improved, but it is conceptually sufficient.

Throughout the paper, we will use the following conventions.

- By a κ -*model*, we shall mean a transitive class which satisfies

$$Z \models \ulcorner \text{every set can be well-ordered} \urcorner,$$

which contains κ , and which is closed for definable κ -sequences. This closure is expressible as a first-order axiom schema (it is a weakening of the replacement).

- By a *small κ -model*, we shall mean a transitive set which is κ -model.
- $\{\theta : V_\theta \prec_{1000} V\}$ is a club class. $\{\theta : \text{cof } \theta > \lambda \wedge V_\theta \prec_{1000} V\}$ is a cofinal class. For a large enough λ , any large enough element θ of the previous class shall be referred to simply as *large enough*. For such a θ , V_θ is a small λ -model (and it is in fact closed for all λ -sequences, not only the definable ones).
- By a *countable hull* of a model $(N, \in, \dots)$, we mean an elementary embedding $j : (M, \in, \dots) \prec (N, \in, \dots)$, with M transitive and countable.
- All well-founded, extensional, set-like structures are implicitly identified with their transitive collapses.

1 Linear Iterability

The purpose of this section is to introduce the notion of iterability in the simplest case, i.e. that of a linear iteration via single measure. Starting from any $(B, E, U) \models \text{ZFC} + U$ is an ultrafilter, working within it, we may construct the ultrapower $\text{Ult}(B, U)$, with corresponding E^* and U^* , as well as the canonical embedding $j : (B, E, U) \prec (\text{Ult}(B, U), E^*, U^*)$. This construction can be iterated in the obvious manner, while at the limit steps, we take direct limit of the previously constructed models and embeddings. The model obtained in the α^{th} -step is denoted by $\text{Ult}^\alpha(B, U)$. The case of a transitive model $B = M$ is of the particular interest. We are interested whether the iterates of M are well-founded/transitive.

Definition 1.1. $(M, \in, \mu)$ is α -iterable if every $M_\beta := \text{Ult}^\beta(M, \mu)$ is well-founded ($\beta < \alpha$). M is iterable if it is α -iterable for every α .

We consider the case when μ is countably complete¹ ultrafilter in M and give an iterability criterion for countable M . Afterwards, we will deduce the iterability in the case $M = V$, which then can be relativized to any transitive model $M \models \text{ZFC}$, yielding its Ord^M -iterability. We start with the following two lemmas.

Lemma 1.2. *A countable transitive $M \models \text{ZFC}$ is iterable via $\mu \in M$ if and only if it is α -iterable, for every $\alpha < \omega_1$.*

Proof. For the non-trivial direction, assume that M is not iterable. This fact can be expressed as a Σ_1 -statement in the parameter M . Since $M \in H_{\omega_1}$ and $H_{\omega_1} \prec_1 V$, we have non-iterability of M in H_{ω_1} . This is then witnessed by some ordinal $\alpha < \omega_1$, thus ensuring non- α -iterability. $\square$

¹Note that if μ is not countably complete in M , then the ultrapower $\text{Ult}(M, \mu)$ (which can be computed in M) is not well-founded, thus witnessing the failure of 2-iterability.

Lemma 1.3 (1-step lemma). *Let $N \models \text{ZFC}$ be a transitive set, $N^\omega \subseteq N$, and ν a countably complete ultrafilter in N . Let $\sigma : (M, \in, \mu) \prec (N, \in, \nu)$, where M is transitive and countable. Let $\pi : M \prec \text{Ult}(M, \mu)$ be the canonical embedding. Then there is $k : \text{Ult}(M, \mu) \prec N$ such that the following diagram commutes.*

$$\begin{array}{ccc} & N & \\ \sigma \uparrow & \swarrow k & \\ M & \xrightarrow{\pi} & \text{Ult}(M, \mu) \end{array}$$

We say that $\text{Ult}(M, \mu)$ is *realizable* in N .

Proof. $\sigma[\mu] \subseteq \nu$ is countable, thus $\sigma[\mu] \in N$ and $\bigcap \sigma[\mu] \in \nu$. Let $s \in \bigcap \sigma[\mu]$ and define $k([f]_\mu) := \sigma(f)(s)$. We need to show that k is well defined and elementary, which will follow once we verify that for every $\phi(x \dots)$,

$$\text{Ult}(M, \mu) \models \phi([f] \dots) \implies N \models \phi(\sigma(f)(s) \dots).$$

To see this, assume that $\text{Ult}(M, \mu) \models \phi([f] \dots)$. This implies $A := \{s : \phi(f(s) \dots)\} \in \mu$. Thus, $\sigma(A) \in \sigma[\mu]$, ensuring $s \in \sigma(A)$, which translates to $N \models \phi(\sigma(f)(s) \dots)$.

To check the commutativity of the diagram, let $x \in M$ be arbitrary. We compute the following:

$$k(\pi(x)) = k([c_x]) = \sigma(c_x)(s) = c_{\sigma(x)}(s) = \sigma(x).$$

□

Theorem 1.4. *Let $N \models \text{ZFC}$ be a transitive set, $N^\omega \subseteq N$, ν a countably complete ultrafilter in N . Let $\sigma : (M, \in, \mu) \prec (N, \in, \nu)$, where M is transitive and countable. Then M is iterable (via μ).*

Proof. By recursion, we can construct embeddings $\sigma_\alpha : (M_\alpha, \in, \mu_\alpha) \prec (N, \in, \nu)$ such that all the diagrams

$$\begin{array}{ccc} & N & \\ \sigma_\alpha \uparrow & \swarrow \sigma_\beta & \\ M_\alpha & \xrightarrow{j_{\alpha, \beta}} & M_\beta \end{array}$$

($\alpha < \beta$). The embedding σ_α ensures the well-foundedness of M_α . The construction is done in the obvious manner: $\sigma_0 := \sigma$, $\sigma_{\alpha+1}$ is obtained from σ_α by applying the 1-step lemma, while σ_λ for limit λ is obtained from the universal property of the direct limit. □

Theorem 1.5. *If μ is a countably complete ultrafilter, then V is iterable via μ .*

Proof. Assume otherwise. Then there is a minimal λ for which M_λ is ill-founded. The countable completeness of μ ensures that well-foundedness is preserved at successor steps, so λ is limit. The ill-foundedness of M_λ is witnessed by sequences $(\alpha_n)_n \in \lambda^\omega$ and $(\xi_n)_n$ such that $(j_{\alpha_n, \lambda}(\xi_n))_n$ is strictly decreasing. There is a large enough θ such that

$$V_\theta \models V \text{ is } \lambda\text{-iterable via } \mu \wedge \text{Ult}^\lambda(V, \mu) \text{ is ill-founded.}$$

Let $\sigma : (M, \in, \nu, \rho) \prec (V_\theta, \in, \mu, \lambda)$ be a transitive hull. We have

$$M_\theta \models V \text{ is } \rho\text{-iterable via } \nu \wedge \text{Ult}^\rho(V, \nu) \text{ is ill-founded.}$$

However, we have sufficient conditions for reasoning of the previous theorem to go through, starting from the embedding $\sigma : M \prec V_\theta$ and for $\rho + 1$ steps. In other words, we obtain $\sigma_\rho : M_\rho \prec V_\theta$, implying that $M_\rho = \text{Ult}^\rho(M, \nu)$ is well-founded, which is a contradiction. $\square$

We shall now apply the theorem 1.4 to obtain Σ_3^1 generic absoluteness. Before we do so, let us recall two well known absoluteness theorems (see Introduction of [6]). *Mostowski's absoluteness theorem* states that arithmetical Σ_1^1 sentences (with parameters in $\mathbb{R}^M$) are absolute between M and V . *Shoenfield's absoluteness theorem* states that every Σ_2^1 sentence (with parameters in $\mathbb{R}^M$) is absolute between M and V , provided that $\omega_1^V \subseteq M$.

Theorem 1.6 (Σ_3^1 generic absoluteness). *Assume that there is a proper class of measurable cardinals. Let ϕ be a Σ_3^1 sentence (with parameters in $\mathbb{R}^V$), let $\mathbb{P}$ be a forcing poset and let $g \subseteq \mathbb{P}$ be V -generic. Then ϕ is absolute between V and $V[g]$.*

Proof. ϕ is of the form $\exists x \psi(x)$, for Π_2^1 formula $\psi(x)$. If $V \models \phi$, then $V \models \psi(a)$ for some $a \in \mathbb{R}^V$, so we may apply Shoenfield's theorem, yielding $V[g] \models \psi(a)$. This of course means $V[g] \models \phi$.

Let us now assume that $V[g] \models \phi$. This means that for some $p \in g$, $V \models (p \Vdash_{\mathbb{P}} \phi)$. Choose a measurable λ such that $\mathbb{P} \in V_\lambda$, an ultrafilter ν witnessing this measurability, and an inaccessible $\theta > \lambda$. Let

$$\sigma : (M, \in, \mu, \kappa, q, \mathbb{Q}) \prec (V_\theta, \in, \nu, \lambda, p, \mathbb{P}),$$

where M is countable and transitive. In particular, $M \models (q \Vdash_{\mathbb{Q}} \phi)$.

By the theorem 1.4, we have the transitive iterate M_{ω_1} of M and the embedding $j = j_{0, \omega_1} : M \prec M_{\omega_1}$. Since μ is a measure on κ , the rank V_κ^M remains fixed throughout the iteration. This means that $j(\mathbb{Q}) = \mathbb{Q}$, $j(q) = q$, $M_{\omega_1} \models (q \Vdash_{\mathbb{Q}} \phi)$ and $P(\mathbb{Q}) \cap M_{\omega_1} = P(\mathbb{Q}) \cap V_\kappa^{M_{\omega_1}} = P(\mathbb{Q}) \cap V_\kappa^M$. Thus, $P(\mathbb{Q}) \cap M_{\omega_1}$ is countable, so there is an M_{ω_1} -generic filter $h \subseteq \mathbb{Q}$, $h \ni q$.

We can now conclude that $M_{\omega_1}[h] \models \phi$, i.e. $M_{\omega_1}[h] \models \psi(a)$, for some $a \in \mathbb{R} \cap M_{\omega_1}[h]$. Since $\omega_1 \subseteq M_{\omega_1} \subseteq M_{\omega_1}[h]$, we can apply Shoenfield's theorem once again. This yields $V \models \psi(a)$, i.e. $V \models \phi$. $\square$

2 Extenders I

Extenders are generalization of measures which provide for more sophisticated ultrapower construction. The extenders that we study are also called *short* extenders and they will play crucial rule throughout this thesis. Notation-wise, $[\lambda]^{<\omega}$ denotes the set of all finite subsets of λ and for $s \in [\lambda]^{<\omega}$ and $i < |s|$, we denote by s_i the i^{th} element of s (in the ordinal ordering).

Definition 2.1. Let M be a κ -model. An M -pre-extender on κ of length λ ($\kappa < \lambda$ limit ordinals) is a family $E = (E_a : a \in [\lambda]^{<\omega})$ satisfying the following properties.

- Every E_a is an M -ultrafilter on $[\kappa]^{|a|}$, i.e. an ultrafilter in the algebra $P([\kappa]^{|a|}) \cap M$. We define projections $\pi_{b,a}$ for $a \subseteq b \in [\lambda]^{<\omega}$ in the following manner: if $b = \{\beta_0 < \dots < \beta_{n-1}\}$ and $a = \{\beta_{i_0} < \dots < \beta_{i_{m-1}}\}$, then

$$\pi_{b,a} : [\kappa]^{|b|} \rightarrow [\kappa]^{|a|} : \{\xi_0 < \dots < \xi_{n-1}\} \mapsto \{\xi_{i_0} < \dots < \xi_{i_{m-1}}\}.$$

For $f : [\kappa]^{|a|} \rightarrow \kappa$, we also define a function

$$f^{b,a} : [\kappa]^{|b|} \rightarrow \kappa : s \mapsto f(\pi_{b,a}(s)).$$

- Every E_a is M - κ -complete, i.e. $\bigcap_{\alpha < |A|} A_\alpha \in E_a$ for every $A \in E_a^{<\kappa} \cap M$.
- E is M -normal, i.e. for every $a \in [\lambda]^{<\omega}$, $f \in \kappa^{[\kappa]^{|a|}} \cap M$, and $i < |a|$ with $\{s \in [\kappa]^{|a|} : f(s) < s_i\}$, there is $\beta < \lambda$ with $b := a \cup \{\beta\}$ and j witnessing $\beta = b_j$, such that $\{t \in [\kappa]^{|b|} : f^{b,a}(t) = t_j\} \in E_b$.
- E is coherent, i.e. for $a \subseteq b \in [\lambda]^{<\omega}$,

$$(\forall A \in E_b) \pi_{b,a}[A] \in E_a.$$

Let E be an M -pre-extender. We want to define the ultrapower $\text{Ult}(M, E)$ of M via E . For every $a \in [\lambda]^{<\omega}$, we can construct the ultrapower $\text{Ult}(M, E_a)$. We naturally identify $\text{Ult}(M, E_\emptyset)$ and M . For $a \subseteq b$, we define mappings

$$j_{a,b}^{M,E} : \text{Ult}(M, E_a) \rightarrow \text{Ult}(M, E_b) : [f]_{E_a}^M \mapsto [f^{b,a}]_{E_b}^M.$$

This mapping is correctly defined and elementary. For otherwise, there would be a formula $\phi(x \dots)$ and a function f such that

$$\text{Ult}(M, E_a) \models \phi([f] \dots), \quad \text{Ult}(M, E_b) \models \neg \phi([f^{b,a}] \dots);$$

applying Łoś's theorem would yield

$$\{s : M \models \phi(f(s) \dots)\} \in E_a, \quad \{t : M \models \neg \phi(f(\pi_{b,a}(t) \dots))\} \in E_b,$$

which is in contradiction with the coherence property, since acting with $\pi_{b,a}$ on the second set produces exactly the complement of the first one.

Furthermore, for $a \subseteq b \subseteq c$ in $[\lambda]^{<\omega}$, it is a straightforward matter to verify $\pi_{b,a} \circ \pi_{c,b} = \pi_{c,a}$, so we can compute

$$j_{b,c}(j_{a,b}([f]_{E_a})) = j_{b,c}([f \circ \pi_{b,a}]_{E_b}) = [f \circ \pi_{b,a} \circ \pi_{c,b}]_{E_c} = [f \circ \pi_{c,a}]_{E_c}$$

(for $f : [\kappa]^{|a|} \rightarrow \kappa$ with $f \in M$). Thus, we have constructed a system

$$(\text{Ult}(M, E_a), j_{a,b} : a \subseteq b \in [\lambda]^{<\omega}).$$

of models and elementary embeddings. We define $\text{Ult}(M, E)$ as the direct limit of this system, denoting the obtained embeddings by $j_a : \text{Ult}(M, E_a) \prec \text{Ult}(M, E)$, and in particular $j_E := j_\emptyset : M \prec \text{Ult}(M, E)$. We proceed to go through the construction of the direct limit in order to get a more explicit description of $\text{Ult}(M, E)$.

Let $D := \{(a, f) : a \in [\lambda]^{<\omega}, f : [\kappa]^{|a|} \rightarrow \kappa, f \in M\}$. If we identify the pairs of the with the same first coordinate a modulo E_a , we obtain exactly the disjoint union $\bigsqcup_a \text{Ult}(V, E_a)$. We identify (a, f) and (b, g) exactly when for some/any $c \supseteq a, b$, $j_{a,c}([f]) = j_{b,c}([g])$. (We may take, for example, $c := a \cup b$.) The relation obtained in this way is denoted by $=^D$. After unraveling the definitions, we get the following:

$$(a, f) =^D (b, g) \iff \{s : f^{c,a}(s) = f^{c,b}(s)\} \in E_c.$$

Analogously as for $=^D$, we can define

$$(a, f) \in^D (b, g) : \iff j_{a,c}([f]) \in j_{b,c}([g]) \iff \{s : f^{c,a}(s) \in g^{c,b}(s)\} \in E_c,$$

for some/any $c \supseteq a, b$. It is now clear that we can take $(\text{Ult}(M, E), \in^E) := (D, \in^D) / =^D$ and $j_a : [f] \mapsto (a, f) / =^D$. In particular, $j_E : M \prec \text{Ult}(M, E)$ is defined as $j_E(x) = [\emptyset, \emptyset \mapsto x]$. We have the following version of Łoś's theorem.

Theorem 2.2 (Łoś). *For a formula $\phi(\bar{x})$, finite subsets a_i of λ , functions $f_i : [\kappa]^{|a_i|} \rightarrow \kappa$ and any $c \supseteq a_0, \dots, a_{n-1}$, we have*

$$\begin{aligned} \text{Ult}(M, E) \models \phi([a_0, f_0], \dots, [a_{n-1}, f_{n-1}]) \\ \iff \{s : \phi(f_0^{c,a_0}(s), \dots, f_{n-1}^{c,a_{n-1}}(s))\} \in E_c. \end{aligned}$$

Proof. Noting that $[a_i, f_i] = j_{a_i}([f_i]) = j_c(j_{a_i,c}([f_i])) = j_c([f_i^{c,a_i}])$, we see that both sides are equivalent to

$$\text{Ult}(M, E_c) \models \phi([f_0^{c,a_0}], \dots, [f_{n-1}^{c,a_{n-1}}]),$$

the one on the left by elementarity of j_c , while the one on the right by Łoś's theorem for ultrafilters. $\square$

Definition 2.3. Let M be a κ -model. An M -extender on κ is an M -pre-extender E on κ for which $\text{Ult}(M, E)$ is well-founded.

An M -extender on κ of length λ is referred to as “ M -(κ, λ)-extender” (and similarly for pre-extenders).

Theorem 2.4. *Let M be a κ -model and let E be a M -(κ, λ)-extender. Then the following holds.*

1. $[\{\alpha\}, \{\xi\} \mapsto \xi] = \alpha$ for $\alpha < \lambda$.
2. $[a, s \mapsto s_i] = a_i$ and $[a, s \mapsto s] = a$ for $a \in [\lambda]^{<\omega}$.
3. $[a, f] = j_E^M(f)(a)$ for $a \in [\lambda]^{<\omega}$ and $f : [\kappa]^{|a|} \rightarrow \kappa$ with $f \in M$.
4. $\text{crit } j_E^M = \kappa$ and $j_E^M(\kappa) \geq \lambda$.

Proof.

1. Let us define a map $F : \lambda \rightarrow \text{Ord}$ by setting $F(\alpha) := [\{\alpha\}, \{\xi\} \mapsto \xi]$. We want to verify that F is increasing, i.e. that for every $\alpha < \beta < \lambda$,

$$\{s \in [\kappa]^2 : (\{\xi\} \mapsto \xi)^{\{\alpha, \beta\}, \{\alpha\}}(s) < (\{\xi\} \mapsto \xi)^{\{\alpha, \beta\}, \{\beta\}}(s)\} \in E_{\{\alpha, \beta\}}.$$

However, this just reduces to $\{s : s_0 < s_1\} = [\kappa]^2 \in E_{\{\alpha, \beta\}}$.

Furthermore, $\text{im } F$ is an initial segment of Ord , i.e. an ordinal. To see this, take any $\alpha < \lambda$ and any $[b, g] \in [\{\alpha\}, \{\xi\} \mapsto \xi]$. W.l.o.g. we may assume that $\alpha \in b$ and $\alpha = b_i$, so we have

$$\{s : g(s) \in (\{\xi\} \mapsto \xi)^{b, \{\alpha\}}(s)\} \in E_b,$$

which translates to $\{s : g(s) < s_i\} \in E_b$. M -normality implies existence of some $c \supseteq b$ and $k < |c|$ satisfying $\{s : g^{c, b}(s) = s_k\} \in E_c$. This last relation translates to $\{s : g^{c, b}(s) \in (\{\xi\} \mapsto \xi)^{c, \{c_k\}}(s)\} \in E_c$, verifying that $[b, g] = [\{c_k\}, \{\xi\} \mapsto \xi] = F(c_k) \in \text{im } F$.

Thus, F is an order isomorphism between ordinals, implying that $F = \text{id}_\lambda$. Consequently, $\alpha = F(\alpha) = [\{\alpha\}, \{\xi\} \mapsto \xi]$ for every $\alpha < \lambda$.

2. Notice that $\{s : (s \mapsto s_i)(s) = (\{\xi\} \mapsto \xi)^{a, a_i}(s)\} = \{s : s_i = s_i\} \in E_a$, implying $[a, s \mapsto s_i] = [\{a_i\}, \{\xi\} \mapsto \xi] = a_i$. Further, denoting $n := |a|$, we notice that for every $s \in [\kappa]^n$, $s = \{s_0, \dots, s_{n-1}\}$, so by Łoś's theorem

$$[a, s \mapsto s] = \{[a, s \mapsto s_0], \dots, [a, s \mapsto s_{n-1}]\} = \{a_0, \dots, a_{n-1}\} = a.$$

3. $j_E(f)(a) = [\emptyset, \emptyset \mapsto f]([a, s \mapsto s]) = [a, s \mapsto f(s)] = [a, f]$, where the second equality follows from Łoś's theorem.
4. Let $\alpha < \kappa$ and let us verify $j_E^M(\alpha) = \alpha$. $j_E^M(\alpha) \geq \alpha$ is a general fact, so let us consider the other inequality. Suppose that $[a, f] \in j_E^M(\alpha) = [\emptyset, \emptyset \mapsto \alpha]$. This means that $\{s : f(s) \in \alpha\} \in E_a$. It is now implied by M - κ -completeness that $\{s : f(s) = \beta\} \in E_a$ for some $\beta < \alpha$, which means $[a, f] = [\emptyset, \emptyset \mapsto \beta] = \beta < \alpha$. This shows that $j_E(\alpha) \leq \alpha$.

It remains to show that $j_E^M(\kappa) \geq \lambda$. It suffices to verify that $j_E^M(\kappa) > \alpha$ for any $\alpha < \lambda$.

$$\begin{aligned}
\alpha < j_E^M(\kappa) &\iff [\{\alpha\}, \{\xi\} \mapsto \xi] < [\emptyset, \emptyset \mapsto \kappa] \in E_a \\
&\iff \{s \in [\kappa]^1 : (\{\xi\} \mapsto \xi)(s) < (\emptyset \mapsto \kappa)^{\{\alpha\}, \emptyset}(s)\} \in E_a \\
&\iff \{s \in [\kappa]^1 : s_0 < \kappa\} \in E_a \\
&\iff [\kappa]^1 \in E_a \\
&\iff \top.
\end{aligned}$$

□

We have established that extenders produce elementary embeddings. Let us show that one can also obtain extenders starting from an elementary embedding.

Proposition 2.5. *Let M be a κ -model, let N be transitive, let $j : M \prec N$ be an embedding with $\text{crit } j = \kappa$, and let $\lambda \leq j(k)$. Let $E = (E_a : a \in [\lambda]^{<\omega})$ be defined by:*

$$X \in E_a \iff a \in j(X)$$

for $a \in [\lambda]^{<\omega}$ and $X \in P([\kappa]^{|a|}) \cap M$. Then E is an M -extender. Moreover, there is k such that $\text{crit } k \geq \lambda$ and the following diagram commutes.

$$\begin{array}{ccc}
& N & \\
& \nwarrow k & \\
j \uparrow & & \\
M & \xrightarrow{j_E^M} & \text{Ult}(M, E)
\end{array}$$

($\text{Ult}(M, E)$ is said to be j -realizable.)

Proof. It is a matter of routine to verify that E is a family of M -ultrafilters (on the appropriate sets) and that they are M - κ -complete. To verify coherence, assume that $a \subseteq b$ are finite subsets of λ and that $A \in E_b$. This means that $b \in j(A)$, so acting by $\pi_{b,a}$ yields

$$a = \pi_{b,a}(b) \in \pi_{b,a}[j(A)] = j(\pi_{b,a}[A]),$$

showing that $\pi_{b,a}[A] \in E_a$.

Let us verify normality. Let $a \in [\lambda]^{<\omega}$, $i < |a|$, and $f : [\kappa]^{|a|} \rightarrow \kappa$ with $f \in M$ and $\{s : f(s) < s_i\} \in E_a$. This means that $a \in j(\{s : f(s) < s_i\})$, implying $j(f)(a) < a_i$. We shall take $\beta := j(f)(a)$, $b := a \cup \{\beta\}$, and $k < \omega$ satisfying $\beta = b_k$. It follows that

$$b \in \{s : j(f)(\pi_{b,a}(s)) = s_k\} = j(\{s : f^{b,a}(s) = s_k\}),$$

i.e. $\{s : f^{b,a}(s) = s_k\} \in E_b$, as required.

The fact that E is an M -extender will follow from the construction of an embedding k . We set

$$k([a, f]_E^M) := j(f)(a) \quad (a \in [\lambda]^{<\omega}, f : [\kappa]^{|a|} \rightarrow \kappa, f \in M).$$

Let us check that this mapping is well defined and elementary. If $\text{Ult}(M, E) \models \phi([a_0, f_0], \dots, [a_{n-1}, f_{n-1}])$, we have $\{s : \phi^M(f_0^{b, a_0}(s), \dots)\} \in E_b$ (for $b := \bigcup_{i < n} a_i$). By the definition of E , it follows that

$$b \in j(\{s : \phi^M(f_0^{b, a_0}(s), \dots)\}) = \{s : \phi^N(j(f)^{j(b), j(a_0)}(s), \dots)\},$$

implying $N \models \phi(j(f)(a_0), \dots)$. For the commutativity of the diagram, we simply have

$$(k \circ j_E)(x) = k([\emptyset, \emptyset \mapsto x]) = j(\emptyset \mapsto x)(\emptyset) = (\emptyset \mapsto j(x))(\emptyset) = j(x).$$

We also have

$$k(\alpha) = k([\{\alpha\}, s \rightarrow s_0]) = j(s \rightarrow s_0)(\{\alpha\}) = \alpha$$

for every $\alpha < \lambda$. □

The extender from the previous proposition is said to be *derived from j at λ* .

Proposition 2.6. *Let E be an M -(κ, λ)-extender. Then E is the extender derived from j_E^M at λ .*

Proof. Let $a \in [\lambda]^{<\omega}$, $n := |a|$, and $X \subseteq [\kappa]^n$.

$$\begin{aligned} a \in j_E(X) &\iff [a, s \mapsto s] \in [\emptyset, \emptyset \mapsto X] \\ &\iff \{s \in [\kappa]^n : (s \mapsto s)(s) \in (\emptyset \mapsto X)^{a, \emptyset}(s)\} \in E_a \\ &\iff \{s \in [\kappa]^n : s \in X\} \in E_a \\ &\iff X \in E_a. \end{aligned}$$

□

Note that V -(pre-)extenders are referred to simply as (pre-)extenders.

Definition 2.7. A pre-extender E is *complete* if for every family $(a_i : i < \omega)$ of elements of $[\lambda]^{<\omega}$ and every family $(A_i : i < \omega) \in \prod_{i < \omega} E_{a_i}$, there exists an increasing function $f : \bigcup_{i < \omega} a_i \rightarrow \kappa$ such that

$$(\forall i < \omega) f[a_i] \in A_i.$$

Theorem 2.8. *A pre-extender E is an extender if and only if it is complete.*

Proof. Let us assume that E is not complete and show that $\text{Ult}(V, E)$ is ill-founded. There are finite subsets $(a_i)_{i < \omega}$ of λ and sets $A_i \in E_{a_i}$ ($i < \omega$) such that

$$(\forall f : \bigcup_{i < \omega} a_i \rightarrow \kappa) (\exists i < \omega) (f[a_i] \notin A_i). \quad (1)$$

Up to replacing a_i by $\bigcup_{j \leq i} a_j$ and A_i by $\pi_{\bigcup_{j \leq i} a_j, a_i}^{-1}[A_i]$, we may assume that the sequence $(a_i)_i$ is increasing. Now, if we consider the set $S := \{\sigma : (\exists i <$

$\omega)(\sigma : a_i \rightarrow \kappa \wedge \sigma \text{ is increasing} \wedge \text{im } \sigma \in A_i)\}$, ordered by the reverse inclusion, then (1) is equivalent to the well-foundedness of this order. Let $\rho : S \rightarrow \text{Ord}$ be the rank function.

Let us describe a decreasing sequence of ordinals in $\text{Ult}(V, E)$. First, we note that, up to replacing A_i by $\bigcap_{j \leq i} \pi_{a_i, a_j}^{-1}[A_j]$, we may assume that

$$i < j \implies \pi_{a_j, a_i}[A_j] \subseteq A_i.$$

Fix a natural i . For $s \in [\kappa]^{a_i}$, let $\sigma_s : (a_i, <) \cong (s, <)$; so, $\sigma_s \in S$ whenever $s \in A_i$. Let $g_i : [\kappa]^{a_i} \rightarrow \text{Ord}$ satisfy $g_i(s) = \rho(\sigma_s)$ for $s \in A_i$. Then $([a_i, g_i])_{i < \omega}$ is a sequence of ordinals in $\text{Ult}(V, E)$, and it is decreasing since

$$\begin{aligned} [a_{i+1}, g_{i+1}] < [a_i, g_i] &\iff \{s \in [\kappa]^{a_{i+1}} : g_{i+1}(s) < g_i^{a_{i+1}, a_i}(s)\} \in E_{a_{i+1}} \\ &\iff \{s \in A_{i+1} : g_{i+1}(s) < g_i^{a_{i+1}, a_i}(s)\} \in E_{a_{i+1}} \\ &\iff \{s \in A_{i+1} : \rho(\sigma_s) < \rho(\sigma_{\pi_{a_{i+1}, a_i}(s)})\} \in E_{a_{i+1}} \\ &\iff A_{i+1} \in E_{a_{i+1}}. \end{aligned}$$

Let us now suppose that E is complete and show that $\text{Ult}(V, E)$ is well-founded. Assume otherwise, that there is a $\in^E$ -decreasing sequence $([a_i, g_i])_{i < \omega}$. We have $A_{i+1} := \{s \in [\kappa]^{a_{i+1}} : g_{i+1}(s) \in g_i^{a_{i+1}, a_i}(s)\} \in E_{a_{i+1}}$. By completeness, there is an increasing $f : \bigcup_{0 < i < \omega} A_i \rightarrow \kappa$ such that $f[a_i] \in A_i$. However, this yields $g_{i+1}(f[a_{i+1}]) \in g_i^{a_{i+1}, a_i}(f[a_{i+1}]) = g_i(f[a_i])$ for every $i < \omega$, which is a contradiction. $\square$

We can also obtain linear iterability analogous to that in the case of ultrafilters. We do not repeat the definition of iterability and we note that arguments for the following theorem are almost identical to those in the previous section. There is a difference in the proof of the 1-step lemma, so we will write it down here.

Lemma 2.9 (1-step lemma). *Suppose that*

- a. N is a μ -model;
- b. F is a (μ, ν) -pre-extender N ;
- c. $j : (M, \in, E, \kappa, \lambda) \prec (N, \in, F, \mu, \nu)$ is a countable hull;
- d. $j \in N$ and $|j|^N = \aleph_0$.

Then F is an extender in N if and only if there is $k : (\text{Ult}(M, E), \in^{\text{Ult}(M, E)}) \prec N$ such that the following diagram commutes.

$$\begin{array}{ccc} & N & \\ j \uparrow & & \swarrow k \\ M & \xrightarrow{j_E} & \text{Ult}(M, E) \end{array}$$

This is another example of realizability. The condition d. is satisfied if e.g. $N^\omega \subseteq N$.

Proof. If there is such a k , then $\text{Ult}(M, E)$ is well-founded, implying that E is an extender in M , and by elementarity, F is an extender in N . Let us now assume that F is an extender in N . In particular, all of its coordinates are countably complete. Let $A_a := \bigcap j[E_a] \in F_{j(a)}$. Enumerating $[\lambda]^{<\omega} = \{a_n : n < \omega\}$, let $f : \bigcup_{n < \omega} j(a_n) \rightarrow \text{Ord}$ be an increasing function satisfying $f[j(a_n)] \in A_{a_n}$ ($n < \omega$). We define $k([b, g]) := j(g)(f[j(b)])$.

Let us verify that k is correctly defined and elementary. To that end, assume that $\text{Ult}(M, E) \models \phi([b_0, g_0], \dots, [b_{n-1}, g_{n-1}])$. By Łoś's theorem, we have $A := \{s : \phi^M(g_0^{c, b_0}(s), \dots, g_{n-1}^{c, b_{n-1}}(s))\} \in E_c$, where $c := \bigcup_{i < n} b_i$. By the choice of f , we have $f[j(c)] \in j(A)$, implying

$$N \models \phi(j(g_0)^{j(c), j(b_0)}(f[j(c)]), \dots, j(g_{n-1})^{j(c), j(b_{n-1})}(f[j(c)])),$$

where $j(c) = \bigcup_{i < n} j(b_i)$. This exactly means that

$$N \models \phi(j(g_0)(f[j(b_0)]), \dots, j(g_{n-1})(f[j(b_{n-1}]))).$$

To see that the diagram commutes, we compute

$$k(j_E(x)) = k([\emptyset, \emptyset \mapsto x]) = j(\emptyset \mapsto x)(f[\emptyset]) = (\emptyset \mapsto j(x))(\emptyset) = j(x).$$

□

Lemma 2.10. *Let M be a countable κ -model and let E be an extender on κ in M . Then M is iterable via E if and only if it is α -iterable for every $\alpha < \omega_1$.*

Theorem 2.11. *Let N be an ω -closed μ -model, let F be an extender on μ in N , and let $j : (M, \in, E) \prec (N, \in, F)$ be a countable hull. Then M is iterable via E .*

Theorem 2.12. *Let E be an extender. Then V is iterable via E .*

3 Large Cardinals

In this section, we elaborate on how extenders are used to capture large cardinal notions between a measurable and a superstrong cardinal. We also introduce the notion of a Woodin cardinal which plays a crucial role in the study of determinacy.

Proposition 3.1. *A cardinal κ is measurable if and only if for every $\lambda > \kappa$ there is a (κ, λ) -extender.*

Proof. If κ is measurable, as witnessed by some U , then V is iterable via U , so we can take the λ^{th} iterate of M and the corresponding embedding $j : V \prec M$. It is now clear that $\text{crit } j = \kappa$ and $j(\kappa) \geq \lambda$, so we can derive a (κ, λ) -extender from j . On the other hand, if there is a (κ, λ) -extender E , then there is a non-trivial $j_E : V \prec \text{Ult}(V, E)$ with $\text{crit } j_E = \kappa$, so κ is measurable. □

Stronger large cardinal notions are obtained by imposing additional conditions on extenders.

Definition 3.2.

- An embedding $j : V \prec M$ is α -strong if $V_\alpha \subseteq M$.
- An extender E is α -strong if $j_E : V \prec \text{Ult}(V, E)$ is α -strong.
- A (κ, λ) -extender is *strong* if it is λ -strong.
- A cardinal κ is λ -strong if there is a λ -strong embedding $j : V \prec M$ with $\text{crit } j = \kappa$ and $j(\kappa) \geq \lambda$.
- A cardinal κ is *strong* if it is λ -strong for every $\lambda > \kappa$.

Definition 3.3.

- An embedding $j : V \prec M$ with $\text{crit } j = \kappa$ is *superstrong* if $V_{j(\kappa)} \subseteq M$.
- An extender E on κ is *superstrong* if $j_E : V \prec \text{Ult}(V, E)$ is superstrong.
- A cardinal κ is *superstrong* if there is a superstrong embedding with the critical point κ .

Theorem 3.4. *A cardinal κ is λ strong if and only if there is a strong (κ, λ) -extender. A cardinal κ is superstrong if and only if there is a superstrong extender on κ .*

Proof. If κ is λ -strong, then, for $j : V \prec M$ witnessing this fact, we may derive the (κ, λ) -extender E . Since $\text{Ult}(V, E)$ and M agree up to λ , we have $V_\lambda = V_\lambda^M = V_\lambda^{\text{Ult}(V, E)}$, so E is strong. On the other hand, if some E is a strong (κ, λ) -extender, then $j : V \prec \text{Ult}(V, E)$ witnesses λ -strongness of κ . The case of a superstrong cardinal is completely analogous. $\square$

Definition 3.5. The *strength* of an embedding $j : V \prec M$ is the supremum/maximum of all the λ for which j is λ -strong. The *strength* of an extender E is the strength of the embedding j_E which it defines.

Let κ be the critical point of an embedding j and let $\lambda := \sup_{n < \omega} j^n(\kappa)$. By Kunen's Inconsistency, $V_{\lambda+2} \not\subseteq M$, implying that

$$\text{strength } j \leq \sup_{n < \omega} j^n(\kappa) + 1.$$

However, for the embeddings coming from (short) extenders, this boundary is much lower: namely, the length of the extender. This means that the greatest possible strength of an extender is achieved when that extender is strong.

Lemma 3.6. *If E is a (κ, λ) -extender, then j_E is $(\kappa + 1)$ -strong.*

Proof. We know that $j_E \upharpoonright V_\kappa = \text{id}_{V_\kappa}$, which implies $j_E(X) \cap V_\kappa = X$ for all $X \in V_{\kappa+1}$ and concludes the proof. $\square$

Proposition 3.7. *If E is a (κ, λ) -extender, then $E \notin \text{Ult}(V, E)$. In particular, j_E is not $(\lambda + 1)$ -strong.*

Proof. Functions $\kappa \rightarrow \kappa^+$ can be coded as elements of $V_{\kappa+1}$, implying that $(\kappa(\kappa^+))^{\text{Ult}(V, E)} = \kappa(\kappa^+)$.

If it were the case that $E \in \text{Ult}(V, E)$, it would follow that $j_E(\kappa^+)$ can be correctly computed in $\text{Ult}(V, E)$ as the order type of

$$\left(\bigcup_{a \in [\lambda]^{<\omega}} \{a\} \times {}^{[\kappa]^{|a|}} \kappa \right) / \equiv^E$$

ordered by $<^E$. Working in $\text{Ult}(V, E)$, we can then bound the cardinality of $j_E(\kappa^+)$ by $|\lambda|^{2^\kappa}$. Since $\kappa < j_E(\kappa)$ and $j_E(\kappa)$ is a strong limit in $\text{Ult}(V, E)$, we conclude that $|j_E(\kappa^+)|^{\text{Ult}(V, E)} \leq |j_E(\kappa)|^{\text{Ult}(V, E)}$, which is a contradiction.

Thus, $E \notin \text{Ult}(V, E)$. Since E can be coded by an element of $V_{\lambda+1}$, we conclude that $V_{\lambda+1} \not\subseteq \text{Ult}(V, E)$ and that j_E is not $(\lambda + 1)$ -strong. $\square$

We now proceed to study Woodin cardinals.

Definition 3.8. A cardinal δ is *Woodin for $f : \delta \rightarrow \delta$* if there is an embedding $j : V \prec M$ with $\kappa := \text{crit } j < \delta$, $f[\kappa] \subseteq \kappa$, and $V_{j(f)(\kappa)} \subseteq M$. δ is *Woodin* if it is Woodin for every $f : \delta \rightarrow \delta$.

In the definition, we quantify over a class-sized object j . To rectify this, notice that $j[\kappa] \subseteq \kappa$ implies $\lambda := j(f)(\kappa) + \omega < j(\kappa)$, which allows us to derive a (κ, λ) -extender E from j . We may apply proposition 2.5, obtaining k with $\text{crit } k \geq \lambda$ which completes the diagram.

$$\begin{array}{ccc} V & \xrightarrow{j} & M \\ & \searrow j_E & \nearrow k \\ & \text{Ult}(V, E) & \end{array}$$

We have $V_{j(f)(\kappa)} \subseteq \text{Ult}(V, E)$ and

$$k(j(f)(\kappa)) = j(f)(\kappa) = k(j_E(f))(\kappa) = k(j_E(f)(\kappa)),$$

i.e. $j(f)(\kappa) = j_E(f)(\kappa)$. This means that we may have assumed that $j = j_E$, so the quantification should be considered to be over extenders E with the critical point $< \delta$.

We first establish the upper bound for this new large cardinal notion. We require the following lemma.

Lemma 3.9. *Let N_0, N_1 be κ -models agreeing up to $\kappa + 1$ and let F be both N_0 -extender and N_1 -extender on κ . Then $j_F^{N_0}(\kappa) = j_F^{N_1}(\kappa)$ and $\text{Ult}(N_0, F)$ and $\text{Ult}(N_1, F)$ agree up to $j_{N_0, F}(\kappa) + 1$.*

Proof. Let $v([a, f]_{N_0, F}) := [a, f]_F^{N_1}$ for $a \in [\lambda]^{<\omega}$ and $f \in ([\kappa]^{|a|} \rightarrow V_{\kappa+1}) \cap N_0 = ([\kappa]^{|a|} \rightarrow V_{\kappa+1}) \cap N_1$ (these sets being equal since such function can be coded as elements of $V_{\kappa+1}$). It is a matter of routine to verify that

$$v : N_0 \upharpoonright (j_{N_0, F}(\kappa) + 1) \rightarrow N_1 \upharpoonright (j_F^{N_1}(\kappa) + 1)$$

is a correctly defined bijection agreeing with $\in$, which is then necessarily the identity. Thus, $N_0 \upharpoonright (j_{N_0, F}(\kappa) + 1) = N_1 \upharpoonright (j_F^{N_1}(\kappa) + 1)$, as required. $\square$

Proposition 3.10. *A superstrong cardinal is Woodin. Moreover, Woodin cardinals are stationary below a superstrong cardinal.*

Proof.

- 1° Let δ be superstrong and let $j : V \prec M$ witness this fact.
- 2° Let $f : \delta \rightarrow \delta$ be arbitrary. We want to show that there is $\kappa < \delta$ closed for f and an embedding $k : V \prec N$ with $\text{crit } k = \kappa$ and $V_{k(f)(\kappa)} \subseteq N$.
- 3° It suffices to show that there is $\kappa < j(\delta)$ closed for $j(f)$ and an embedding $k : M \prec N$ in M satisfying $\text{crit } (k) = \kappa$ and $M \upharpoonright k(j(f))(\kappa) \subseteq N$.
- 4° We will show that $\kappa := \delta$ is as required.
- 5° For $\alpha < \delta$, we have $j(f)(\alpha) = j(f)(j(\alpha)) = j(f(\alpha)) = f(\alpha) < \delta$. In other words, δ is closed for $j(f)$.
- 6° Let $\lambda := j(f)(\delta) + \delta + \omega$.
- 7° We have $\lambda < j(\delta)$, since $j(f) : j(\delta) \rightarrow j(\delta)$.
- 8° Let E be the (δ, λ) -extender derived from j and let $l : \text{Ult}(V, E) \prec M$ with $\text{crit } l \geq \lambda$ complete the diagram

$$\begin{array}{ccc} & M & \\ & \uparrow j & \nwarrow l \\ V & \xrightarrow{j_E} & \text{Ult}(V, E) \end{array}$$

- 9° We have $E \in V_{j(\delta)} = M \upharpoonright j(\delta)$. We will show that $k := j_E^M : M \prec \text{Ult}(M, E)$ is as required.
- 10° $k(j(f))(\delta) = (k(j(f)) \upharpoonright k(\delta))(\delta) = k(j(f) \upharpoonright \delta)(\delta) = k(f)(\delta) = j_E^M(f)(\delta) = j_E(f)(\delta)$. The last equality is due to lemma 3.9.
- 11° We have $l(j_E(f)(\delta)) = (l \circ j_E)(f)(l(\delta)) = j(f)(\delta) < \lambda$, implying $j_E(f)(\delta) < \lambda < j(\delta)$.

12° $N \restriction k(j(f))(\delta) = \text{Ult}(M, E) \restriction j_E^M(f)(\delta) = \text{Ult}(V, E) \restriction j_E(f)(\delta) = M \restriction j_E(f)(\delta) = M \restriction k(j(f))(\delta)$. The first equality is due to 10°, the second one is due to lemma 3.9, the third one is due to $\text{crit } l \geq \lambda > j_E(f)(\delta)$, and the fourth one is again 10°.

13° It remains to verify that Woodin cardinals are stationary below δ .

14° Since δ is Woodin and $V_{\delta+1} = M \restriction (\delta + 1)$, we have $M \models \ulcorner \delta \text{ is Woodin} \urcorner$. This implies that the set of Woodin cardinals below δ is measure-one with respect to the normal measure on δ derived from j .

□

Let us now give lower bounds for Woodinness.

Proposition 3.11. *A Woodin cardinal δ is inaccessible and measurable cardinals are cofinal below it.*

Proof. To show that δ is regular, assume that there are $\lambda < \delta$ and a cofinal $f : \lambda \rightarrow \delta$. We may assume $f(0) > \lambda$. Let us extend f to $\bar{f} : \delta \rightarrow \delta$ by 0's and apply the definition of a Woodin cardinal. This yields $\kappa < \delta$ such that $\bar{f}[\kappa] \subseteq \kappa$. Since $f(0) > \lambda$, we have $\kappa > \lambda$, but then $\kappa \geq \sup f[\lambda] = \delta$, which is a contradiction.

To show that δ is a strong limit, it suffices to show that measurable cardinals are cofinal below it. Given $\alpha < \delta$, let j witness that δ is Woodin for $\delta \ni \xi \rightarrow \alpha + 1 \in \delta$. Then $\kappa := \text{crit } j$ is a measurable cardinal closed for this function, implying $\kappa > \alpha$.

□

Proposition 3.12. *If δ is Woodin, then*

$$V_\delta \models \ulcorner \text{strong cardinals are cofinal in Ord} \urcorner.$$

Proof. Suppose not, i.e. that there is an $\alpha < \delta$ such that for every $\beta > \alpha$ there is the least $\eta_\beta \in (\beta, \delta)$ such that β is not η_β -strong. Let us define

$$f(\beta) = \begin{cases} \alpha + 1, & \beta \leq \alpha \\ \eta_\beta + \omega, & \alpha < \beta < \delta \end{cases}$$

and let $j : V \prec M$, $\text{crit } j =: \kappa$, witness that δ is Woodin for f .

We have $V_{j(f)(\kappa)} \subseteq M$, where $j(f)(\kappa) = \eta + \omega$ and

$$M \models \ulcorner \kappa \text{ is not } \eta\text{-strong} \urcorner.$$

Let E be (κ, η) -extender derived from j ; $\text{Ult}(V, E) \restriction \eta = M \restriction \eta = V_\eta$ and $E \in M$. By applying proposition 3.9, we get $j_E^M(\kappa) = j_E^V(\kappa) \geq \eta$ and $\text{Ult}(M, E) \restriction \eta = \text{Ult}(V, E) \restriction \eta = M \restriction \eta$. This means that κ is η -strong in M , which is a contradiction.

□

In the remainder of the section, we describe reformulations of Woodinness.

Definition 3.13. Let δ be inaccessible and let A be a set of ordinals. We say that $\kappa < \delta$ is A -reflecting up to δ if for every $\lambda < \delta$ there is a $j : V \prec M$ with $\text{crit } j = \kappa$, $j(\kappa) > \lambda$, and $j(A \cap \kappa) \cap \lambda = A \cap \lambda$.

In other words, we can reconstruct $A \cap \delta$ from $A \cap \kappa$ using the appropriate embeddings.

Proposition 3.14. Let δ be inaccessible. Then δ is Woodin if and only if for every $A \subseteq \delta$ there is a $\kappa < \delta$ which is A -reflecting up to δ .

Proof. Let us first consider the direction $(\Rightarrow)$. Assume that there is an $A \subseteq \delta$ such that

$$(\forall \kappa < \delta)(\exists \lambda < \delta) \neg (\exists i : V \prec N)(\text{crit } i = \kappa \wedge i(\kappa) > \lambda \wedge i(A \cap \kappa) \cap \lambda = A \cap \lambda).$$

For every $\kappa < \delta$ choose $\lambda(\kappa) > \kappa$ minimal witnessing the previous fact and set $f(\kappa) := \lambda(\kappa) + \omega 2$. Since δ is Woodin, there is a $j : V \prec M$ with $\kappa := \text{crit } j < \delta$, $f[\kappa] \subseteq \kappa$, $V_{j(f)(\kappa)} \subseteq M$.

We now have that $\lambda^M(\kappa) < j(\delta)$ is the minimal λ witnessing

$$M \models \neg (\exists i : V \prec N)(\text{crit } i = \kappa \wedge i(\kappa) > \lambda \wedge i(j(A) \cap \kappa) \cap \lambda = j(A) \cap \lambda) \quad (2)$$

and that $j(f)(\kappa) = \lambda^M(\kappa) + \omega 2$. Let us denote $\lambda := \lambda^M(\kappa)$ and let E be $(\kappa, \lambda + \omega)$ -extender derived from j . There is $k : V \prec \text{Ult}(V, E)$ such that the following diagram commutes and $\text{crit } k \geq \lambda + \omega$.

$$\begin{array}{ccc} V & \xrightarrow{j} & M \\ & \searrow j_E^V & \nearrow k \\ & \text{Ult}(V, E) & \end{array}$$

We have $E \in V_{\lambda + \omega 2} = V_{j(f)(\kappa)} \subseteq M$, so, working in M , we have $j_E^M : M \prec \text{Ult}(M, E)$ with $\text{crit } j_E^M = \kappa$ and $j_E^M(\kappa) \geq \lambda + \omega > \lambda$. We now have

$$j_E^M(j(A) \cap \kappa) \cap \lambda = j_E^M(A \cap \kappa) \cap \lambda = j(A \cap \kappa) \cap \lambda, \quad (3)$$

which is a contradiction with (2).

Let us elaborate (3) further. The first equation follows from $\text{crit } j = \kappa$. For the second one, notice first that for $B \in P^M(\kappa) = P(\kappa)$ and $\alpha < \lambda$,

$$\begin{aligned} \alpha \in j_E^M(B) &\iff [\{\alpha\}, s \mapsto s_0]_E^M \in [\emptyset, \emptyset \mapsto B]_E^M \iff \{s \in [\kappa]^1 : s_0 \in B\} \in E_{\{\alpha\}} \\ &\iff [\{\alpha\}, s \mapsto s_0]_E^V \in [\emptyset, \emptyset \mapsto B]_E^V \iff \alpha \in j_E^V(B). \end{aligned}$$

This means that $j_E^M(B) \cap \lambda = j_E^V(B)$. In particular, $j_E^M(A \cap \kappa) \cap \lambda = j_E^V(A \cap \kappa) \cap \lambda$. Now, since $\text{crit } k \geq \lambda + \omega$, we have $j_E^V \upharpoonright V_{\kappa+1} = j \upharpoonright V_{\kappa+1}$, implying $j_E^V(A \cap \kappa) = j(A \cap \kappa)$ and concluding the argument.

We now consider the direction ($\Leftarrow$). First, inaccessible cardinals are cofinal in δ . For, given $\alpha < \delta$, we there is a α -reflecting $\kappa < \delta$. This is witnessed by some $j : V \prec M$ for $\lambda := \alpha + 1$, so $j(\alpha \cap \kappa) \cap \lambda = \{\alpha\} \cap \lambda \neq \emptyset$, implying $\kappa > \alpha$ and it is measurable.

Let $f : \delta \rightarrow \delta$ be arbitrary. Let us denote by $\Gamma = \langle \cdot, \cdot \rangle : \text{Ord} \times \text{Ord} \rightarrow \text{Ord}$ the Gödel's pairing function. For given $\alpha < \delta$, choose a binary relation E_α on $\beth_{f(\alpha)}$ such that $(\beth_{f(\alpha)}, E_\alpha) \cong (V_{\omega+f(\alpha)}, \in)$. Let

$$A := \{\langle i, \langle \alpha, \beta \rangle \rangle : (i = 0 \wedge \alpha < \delta \wedge \beta \in \Gamma[E_\alpha]) \vee (i = 1 \wedge \alpha < \delta \wedge \beta = f(\alpha))\}$$

and let κ be A -reflecting. For $\alpha < \delta$, let $\lambda_\alpha > \max\{\alpha, \sup\{f(\beta) : \beta \leq \alpha\}\}$ be the minimal inaccessible and fix $j_\alpha : V \prec M_\alpha$ satisfying $\text{crit } j_\alpha = \kappa$, $j_\alpha(\kappa) > \lambda_\alpha$, and $j_\alpha(A \cap \kappa) \cap \lambda_\alpha = A \cap \lambda_\alpha$.

For $\alpha < \delta$ and $(\beta, \gamma) \in E_\alpha$, we have

$$\langle 0, \langle \alpha, \langle \beta, \gamma \rangle \rangle \rangle \in A \cap \lambda_\alpha = j_\alpha(A \cap \kappa) \cap \lambda_\alpha.$$

This means that

$$E_\alpha = \{(\beta, \gamma) : \langle 0, \langle \alpha, \langle \beta, \gamma \rangle \rangle \rangle \in j_\alpha(A \cap \kappa)\} \in M_\alpha$$

and consequently its transitive collapse is also in M_α . We conclude that $V_{f(\alpha)} \subseteq V_{\omega+f(\alpha)} \subseteq M_\alpha$.

If we set $j := j_\kappa : V \prec M_\kappa =: M$, we have $\text{crit } j = \kappa$ and $V_{f(\kappa)} \subseteq M$, so we are done if we show that (1) $j(f)(\kappa) = f(\kappa)$ and (2) $f[\kappa] \subseteq \kappa$. For $\alpha \leq \kappa$, we have

$$\langle 1, \langle \alpha, f(\alpha) \rangle \rangle \in A \cap \lambda_\kappa = j(A \cap \kappa) \cap \lambda_\kappa,$$

i.e. $\langle 1, \langle \alpha, f(\alpha) \rangle \rangle \in j(A)$, so the elementarity of j applied to the definition of A implies $f(\alpha) = j(f)(\alpha)$. For (1), simply set $\alpha := \kappa$. For (2), we have

$$\lambda_\kappa > f(\alpha) = j(f)(\alpha) = j(f)(j(\alpha)) = j(f(\alpha))$$

for $\alpha < \kappa$. Since there are no fixed points of j in the interval $[\kappa, \lambda_\kappa)$, we have $f(\alpha) < \kappa$. $\square$

Proposition 3.15. *If δ is Woodin, then for every $f : \delta \rightarrow \delta$ there is an extender $E \in V_\delta$ such that $\kappa := \text{crit } E < \delta$, $f[\kappa] \subseteq \kappa$, and $V_{j_E(f)(\kappa)} \subseteq \text{Ult}(V, E)$.*

Proof. We continue the previous proof. Let E be the (κ, λ_κ) -extender derived from j and k the factor embedding. We need only to show that $V_{j_E(f)(\kappa)} \subseteq \text{Ult}(V, E)$. Since we know that $V_{f(\kappa)} \subseteq M$, $M \restriction \lambda_\kappa = \text{Ult}(V, E) \restriction \lambda_\kappa$, and $f(\kappa) < \lambda_\kappa$, it suffices to verify $j_E(f)(\kappa) = f(\kappa)$. We have

$$f(\kappa) = j(f)(\kappa) = k(j_E(f))(k(\kappa)) = k(j_E(f)(\kappa)).$$

Since the left side is $< \lambda_\kappa$, the same must be true for $j_E(f)(\kappa)$, but then $f(\kappa) = k(j_E(f)(\kappa)) = j_E(f)(\kappa)$. $\square$

Proposition 3.16. *The statement “ δ is Woodin for f ” is absolute for transitive models containing $V_\delta \cup \{f\}$. In particular, “ δ is Woodin” is downwards absolute for such models.*

Proof. Immediately from the previous proposition. $\square$

In the proposition 3.14, in order to show that the cardinal is Woodin, we assumed beforehand that it is inaccessible. This assumption can be omitted if we strengthen the notion of reflecting by requiring certain strength of the appropriate embeddings.

Definition 3.17. Let δ be a cardinal and A a set of ordinals. $\kappa < \delta$ is said to be *strong for A up to δ* if for every $\lambda < \delta$ there exists $j : V \prec M$ such that $\text{crit } j = \kappa$, $j(\kappa) > \lambda$, $V_\lambda \subseteq M$, and $A \cap \lambda = j(A \cap \kappa) \cap \lambda$.

Proposition 3.18. *δ is Woodin if and only if for every $A \subseteq \delta$ there is $\kappa < \delta$ which is strong for A up to δ .*

Proof. The implication $(\Rightarrow)$ is completely analogous to the same direction of the proposition 3.14. Let us consider the implication $(\Leftarrow)$. Let $f : \delta \rightarrow \delta$ be arbitrary and set $A := \Gamma[f]$. Let κ be strong for A up to δ and let $\lambda \in (\Gamma(\kappa, \sup\{f(\alpha) : \alpha \leq \kappa\}), \delta)$. We choose $j : V \prec M$ such that $\text{crit } j = \kappa$, $j(\kappa) > \lambda$, $V_\lambda \subseteq M$, and $A \cap \lambda = j(A \cap \kappa) \cap \lambda$. In particular, we have $V_{f(\kappa)} \subseteq M$, so we will be done if we show $f[\kappa] \subseteq \kappa$ and $j(f)(\kappa) = f(\kappa)$.

For $\alpha \leq \kappa$, we have

$$\Gamma(\alpha, f(\alpha)) \in A \cap \lambda = j(A \cap \kappa) \cap \lambda.$$

If $\alpha < \kappa$, we have

$$j(\alpha) = \alpha \in \text{dom } \Gamma^{-1}[j(A \cap \kappa)],$$

i.e. $\alpha \in \Gamma^{-1}[A \cap \kappa] = f \cap \kappa \times \kappa$, so $f(\alpha) < \kappa$. If $\alpha := \kappa$, we have $\Gamma(\kappa, f(\kappa)) \in j(A \cap \kappa) \cap \lambda$. On the other hand,

$$(\forall \alpha < \kappa) \Gamma(\alpha, f(\alpha)) \in A \cap \kappa,$$

implying that

$$(\forall \alpha < j(\kappa)) \Gamma(\alpha, j(f)(\alpha)) \in j(A \cap \kappa).$$

Setting $\alpha := \kappa$, we get $\Gamma(\kappa, j(f)(\kappa)) \in j(A \cap \kappa)$, so we can conclude that $j(f)(\kappa) = \kappa$. $\square$

Reasoning similarly as before, we can show that extenders can be picked in V_δ , i.e. that the Woodiness is equivalent to

$$(\forall A \subseteq \delta)(\exists \kappa < \delta)(\forall \lambda < \delta)(\exists E \in V_\delta \text{ strong } (\kappa, \lambda)\text{-extender})(A \cap \lambda = j(A \cap \kappa) \cap \lambda).$$

We also have the following proposition, whose proof is omitted since it does not contain anything that has not already been seen.

Proposition 3.19. *δ is Woodin if and only if for every $A \subseteq V_\delta$ there is $\kappa < \delta$ satisfying*

$$(\forall \lambda < \delta)(\exists E \in V_\delta \text{ strong } (\kappa, \lambda)\text{-extender})(A \cap V_\lambda = j(A \cap V_\kappa) \cap V_\lambda).$$

4 Extenders II

In this section, we study when an extender chosen in one model is an extender applicable to some other model. Also, we study the closure of Ult for sequences (of some length).

Proposition 4.1. *Suppose that:*

- a. N_i are μ -models ($i = 0, 1$);
- b. $j_i : (M_i, \in, \kappa) \prec (N_i, \in, \mu)$ are countable hulls ($i = 0, 1$);
- c. $j_1 \in M_1$ and $|j_1|^{M_1} = \aleph_0$;
- d. M_0 and M_1 agree up to $\kappa + 1$;
- e. $j_0 \upharpoonright (M_0 \upharpoonright (\kappa + 1)) = j_1 \upharpoonright (M_1 \upharpoonright (\kappa + 1))$;
- f. E is a (κ, λ) -extender in M_1 .

Then E is M_0 -extender and there is an embedding k such that the following diagram commutes.

$$\begin{array}{ccc}
 N_0 & & N_1 \\
 j_0 \uparrow & \swarrow k & \uparrow j_1 \\
 M_0 & \xrightarrow{j_E^{M_0}} & \text{Ult}(M_0, E) \\
 & & M_1
 \end{array}$$

The condition c. is satisfied if e.g. $N_1^\omega \subseteq \omega$.

Proof. The fact that E is M_0 -pre-extender follows from $P(\kappa) \cap M_0 = P(\kappa) \cap M_1$, while the well-foundedness of $\text{Ult}(M_0, E)$ follows from the realizability (i.e. from the existence of k). To define k , we first recall the mapping constructed in the proof of lemma 2.9, as applied to j_1 . Denoting it by k_1 , we have $k_1([a, f]_E^{M_1}) = j_1(f)(g[a])$, where $g : \lambda \rightarrow \text{Ord} \cap N_1$ is an increasing function in N_1 satisfying $g[a] \in \bigcap j_1[E_a]$ for $a \in [\lambda]^{<\omega}$.

$$\begin{array}{ccc}
 N_1 & & \\
 j_1 \uparrow & \swarrow k_1 & \\
 M_1 & \xrightarrow{j_E^{M_1}} & \text{Ult}(M_1, E)
 \end{array}$$

Similarly to it, we define $k([a, f]_{M_0, E}) := j_0(f)(g[a])$ for $a \in [\lambda]^{<\omega}$ and $f \in M_0^\kappa \cap M_0$. We note that the value on the right side is well defined, since $g[a] \in \text{dom}(j_1(f)) = [j_1(\kappa)]^{|a|} = [j_0(\kappa)]^{|a|} = \text{dom}(j_0(f))$. Let us verify that

the definition of k does not depend on the representative (a, f) and that k is elementary.

We have the following reasoning.

$$\text{Ult}(M_0, E) \models \phi([a_0, f_0]_{M_0, E}, \dots, [a_{n-1}, f_{n-1}]_{M_0, E}) \quad (4)$$

$$\implies \{s : M_0 \models \phi(f_0^{b, a_0}(s), \dots, f_{n-1}^{b, a_{n-1}}(s))\} \in E_b \quad (5)$$

$$\implies j_1(\{s : M_0 \models \phi(f_0^{b, a_0}(s), \dots, f_{n-1}^{b, a_{n-1}}(s))\}) \in j_1[E_b] \quad (6)$$

$$\implies j_0(\{s : M_0 \models \phi(f_0^{b, a_0}(s), \dots, f_{n-1}^{b, a_{n-1}}(s))\}) \in j_1[E_b] \quad (7)$$

$$\implies g[b] \in j_0(\{s : M_0 \models \phi(f_0^{b, a_0}(s), \dots, f_{n-1}^{b, a_{n-1}}(s))\}) \quad (8)$$

$$\implies N_0 \models \phi(j_0(f_0)(g[a_0]), \dots, j_0(f_{n-1})(g[a_{n-1}])) \quad (9)$$

The formula (3) follows by Łoś's theorem, where $b = \bigcup_{i < n} a_i$. For (4), we notice that the set in (3) is a subset of $[\kappa]^{[b]}$, so it can be coded as an element of $M_0 \upharpoonright (\kappa + 1) = M_1 \upharpoonright (\kappa + 1)$ and we can apply j_1 . For (5), we simply use the previous observation and the fact that $j_0 \upharpoonright (M_0 \upharpoonright (\kappa + 1)) = j_1 \upharpoonright (M_1 \upharpoonright (\kappa + 1))$. For (6), we use the choice of g , while (7) follows by elementarity of j_0 and the fact that g and $j_0 \upharpoonright \text{Ord}^{M_0}$ are order preserving.

Note that the commutativity of the diagram is obvious:

$$k(j_{M_0, E}(x)) = k([\emptyset, \emptyset \mapsto x]_{M_0, E}) = j_0(\emptyset \mapsto x)(j_1(\emptyset)) = (\emptyset \mapsto j_0(x))(\emptyset) = j_0(x).$$

□

We have the following corollary.

Proposition 4.2. *Suppose that:*

- a. N_0, N_1 are small μ -models;
- b. $N_0 \upharpoonright (\mu + 1) = N_1 \upharpoonright (\mu + 1)$;
- c. $N_1^\omega \subseteq N_1$;
- d. F be a (μ, ν) -extender in N_1 .

Then F is an N_0 -extender on κ .

Proof. It is obvious that F is an N_0 -pre-extender, so let us verify the well-foundedness. Let θ be large enough and let

$$j : (W, \in, M_0, M_1, \kappa, E) \prec (V_\theta, \in, N_0, N_1, \mu, F)$$

be a countable hull. We have that M_0 and M_1 are transitive countable sets and we have mappings $j_i := j \upharpoonright M_i : M_i \rightarrow N_i$ ($i = 0, 1$). Let us verify that they are elementary:

$$\begin{aligned} M_i \models \phi(x, \dots) &\iff W \models \phi^{M_i}(x, \dots) \\ &\iff V_\theta \models \phi^{N_i}(j(x), \dots) \\ &\iff N_i \models \phi((j \upharpoonright M_i)(x), \dots). \end{aligned}$$

Obviously M_0 and M_1 agree up to $\kappa + 1$, $j_0 \upharpoonright (M_0 \upharpoonright (\kappa + 1)) = j_1 \upharpoonright (M_1 \upharpoonright (\kappa + 1))$ (since they are restrictions of j), and E is an extender on κ in M_1 , so we may apply the previous proposition. This yields the well-foundedness of $\text{Ult}(M_0, E)$, so we have

$$W \models \ulcorner \text{Ult}(M_0, E) \text{ is well-founded} \urcorner;$$

by elementarity of j , we have

$$V_\theta \models \ulcorner \text{Ult}(N_0, F) \text{ is well-founded} \urcorner,$$

i.e. $\text{Ult}(N_0, F)$ is well-founded. □

Proposition 4.3. *Suppose that:*

- a. N_0, N_1 are transitive models of ZFC (possibly proper classes);
- b. $\mu \in N_0 \cap N_1$ and $N_0 \upharpoonright (\mu + 1) = N_1 \upharpoonright (\mu + 1)$;
- c. $N_1^\omega \subseteq N_1$;
- d. F is a (μ, ν) -extender in N_1 .

Then F is an N_0 -extender on μ .

Proof. We need to verify well-foundedness. We may assume w.l.o.g. that N_1 is a set: otherwise, we can replace it by $N_1 \upharpoonright \theta_1$ for a large enough θ_1 . Ordinals $\theta_0 \in \text{Ord}^{N_0}$ which are sufficiently large, of sufficiently large cofinality, and for which $N_0 \upharpoonright \theta_0$ is sufficiently elementary in N_0 are unbounded in Ord^{N_0} . By the previous proposition, every $\text{Ult}(N_0 \upharpoonright \theta_0, F)$ is well-founded, implying that $\text{Ult}(N_0, F)$ is well-founded as well. □

We now describe a situation in which the ultrapower is closed for α -sequences.

Proposition 4.4. *Suppose that:*

- a. N_0, N_1 are either small κ -models or transitive models of ZFC;
- b. $N_0 \upharpoonright (\kappa + 1) = N_1 \upharpoonright (\kappa + 1)$;
- c. $N_1 \models \ulcorner \lambda \text{ is inaccessible} \urcorner$;
- d. $N_1 \models \ulcorner F \text{ is a strong } (\kappa, \lambda)\text{-extender} \urcorner$;
- e. $\alpha \in [\omega, \kappa]$;
- f. N_0 and N_1 are α -closed.

Then F is an N_0 -extender and $\text{Ult}(N_0, F)$ is α -closed.

Proof. It has been already established that F is N_0 -(κ, λ)-extender. Let $([a_\xi, f_\xi] : \xi < \alpha)$ be a sequence of elements of $\text{Ult}(N_0, F)$. On one hand, we have

$$(a_\xi : \xi < \alpha) \in N_1 \cap V_\lambda = V_\lambda^{N_1} = V_\lambda^{\text{Ult}(N_1, F)} = V_\lambda^{\text{Ult}(N_0, F)} \subseteq \text{Ult}(N_0, F)$$

(the last equality being a consequence of lemma 3.9). On the other hand, we have

$$(j_{N_0, F}(f_\xi) : \xi < \alpha) = j_{N_0, F}([a_\xi : \xi < \alpha]) \restriction \alpha \in \text{Ult}(N_0, F).$$

These two together yield

$$([a_\xi, f_\xi] : \xi < \alpha) = (j_{N_0, F}(f_\xi)(a_\xi) : \xi < \alpha) \in \text{Ult}(N_0, F).$$

□

Next, we describe the copying construction.

Proposition 4.5. *Suppose that:*

- a. N_0, N_1 are small μ -models;
- b. $j_i : (M_i, \in, \kappa) \prec (N_i, \in, \mu)$ are countable hulls ($i = 0, 1$);
- c. $M_0 \restriction (\kappa + 1) = M_1 \restriction (\kappa + 1)$;
- d. $j_0 \restriction (M_0 \restriction (\kappa + 1)) = j_1 \restriction (M_1 \restriction (\kappa + 1))$;
- e. $M_1 \models \ulcorner E \text{ is a } (\kappa, \lambda)\text{-extender} \urcorner$;
- f. E is M_0 -extender;
- g. $F := j_1(E)$ is N_0 -extender.

Then there is k such that the following diagram commutes and $k \restriction [\lambda]^{<\omega} = j_1 \restriction [\lambda]^{<\omega}$.

$$\begin{array}{ccc} N_0 & \xrightarrow{j_F^{N_0}} & \text{Ult}(N_0, F) \\ j_0 \uparrow & & \uparrow k \\ M_0 & \xrightarrow{j_E^{M_0}} & \text{Ult}(M_0, E) \\ & & \uparrow j_1 \\ & & M_1 \end{array}$$

Proof. Let $k([a, f]_E^{M_0}) := [j_1(a), j_0(f)]_F^{N_0}$ for $a \in [\lambda]^{<\omega}$, $f \in M_0^\kappa \cap M_0$. First, the right side makes sense: $j_1(a) \in [j_1(\lambda)]^{<\omega}$ and $j_1(\lambda)$ is the length of F ; $\text{dom } j_0(f) = j_0([\kappa]^{|a|}) = [j_0(\kappa)]^{|a|} = [j_1(\kappa)]^{|j_1(a)|}$. Let us verify that the definition of k does not depend on the choice of (a, f) and that k is elementary.

We have the following reasoning.

$$\begin{aligned}
& \text{Ult}(M_0, E) \models \phi([a_0, f_0]_{M_0, E} \dots) & (10) \\
\iff & \{s : M_0 \models \phi(f_0^{b, a_0}(s) \dots)\} \in E_b & (11) \\
\iff & j_1(\{s : M_0 \models \phi(f_0^{b, a_0}(s) \dots)\}) \in j_1(E_b) & (12) \\
\iff & j_0(\{s : M_0 \models \phi(f_0^{b, a_0}(s) \dots)\}) \in F_{j_1(b)} & (13) \\
\iff & \{s : N_0 \models \phi(j_0(f_0)^{j_0(b), j_0(a_0)}(s) \dots)\} \in F_{j_1(b)} & (14) \\
\iff & \{s : N_0 \models \phi(j_0(f_0)^{j_1(b), j_1(a_0)}(s) \dots)\} \in F_{j_1(b)} & (15) \\
\iff & \text{Ult}(N_0, F) \models \phi([j_1(a), j_0(f)]_{N_0, F} \dots). & (16)
\end{aligned}$$

Equivalence (9) is Loś's theorem, with $b := \bigcup_i a_i$, while (10) is due to elementarity of j_1 and fact that the set is an element of $M_0 \upharpoonright (\kappa + 1) = M_1 \upharpoonright (\kappa + 1)$. The equivalence (11) comes from the agreement of j_0 and j_1 , (12) is due to elementarity of j_0 , while (13) is a consequence of the fact that j_0 and j_1 preserve order of ordinals. The equivalence (14) is again Loś's theorem, while keeping in mind that $j_1(b) = \bigcup_i j_1(a_i)$.

Let us verify commutativity of the diagram:

$$\begin{aligned}
k(j_E^{M_0}(x)) &= k([\emptyset, \emptyset \mapsto x]_E^{M_0}) = [j_1(\emptyset), j_0(\emptyset \mapsto x)]_F^{N_0} \\
&= [\emptyset, \emptyset \mapsto j_0(x)]_F^{N_0} = j_F^{N_0}(j_0(x)).
\end{aligned}$$

To show the agreement between k and j_1 , we compute for $a \in [\lambda]^{<\omega}$:

$$k(a) = k([a, s \mapsto s]_E^{M_0}) = [j_1(a), j_0(s \mapsto s)]_F^{N_0} = [j_1(a), s \mapsto s]_F^{N_0} = j_1(a).$$

□

5 Iteration trees

In this section, we introduce the iteration trees, which play a crucial role in the proof of projective determinacy. We prove some general facts about them as well. All the iterations will be *normal*, i.e. the extenders are applied to the first model to which they can be applied and the lengths of the extenders increase. Moreover, we will require that all the extenders are strong and of inaccessible length in the model in which they are chosen. This is all included in the main definition.

Definition 5.1. An *iteration* (or *iteration tree*) of length $\alpha > 0$ on M is a partial order $\triangleleft$ on α together with a sequence $E = (E_\xi : \xi < \alpha)$ satisfying the following properties, for uniquely determined sequences $(M_\xi : \xi < \alpha)$ and $(j_{\xi, \eta} : \xi \triangleleft \eta < \alpha)$.

1. $\triangleleft \subseteq <$, initial segments of $\triangleleft$ are linearly ordered, and 0 is the $\triangleleft$ -minimum. For limit $\xi < \alpha$, the set $\{\eta : \eta \triangleleft \xi\}$ is club in ξ .

2. M_0 is a transitive model of (a sufficient fragment of) ZFC.
3. Let ξ be such that $\xi + 1 < \alpha$ and let η be the $\triangleleft$ -predecessor of $\xi + 1$. Then:
 - $M_\xi \models \ulcorner E_\xi \text{ is a strong } (\kappa_\xi, \lambda_\xi)\text{-extender} \urcorner$;
 - $M_\xi \models \ulcorner \lambda_\xi \text{ is inaccessible} \urcorner$;
 - E_ξ is an M_η -extender;
 - E_ξ is not an $M_{\eta'}$ -extender for $\eta' < \eta$;
 - $M_{\xi+1} = \text{Ult}(M_\eta, E_\xi)$;
 - $j_{\eta, \xi+1}$ is the ultrapower embedding, while $j_{\zeta, \xi+1} = j_{\eta, \xi+1} \circ j_{\zeta, \eta}$ for $\zeta \triangleleft \eta$.
4. $\lambda_\xi < \lambda_\eta$ for $\xi < \eta < \eta + 1 < \alpha$
5. Let $\xi < \alpha$ be limit. The direct limit of the system $(M_\eta, j_{\eta, \zeta} : \eta \triangleleft \zeta \triangleleft \xi)$ is well-founded, M_ξ is equal to its transitive collapse, and $(j_{\eta, \xi} : \eta \triangleleft \xi)$ are the direct limit embeddings.

Due to condition 1, the structure $(\alpha, \triangleleft)$ is a tree with root 0 and without branching at limit levels. Its order agrees with the order of ordinals. For an iteration $\mathcal{T}$, we denote by $E_\xi^{\mathcal{T}}$ and $M_\xi^{\mathcal{T}}$ extenders and models of that iteration, omitting the superscript $\mathcal{T}$ when it is clear from the context. The similar remark holds for the lengths and critical points of the extenders.

5.1 Iterability

Iterations are deterministically constructed relative to (1) the extender sequence E and (2) the choices of branches at the limit stages. Namely, once $E_\xi \in M_\xi$ is chosen, $\triangleleft$ -predecessor η of $\xi + 1$ uniquely computed as the appropriate minimum (of a non-empty set, since E_ξ is an M_η -extender) and $M_{\xi+1} = \text{Ult}(M_\eta, E_\xi)$. On the other hand, for a limit ξ , once we choose a branch through $\triangleleft \restriction \xi \times \xi$ which is cofinal in ξ , ξ is put at the top of that branch and M_ξ is computed as the appropriate direct limit. Thus, we can formulate the following game for a model M .

The moves of the game are enumerated by ordinals $\xi \in [1, \alpha)$. At a successor step $\xi < \alpha$, player I chooses an extender $E_{\xi-1} \in M_{\xi-1}$ satisfying the point 3 of the definition, whose length is larger than the lengths of all the previous extenders. If I cannot choose such an extender, he loses. This extender is then applied to the smallest $\eta < \xi$ where it can be applied to produce a well-founded model; this model is denoted $M_\xi := \text{Ult}(M_\eta, E_{\xi-1})$. At a limit step $\xi < \alpha$, player II chooses a branch b through the tree $\triangleleft \restriction \xi \times \xi$ thus far constructed, with the requirement that b is club in ξ . If such a branch cannot be picked, II loses. If a branch is picked, the direct limit $(M_\eta, j_{\eta, \zeta} : \eta \triangleleft \zeta \in b)$ is computed; if it is ill-founded, II loses; otherwise, its transitive collapse is defined to be M_ξ and the game continues. If II does not lose at any step $\xi < \alpha$, then II wins. This game is called the α -iteration game on M .

Definition 5.2. M is α -iterable if II has a winning strategy in the α -iteration game on M .

It is obvious that if M is α -iterable and $\beta < \alpha$, then M is also β -iterable. Since player II cannot loose the iteration game if he is not required to make a move, we see that every model is α -iterable for $\alpha \leq \omega$. Since we are interested in the case when II has a winning strategy, the following questions arise immediately.

- a. Is there always a club branch? (We will also use the term *cofinal branch* for a club branch.)
- b. Is there always a cofinal branch which leads to a well-founded direct limit? (We will call such a branch a *well-founded branch*.)
- c. Can there be more than one well-founded branch?

In connection with the first two, we will show in the next section that V is $(\omega + 1)$ -iterable, i.e. for iterations of length ω on V , there always exist a well-founded branch. On question c., we say more in the penultimate subsection.

5.2 Properties of Iterations

We will require the following additional constraint on iterations.

Definition 5.3. An iteration tree $\mathcal{T}$ is said to be *non-overlapping* if for every $\eta + 1 \triangleleft \zeta + 1 < \text{length}(\mathcal{T})$, we have $\lambda_\eta \leq \kappa_\zeta$.

In other words, extenders applied along one branch do not overlap.

Proposition 5.4. Let $\mathcal{T}$ be a non-overlapping iteration of length α . Then for every $\xi < \eta < \alpha$, we have $M_\xi \restriction \lambda_\xi = M_\eta \restriction \lambda_\xi$, while $M_\xi \restriction (\lambda_\xi + 1) \supset M_\eta \restriction (\lambda_\xi + 1)$.

Proof. This is proved by induction on η . The case $\eta = 1$ follows from the facts that E_1 is a strong extender in M_0 (for the equality) and that $E_1 \notin \text{Ult}(M_0, E_1) = M_1$ (for the inequality). We next consider the successor case $\eta = \zeta + 1$.

We have $\xi \leq \zeta < \zeta + 1 = \eta$ and $M_\eta = \text{Ult}(M_\theta, E_\zeta)$, for $\mathcal{T}$ -predecessor θ of $\zeta + 1$. The following holds:

$$M_\eta \restriction \lambda_\zeta = \text{Ult}(M_\theta, E_\zeta) \restriction \lambda_\zeta = \text{Ult}(M_\zeta, E_\zeta) \restriction \lambda_\zeta = M_\zeta \restriction \lambda_\zeta.$$

The second equality is due to lemma 3.9 and the third one is due to the strength of E_ζ . On the other hand,

$$M_\xi \restriction \lambda_\xi = M_\zeta \restriction \lambda_\xi$$

by IH. Since $\lambda_\xi \leq \lambda_\zeta$, this concludes the proof of the equality.

To show the inequality, we consider two cases. The first one is $\xi = \zeta$, in which we have

$$\begin{aligned} M_\eta \restriction (\lambda_\xi + 1) &= \text{Ult}(M_\theta, E_\zeta) \restriction (\lambda_\zeta + 1) \\ &= \text{Ult}(M_\zeta, E_\zeta) \restriction (\lambda_\zeta + 1) \supset M_\zeta \restriction (\lambda_\zeta + 1) = M_\xi \restriction (\lambda_\xi + 1), \end{aligned}$$

where the first equality is trivial, the second one is due to the lemma 3.9, the inequality is due to $E_\zeta \notin \text{Ult}(M_\zeta, E_\zeta)$, and the last equality is also trivial. The second case is $\xi < \zeta$, in which we have

$$\begin{aligned} M_\eta \restriction (\lambda_\xi + 1) &= (M_\eta \restriction \lambda_\zeta) \restriction (\lambda_\xi + 1) = (M_\zeta \restriction \lambda_\zeta) \restriction (\lambda_\xi + 1) \\ &= M_\zeta \restriction (\lambda_\xi + 1) \subset M_\xi \restriction (\lambda_\xi + 1), \end{aligned}$$

where the first and the third equality are due to the fact that $\lambda_\xi < \lambda_\zeta$, the second one is due to the previous paragraph, while the inequality is due to the IH.

It remains to verify the case of a limit η . For this, it suffices to show that for every ζ_0, ζ_1 such that $\xi \triangleleft \zeta_0 \trianglelefteq \zeta_1 \triangleleft \eta$, we have $j_{\zeta_0, \zeta_1} \restriction (M_{\zeta_0} \restriction \lambda_\xi) = \text{id}_{M_{\zeta_0} \restriction \lambda_\xi}$. We proceed by induction on ζ_1 , where the cases of $\zeta_1 = \zeta_0$ and ζ_1 being a limit are obvious. Let us suppose that $\zeta_0 \triangleleft \zeta_1$ and $\zeta_1 = \zeta_2 + 1$. We first consider the case of ζ_0 being the $\triangleleft$ -predecessor of ζ_1 . Since the iteration is non-overlapping, we have $\kappa_{\zeta_2} \geq \lambda_\xi$. This implies $j_{\zeta_0, \zeta_1} = j_{E_{\zeta_2}}^{M_{\zeta_0}}$ is the identity on $M_{\zeta_0} \restriction \lambda_\xi$. If ζ_1 is not the $\triangleleft$ -successor of ζ_0 , we apply the previous conclusion to the $\triangleleft$ -predecessor of ζ_1 and ζ_1 , plus the IH to ζ_0 and that predecessor. $\square$

Definition 5.5. If $\mathcal{T}$ is a non-overlapping iteration, we define

$$M(\mathcal{I}) := \bigcup_{\xi < \alpha} M_\xi \restriction \lambda_\xi, \quad \delta(\mathcal{I}) := \sup\{\lambda_\xi : \xi < \alpha\}.$$

We now want to show that every iteration on an ω -closed model is non-overlapping.

Lemma 5.6. *Let $\mathcal{T}$ be a non-overlapping iteration on M , let $\alpha < \text{length}(\mathcal{T})$, and let $\rho_\alpha := \sup\{\lambda_\beta : \beta + 1 \trianglelefteq \alpha\}$. Then*

$$M_\alpha = \{j_{0, \alpha}(f)(a) : f \in M, a \in [\rho_\alpha]^{<\omega} \cap \text{dom } j_{0, \alpha}(f)\}.$$

Proof.

- 1° This is shown by induction on α .
- 2° The case $\alpha = 0$ is obvious.
- 3° Assume that $\alpha = \beta + 1$ and that the statement holds for ordinals $< \alpha$.
- 4° Let γ be the $\triangleleft$ -predecessor of α and let $E := E_\beta$.
- 5° Since $M_\alpha = \text{Ult}(M_\gamma, E)$, we have

$$M_\alpha = \{j_E(f)(a) : f \in M_\gamma, a \in \lambda_\beta\}.$$

- 6° For $f \in M_\gamma$, we have $f = j_{0,\gamma}(g)(b)$ for some $g \in M$ and $b \in [\rho_\gamma]^{<\omega}$.
- 7° Thus, every $x \in M_\alpha$ is of the form $x = (j_{0,\alpha}(g)(j_E(b)))(a)$ for $a \in \lambda_\beta$, $b \in [\rho_\gamma]^{<\omega}$, $g \in M$.
- 8° Since the iteration is non-overlapping, we have $\text{crit } j_E \geq \rho_\gamma$. This means that $j_E(b) = b$ and $x = (j_{0,\alpha}(g)(b))(a)$.
- 9° Let $c \in [\rho_\alpha]^{<\omega}$ code (b, a) in a Δ_0 manner. Let h be computed in the following manner: decode $c' \in [\rho_\alpha]^{<\omega}$ as (b', a') and return $(g(b'))(a')$. Obviously, $x = j_{0,\alpha}(h)(c)$.
- 10° It remains to consider the case of α being limit.
- 11° Let $x \in M_\alpha$ be arbitrary. Then $x = j_{\beta,\alpha}(y)$ for some $\beta \triangleleft \alpha$ and $y \in M_\beta$.
- 12° By the IH, we have $y = j_{0,\beta}(f)(a)$ for $f \in M$ and $a \in [\rho_\beta]^{<\omega}$.
- 13° This means $x = j_{0,\alpha}(f)(j_{\beta,\alpha}(a))$.
- 14° Since the iteration is non-overlapping, we have $\text{crit } j_{\beta,\alpha} \geq \rho_\beta$, implying $x = j_{0,\alpha}(f)(a)$.

□

Lemma 5.7. *Suppose that:*

- a. M is an ω -closed model;
- b. $\mathcal{T}$ is an iteration on M ;
- c. $\alpha + 1 < \text{length}(\mathcal{T})$;
- d. $\mathcal{T} \restriction \alpha$ is non-overlapping;
- e. β is the $\mathcal{T}$ -predecessor of $\alpha + 1$.

Then $\beta = \min\{\gamma \leq \alpha : \kappa_\alpha < \lambda_\gamma \vee \gamma = \alpha\}$. In particular, $\mathcal{T} \restriction (\alpha + 1)$ is non-overlapping.

Proof.

- 1° If $\gamma < \alpha$ and $\kappa_\alpha \geq \lambda_\gamma$, then E_α is not an M_γ -extender. This follows from 3.9.
- 2° It suffices to show that if γ is the minimal with $\gamma < \alpha$ and $\kappa_\alpha < \lambda_\gamma$, then E_α is an M_γ -extender.
- 3° Let $N := \text{Ult}(M_\gamma, E_\alpha)$. We want to show that N is well-founded.
- 4° Assume otherwise.

5° $\mathcal{T} \upharpoonright \alpha$ is a non-overlapping iteration. If E_α is applied to M_γ , the successor step of the previous proof is applicable here. Thus,

$$N = \{(j_E^{M_\gamma} j_{0,\gamma}^\mathcal{T})(f)(a) : f \in M, a \in [\lambda_\alpha]^{<\omega}\}.$$

6° Let $i := j_E^{M_\gamma} j_{0,\gamma}^\mathcal{T}$. There are sequences $(f_n : n < \omega) \in M^\omega$ and $(a_n : n < \omega) \in ([\lambda_\alpha]^{<\omega})^\omega$ such that

$$(\forall n < \omega)(i(f_{n+1})(a_{n+1}) \in^N i(f_n)(a_n)).$$

7° Since $M^\omega \subseteq M$, we have $(f_n : n < \omega) \in M$ and $(i(f_n) : n < \omega) = "i((f_n)_n) \in N$.

8° Note that $\text{Ult}(M_\alpha, E_\alpha) \upharpoonright (\lambda_\alpha + 1)$ is contained in the well-founded part of N and it is equal to $N \upharpoonright (\lambda_\alpha + 1)$. This is shown in the same manner as the proposition 3.9.

9° Inside N , let θ be large enough and let $k : W \prec V_\theta$ be a transitive hull for $\lambda_\alpha \cup \{i((f_n)_n)\}$ with $|W| = \lambda_\alpha$.

10° It follows that W belongs to the well-founded part of N .

11° On the other hand, $(k^{-1}(i(f_n))(a_n) : n < \omega)$ witnesses ill-foundedness for W . This is a contradiction.

□

Theorem 5.8. *Let M be an ω -closed model and let $\mathcal{T}$ be an iteration on M . Then $\mathcal{T}$ is non-overlapping.*

Proof. This follows by induction on $\text{length}(\mathcal{T})$. The basis is vacuous, the limit step is obvious, while the successor step follows from the previous lemma. □

We now study the closure for α -sequences for iterates of a given model.

Proposition 5.9. *Let $\mathcal{T}$ be an iteration on M of length $\alpha \leq \omega$ and let $\omega \leq \eta \leq \min\{\text{crit}(E_\xi) : \xi < \alpha\}$. If $M^\eta \subseteq M$, then $M_\xi^\eta \subseteq M_\xi$ for every $\xi < \alpha$.*

Proof. We show this by induction on α . The cases $\alpha = 1$ and $\alpha = \omega$ are trivial, so let us assume that $\alpha = n + 2$. Let k be the tree predecessor of $n + 1$. By the inductive hypothesis, M_k and M_n are closed for η -sequences, and by the proposition 4.4, so is $M_{n+1} = \text{Ult}(M_k, E_n)$. □

Closure for ω -sequences of M_ω need not hold: e.g. if V is iterated via single extender E on κ , then $j_{0,\omega}(\kappa) = \bigcup_{n < \omega} j_{0,n}(\kappa)$, while $j_{0,\omega}(\kappa)$ is measurable in M_ω ; i.e. $(j_{0,n}(\kappa) : n < \omega) \in M^\omega - M$. The previous proposition is directly applicable in the case $\eta := \omega$, while the case $\eta := 2^{\aleph_0}$ requires an additional argument.

Proposition 5.10. *Let $\mathcal{T}$ be an iteration on M of length $\alpha \leq \omega$ and let $\eta := 2^{\aleph_0}$. If $M^\eta \subseteq M$, then $M_\xi^\eta \subseteq M_\xi$ for every $\xi < \alpha$.*

Proof. We modify the previous proof in at the inductive step $\alpha = n + 2$. Since M_n is η -closed, we have that $(2^{\aleph_0})^{M_n} = 2^{\aleph_0} = \eta$. With κ_n be inaccessible in M_n , we conclude that $\eta < \kappa_n$. By the IH, M_k is η -closed as well, so we can apply proposition 4.4 as before, concluding that $M_{n+1} := \text{Ult}(M_k, E_n)$ is η -closed. $\square$

5.3 Cofinal Branches

If $\mathcal{T}$ is an iteration of length α with $\text{cof } \alpha > \omega$ and if b, c are two cofinal branches through $\mathcal{T}$, then $b \cap c$ is a club in α . This means that b and c cannot be different from some point on, implying that they are equal. This means that the question c. from above is non-trivial only in the case $\text{cof } \alpha = \omega$. In that case, we have the following.

Theorem 5.11. *Let M is an ω -closed inner model for ZFC and let $\mathcal{T}$ be an iteration on M of limit length. Suppose that there are two distinct cofinal well-founded branches through $\mathcal{T}$. Then*

$$L(M(\mathcal{T})) \models \ulcorner \delta(\mathcal{T}) \text{ is Woodin} \urcorner.$$

Proof. Let $\alpha := \sup(b \cap c) \in b \cap c$ (the membership is due to b, c being closed). α is not maximal in neither b nor c , since they are distinct. We may assume w.l.o.g. that $\min(b - (b \cap c)) < \min(c - (b \cap c))$. Let us define by recursion ordinals β_n and γ_n ($n < \omega$):

- $\beta_0 + 1 := \min(b - (b \cap c))$;
- $\gamma_n + 1 := \min(c - (\beta_n + 2))$;
- $\beta_{n+1} + 1 := \min(b - (\gamma_n + 2))$.

Let us denote by β_n^* and γ_n^* the $\mathcal{T}$ -predecessors of $\beta_n + 1$ and $\gamma_n + 1$, respectively. We introduce the following notation: $\kappa_n := \kappa_{\beta_n}^\mathcal{T}$, $\lambda_n := \lambda_{\beta_n}^\mathcal{T}$, $E_n := E_{\beta_n}^\mathcal{T}$, $\kappa'_n := \kappa_{\gamma_n}^\mathcal{T}$, $\lambda'_n := \lambda_{\gamma_n}^\mathcal{T}$, $E'_n := E_{\gamma_n}^\mathcal{T}$.

We claim that

$$\kappa_n < \lambda'_{n-1} \leq \kappa'_n < \lambda_n \leq \kappa_{n+1} < \lambda'_n$$

for $n \geq 1$. To show $\kappa_n < \lambda'_{n-1}$, we observe that the definition of β_n implies that $\beta_n^* \leq \gamma_{n-1} + 1 < \beta_n + 1$. Since $\beta_n^* \in b - c$ and $\gamma_{n-1} + 1 \in c$, we have $\beta_n^* \neq \gamma_{n-1} + 1$ and consequently $\beta_n^* \leq \gamma_{n-1}$. Since $E_n \in M_{\beta_n}^\mathcal{T}$ is applied to $M_{\beta_n^*}^\mathcal{T}$ and $M_{\beta_n^*}^\mathcal{T}$ and $M_{\beta_n}^\mathcal{T}$ agree exactly up to $\lambda_{\beta_n^*}^\mathcal{T}$, we have $\kappa_n < \lambda_{\beta_n^*}^\mathcal{T} \leq \lambda_{\gamma_{n-1}}^\mathcal{T} = \lambda'_n$. The inequalities $\kappa'_n < \lambda_n$ and $\kappa_{n+1} < \lambda'_n$ are demonstrated analogously, while the other two are due to the fact that the extenders along one branch are non-overlapping.

Notice that

$$\bigcup_{n < \omega} (\beta_n + 1) = \bigcup_{n < \omega} (\gamma_n + 1) = \text{length}(\mathcal{T})$$

(otherwise, this common value would belong to $b \cap c$). Since we have

$$\beta_n + 1 \leq \beta_{n+1}^* < \gamma_n + 1 \leq \gamma_{n+1}^* < \beta_{n+1} + 1$$

(the third inequality is showed in the previous paragraph, while the fifth one is analogous to it), it follows that

$$\bigcup_{n < \omega} \beta_n^* = \bigcup_{n < \omega} \gamma_n^* = \text{length}(\mathcal{T}).$$

Consequently,

$$\delta := \delta(\mathcal{T}) = \bigcup_{n < \omega} \lambda_n = \bigcup_{n < \omega} \lambda'_n = \bigcup_{n < \omega} \kappa_n = \bigcup_{n < \omega} \kappa'_n.$$

We want to show that for $A \in P(\delta) \cap M_b^{\mathcal{T}} \cap M_c^{\mathcal{T}}$, there is $\kappa < \delta$ which is strong for A up to δ and that this fact can be witnessed by extenders in $M(\mathcal{T})$. Let $n \geq 1$, $B \in M_{\beta_n^*}$, $C \in M_{\gamma_n^*}$ be such that $A = j_{\beta_n^*, b}(B) = j_{\gamma_n^*, c}(C)$. We change the enumeration of the sequences in this argument so that the index n becomes 0, but we do not change the notation. For $i < \omega$, set $B_i := j_{\beta_0^*, \beta_i^*}(B)$, $C_i := j_{\gamma_0^*, \gamma_i^*}(C)$, $j_i := j_{E_i}^{M_{\beta_i^*}}$, and $j'_i := j_{E'_i}^{M_{\gamma_i^*}}$. We have

$$\begin{aligned} A \cap \lambda_i &= j_{\beta_0^*, b}(B) \cap \lambda_i = j_{\beta_{i+1}, b}(j_{\beta_0^*, \beta_{i+1}}(B)) \cap \lambda_i = j_{\beta_0^*, \beta_{i+1}}(B) \cap \lambda_i \\ &= j_{\beta_i^*, \beta_{i+1}}(j_{\beta_0^*, \beta_i^*}(B)) \cap \lambda_i = j_{E_i}^{M_{\beta_i^*}}(B_i) \cap \lambda_i = j_i(B_i \cap \kappa_i) \cap \lambda_i. \end{aligned}$$

Similarly, we have

$$A \cap \lambda'_i = j'_i(C_i \cap \kappa'_i) \cap \lambda'_i.$$

It now follows that

$$C_i \cap \kappa'_i = j'_i(C_i \cap \kappa'_i) \cap \kappa'_i = A \cap \kappa'_i = j_i(B_i \cap \kappa_i) \cap \kappa'_i,$$

as well as

$$B_{i+1} \cap \kappa_{i+1} = j_{i+1}(B_{i+1} \cap \kappa_{i+1}) \cap \kappa_{i+1} = A \cap \kappa_{i+1} = j'_i(C_i \cap \kappa'_i) \cap \kappa_{i+1}.$$

In other words, if we set

$$\begin{aligned} X_0 &:= B \cap \kappa_0 \\ Y_n &:= j_n(X_n) \cap \kappa'_n \\ X_{n+1} &:= j'_n(Y_n) \cap \kappa_{n+1}, \end{aligned}$$

we can conclude that $A \cap \kappa'_i = Y_i$. Furthermore, we can obtain extenders $F_n := E_n \upharpoonright [\kappa'_n + \omega]^{<\omega} \in M_{\beta_n} \upharpoonright \lambda_n$ and $F'_n := E'_n \upharpoonright [\kappa_{n+1} + \omega]^{<\omega} \in M_{\gamma_n} \upharpoonright \lambda'_n$ (thus, all of them in $M(\mathcal{T})$) and reformulate the above computation as

$$\begin{aligned} X_0 &:= B \cap \kappa_0 \\ Y_n &:= j_{F_n}^{M_{\beta_n^*}}(X_n) \cap \kappa'_n \\ X_{n+1} &:= j_{F'_n}^{M_{\gamma_n^*}}(Y_n) \cap \kappa_{n+1}. \end{aligned}$$

Noting that $M(\mathcal{T}) \restriction (\kappa_n + 1) = M_{\beta_n^*} \restriction (\kappa_n + 1)$ and $M(\mathcal{T}) \restriction (\kappa'_n + 1) = M_{\gamma_n^*} \restriction (\kappa'_n + 1)$, we can reformulate the computation once more:

$$\begin{aligned} X_0 &:= B \cap \kappa_0 \\ Y_n &:= j_{F_n}^{L(M(\mathcal{T}))}(X_n) \cap \kappa'_n \\ X_{n+1} &:= j_{F'_n}^{L(M(\mathcal{T}))}(Y_n) \cap \kappa_{n+1}. \end{aligned}$$

This yields

$$L(M(\mathcal{T})) \models A \cap \kappa'_i = Y_i = (j_{F_i} \circ j_{F'_{i-1}} \circ j_{F_{i-1}} \circ \cdots \circ j_{F_0})(B \cap \kappa_0).$$

We want to verify that the previous embedding is κ'_i -strong. Once we do this, we can derive the appropriate extender from the above embedding and this will conclude the argument. Thus, we need to show that $M(\mathcal{T}) \restriction \kappa'_i \subseteq N'_i$, where

$$\begin{aligned} N_0 &:= L(M(\mathcal{T})) \\ N'_n &:= \text{Ult}(N_n, F_n) \\ N_{n+1} &:= \text{Ult}(N'_n, F'_n). \end{aligned}$$

This follows from the fact that $N_n \restriction (\kappa_n + \omega) = M(\mathcal{T}) \restriction (\kappa_n + \omega)$ and $N'_n \restriction (\kappa'_n + \omega) = M(\mathcal{T}) \restriction (\kappa'_n + \omega)$ ($n < \omega$), which we show by induction. The basis of the induction is obvious. Let us assume that

$$N_n \restriction (\kappa_n + \omega) = M(\mathcal{T}) \restriction (\kappa_n + \omega) = M_{\beta_n^*} \restriction (\kappa_n + \omega).$$

We can apply prop 3.9 to the pairs $M_{\beta_n^*}, M_{\beta_n}$ and N_n, M_{β_n} and conclude that

$$\begin{aligned} N_n \restriction (\kappa'_n + \omega) &= M_{\beta_{n+1}} \restriction (\kappa'_n + \omega) = \text{Ult}(M_{\beta_n}, F_n) \restriction (\kappa'_n + \omega) \\ &= M_{\beta_n} \restriction (\kappa'_n + \omega) = M(\mathcal{T}) \restriction (\kappa'_n + \omega). \end{aligned}$$

The similar reasoning applies to the step from N'_n to N_{n+1} . $\square$

It is a major open problem, called *Unique Branches Hypothesis*, which asks if there is always a unique cofinal branch. If cofinal branches are unique, the construction of an iteration is deterministic relative to the extender sequence.

5.4 Countable Hulls

Lemma 5.12. *Suppose that:*

- a. N is an ω -closed transitive model;
- b. $\mathcal{U}$ is an iteration of length $\alpha \leq \omega$ on N , with extenders $(F_n)_{n < \alpha}$;
- c. $k_0 = k : (M, \in, E_n : n < \omega) \prec (N, \in, F_n : n < \omega)$ is a countable hull;

Then $\mathcal{T} := (\triangleleft^{\mathcal{U}}, (E_n)_{n < \alpha})$ is an iteration on M of length α and there are embeddings k_n ($n < \alpha$) such that for all $m \triangleleft n$, the following diagram commutes and $k_m \restriction (M_m \restriction \lambda_m^{\mathcal{T}}) = k_n \restriction (M_n \restriction \lambda_m^{\mathcal{T}})$.

$$\begin{array}{ccc}
N_m & \xrightarrow{j_{m,n}^{\mathcal{U}}} & N_n \\
k_m \uparrow & & \uparrow k_n \\
M_m & \xrightarrow{j_{m,n}^{\mathcal{T}}} & M_n
\end{array}$$

Proof. The conclusion follows by inductive application of proposition 4.5. This proposition yields $k_m \upharpoonright \lambda_m^{\mathcal{T}} = k_n \upharpoonright \lambda_n^{\mathcal{T}}$, for $m \triangleleft n$. However, $M_m \upharpoonright \lambda_m^{\mathcal{T}} = M_n \upharpoonright \lambda_m^{\mathcal{T}}$ and $\lambda_m^{\mathcal{T}}$ is inaccessible in M_m , providing the agreement on the whole rank initial segments. $\square$

Under the conditions of the previous lemma, we denote $k^*\mathcal{U} := \mathcal{T}$ and call this tree a *copy of the tree $\mathcal{U}$ by k* .

Lemma 5.13. *Suppose that:*

- a. N is an ω -closed transitive model;
- b. $k = k_0 : M \prec N$ is a countable hull;
- c. $\mathcal{T}$ is an iteration on M of length $\alpha \leq \omega$;

Then $\mathcal{U} := (\triangleleft^{\mathcal{T}}, (k(E_n^{\mathcal{T}}))_{n < \omega})$ is an iteration on N and there are embeddings k_n ($n < \alpha$) such that the following diagram commutes and $k_m \upharpoonright (M_m \upharpoonright \lambda_m^{\mathcal{T}}) = k_n \upharpoonright (M_n \upharpoonright \lambda_m^{\mathcal{T}})$ for all $m \triangleleft n$.

$$\begin{array}{ccc}
N_m & \xrightarrow{j_{m,n}^{\mathcal{U}}} & N_n \\
k_m \uparrow & & \uparrow k_n \\
M_m & \xrightarrow{j_{m,n}^{\mathcal{T}}} & M_n
\end{array}$$

Proof. The proof is analogous to the previous one. $\square$

Under the conditions of the previous lemma, we denote $k_*\mathcal{T} := \mathcal{U}$ and call this tree a *copy of the tree $\mathcal{T}$ by k* .

Proposition 4.1 yields the realizability of one-step iterations. Below we improve this result by passing to the case of an arbitrary finite iteration.

Lemma 5.14. *Let N be a transitive model of ZFC, $N^\omega \subseteq N$, let $k : M \prec N$ be a countable hull, and let $\mathcal{T}$ be an iteration on M of length $n + 1$. Then there is an embedding $l : M_n \prec N$ such that the following diagram commutes.*

$$\begin{array}{ccc}
& N & \\
& \uparrow k & \nwarrow l \\
M & \xrightarrow{j_{0,n}^{\mathcal{T}}} & M_n
\end{array}$$

Proof. Let $k^*\mathcal{T}$ be the copy of $\mathcal{T}$ and k_i ($i < n+1$) the corresponding embeddings. We have the following diagram.

$$\begin{array}{ccc}
N & \xrightarrow{j_{0,n}^{k^*\mathcal{T}}} & N_n \\
\uparrow k & & \uparrow k_n \\
M & \xrightarrow{j_{0,n}^{\mathcal{T}}} & M_n
\end{array}$$

Since $N^\omega \subseteq N$, it follows that $M, M_n, k, k_n \in N$, as well as that $\text{cof Ord}^N > \omega$ and consequently $\{\theta < \text{Ord}^N : N \restriction \theta \prec N\}$ is a club in Ord^N . Choose an element θ of this club which is above $M, M_n, k, k_n, k^*\mathcal{T}$ and denote by $\bar{k} : M \prec N \restriction \theta$ the “restriction” of k . Let $\bar{k}^*\mathcal{T}$ be the copy tree on $N \restriction \theta$, with the embeddings $\bar{k}_i$ ($i < n+1$).

$$\begin{array}{ccc}
N \restriction \theta & \xrightarrow{j_{0,n}^{\bar{k}^*\mathcal{T}}} & N_n \restriction \theta \\
\uparrow \bar{k} & & \uparrow \bar{k}_n \\
M & \xrightarrow{j_{0,n}^{\mathcal{T}}} & M_n
\end{array}$$

Let us enumerate $M_n = \{e_i : i < \omega\}$ and denote by R the set of partial elementary embeddings $\sigma : \{e_i : i < u\} \prec N \restriction \theta$ satisfying

$$(\forall i < u)(\forall x \in M)(j_{0,n}^{\mathcal{T}}(x) = e_i \implies \sigma(e_i) = \bar{k}(x)),$$

ordered by the inclusion. We are attempting to embed M_n into $N \restriction \theta$ such that the triangle commutes.

$$\begin{array}{ccc}
N \restriction \theta & & \\
\uparrow \bar{k} & \nwarrow \sigma & \\
M & \xrightarrow{j_{0,n}^{\mathcal{T}}} & M_n
\end{array}$$

Notice that $R \in N$, so we can act on the whole situation by $j_{0,n}^{k^*\mathcal{T}}$ and get the poset $j_{0,n}^{k^*\mathcal{T}}(R)$ of attempts to embed M_n into $N_n \restriction \theta$ such that the following diagram commutes.

$$\begin{array}{ccc} & N_n \restriction \theta & \\ j_{0,n}^{k^*\mathcal{T}}(\bar{k}) \uparrow & \swarrow \sigma & \\ M & \xrightarrow{j_{0,n}^{\mathcal{T}}} & M_n \end{array}$$

However, we have

$$j_{0,n}^{k^*\mathcal{T}}(\bar{k}) = j_{0,n}^{k^*\mathcal{T}} \circ \bar{k} = j_{0,n}^{\bar{k}^*\mathcal{T}} \circ \bar{k} = \bar{k}_n \circ j_{0,n}^{\mathcal{T}},$$

i.e. $\sigma := \bar{k}_n$ is a solution for the previous diagram. This implies that $j_{0,n}^{k^*\mathcal{T}}(R)$ is ill-founded, but then so is R . We can now conclude that there is $\sigma : M_n \prec N \restriction \theta$ such that $\sigma \circ j_{0,n}^{\mathcal{T}} = \bar{k}$. This translates to an embedding $l : M_n \prec N$ which solves the following diagram.

$$\begin{array}{ccc} & N & \\ k \uparrow & \swarrow l & \\ M & \xrightarrow{j_{0,n}^{\mathcal{T}}} & M_n \end{array}$$

□

6 $(\omega + 1)$ -iterability

The goal of this section is to prove that V is $(\omega + 1)$ -iterable. We also include a proof of the existence of a realizable ω -branch.

6.1 The Proof

We start by studying the ill-foundedness of cofinal branches.

Lemma 6.1. *Let $\mathcal{T}$ be an iteration of length ω on V and let b be a cofinal branch through $\mathcal{T}$ such that $M_b^{\mathcal{T}}$ is ill-founded. Then there are ordinals $(\alpha_n : n \in b)$ such that $j_{m,n}^{\mathcal{T}}(\alpha_m) > \alpha_n$ for every $m \triangleleft n \in b$.*

Proof. We consider the increasing enumeration $b = \{b_n : n < \omega\}$. Let $(j_{b_{n'_i}, b}(\beta'_i) : i < \omega)$ be a decreasing sequence of ordinals in M_b . Let $n_0 := 0$ and choose β_0 such that $j_{0, b_{n'_0}}(\beta_0) > \beta'_0$; define recursively $n_{i+1} := n_i + n'_i + 1$ and $\beta_{i+1} := j_{b_{n'_i}, b_{n_{i+1}}}(\beta'_i)$. In this way, we obtain an increasing sequence of naturals $(n_i : i < \omega)$, starting from 0, and a sequence of ordinals $(\beta_i : i < \omega)$

satisfying $j_{b_{n_i}, b_{n_k}}(\beta_i) > \beta_k$ for all $i < k$. Now, given an arbitrary $n < \omega$, there is a unique i such that $n_i \trianglelefteq n \triangleleft n_{i+1}$; we define $\alpha_{b_n} := \omega j_{n_i, n}(\beta_i) + n_{i+1} - n$. The sequence $(\alpha_m : m \in b)$ is as required. $\square$

The choice of α_n 's in the previous lemma depends on the branch for which they are chosen. The following notion requires that such a choice can be uniformly, depended only on the tree.

Definition 6.2. An iteration tree $\mathcal{T}$ of length ω is said to be *continuously ill-founded* if there are ordinals $\alpha_n \in M_n^{\mathcal{T}}$ ($n < \omega$) satisfying the following condition:

$$(\forall m, n < \omega)(m \triangleleft n \implies j_{m, n}^{\mathcal{T}}(\alpha_m) > \alpha_n).$$

If b is a cofinal branch through $\mathcal{T}$, then $(j_{n, b}^{\mathcal{T}}(\alpha_n) : n \in b)$ is a decreasing ω -sequence of ordinals in $M_b^{\mathcal{T}}$, i.e. it witnesses its ill-foundedness. Thus, every cofinal branch of a continuously ill-founded iteration leads to the ill-founded direct limit. Moreover, decreasing sequences of ordinals witnessing this fact can be assign “continuously” to cofinal branches of the tree, hence explaining the name of the notion.

Lemma 6.3. *There is no continuously ill-founded iteration on V .*

Proof.

- 1° Assume otherwise, as witnessed by some iteration $\mathcal{U}$ and ordinals $(\beta'_n : n < \omega)$.
- 2° Let η be large enough with $\mathcal{U}, (\beta'_n)_n \in V_\eta$.
- 3° Let $\theta > \eta$ be large enough and let $\overline{\mathcal{U}}$ be the iteration on V_θ obtained from $\mathcal{U}$. The sequence $(\beta'_n : n < \omega)$ witnesses the fact that $\overline{\mathcal{U}}$ is continuously ill-founded.
- 4° Note that $j_{0, n}^{\overline{\mathcal{U}}}(\eta) = \eta$ and $\kappa_n^{\overline{\mathcal{U}}} < \lambda_n^{\overline{\mathcal{U}}} < \eta$ (for every $n < \omega$).
- 5° Let β_n be the β'_n th regular cardinal above $j_{0, n}^{\overline{\mathcal{U}}}(\eta) = \eta$ in $M_n^{\overline{\mathcal{U}}}$ for $n < \omega$.
- 6° Sequence $(\beta_n : n < \omega)$ witnesses continuous ill-foundedness of $\overline{\mathcal{U}}$.
- 7° $\lambda_n < \eta < \beta_n$ for every $n < \omega$.
- 8° Let

$$k : (M, \in, \mathcal{T}, \delta, \alpha_n : n < \omega) \prec (V_\theta, \in, \overline{\mathcal{U}}, \eta, \beta_n : n < \omega)$$

be a countable hull. We work with the iteration $\mathcal{T}$ and omit it as a superscript in what follows.

- 9° $(P_n, l_n : n < \omega)$ are constructed by recursion to satisfy the following:
 - a. P_n is a transitive model;
 - b. $l_n : M_n \upharpoonright \alpha_n \prec P_n$;

- c. $l_n \in P_n$ and $|l_n|^{P_n} = \omega$;
- d. $(\forall m < n)(l_n \restriction (M_n \restriction \lambda_m) = l_m \restriction (M_m \restriction \lambda_m))$;
- e. $(\forall n < \omega)(P_{n+1} \in P_n)$.

Once the construction is completed, we will have reached contradiction, as expressed by the last property.

- 10° For $n = 0$, set $P_0 := V_{\beta_0}$ and $l_0 := k \restriction (M \restriction \alpha_0)$.
- 11° Assume that $(P_i, l_i : i \leq n)$ have been defined. Let us define P_{n+1}, l_{n+1} .
- 12° Let m be the $\triangleleft$ -predecessor of $n + 1$.
- 13° There is an embedding l^* such that the diagram commutes and $l^* \restriction \lambda_n = l_n \restriction \lambda_n$.

$$\begin{array}{ccc}
 P_m & \xrightarrow{j_{l_n(E_n)}^{P_m}} & \text{Ult}(P_m, l_n(E_n)) \\
 l_m \uparrow & & \uparrow l^* \\
 M_m \restriction \alpha_m & \xrightarrow{j_{m,n+1}} & \text{Ult}(M_m \restriction \alpha_m, E_n)
 \end{array}
 \qquad
 \begin{array}{ccc}
 P_n & & \\
 l_n \uparrow & & \\
 M_n \restriction \alpha_n & &
 \end{array}$$

(This follows from the proposition 4.5. The mapping denoted by “ $j_{m,n+1}$ ” in the diagram is in fact the restriction of that mapping to $M_m \restriction \alpha_m$.)

- 14° Let $\gamma := j_{m,n+1}(\alpha_m)$. We have

$$\text{Ult}(M_m \restriction \alpha_m, E_n) = \text{Ult}(M_m, E_n) \restriction \gamma = M_{n+1} \restriction \gamma.$$

- 15° Since $M_{n+1} \restriction \lambda_n = M_n \restriction \lambda_n$ and λ_n is inaccessible in M_n , we have that

$$l^* \restriction (M_{n+1} \restriction \lambda_n) = l_n \restriction (M_n \restriction \lambda_n).$$

- 16° Let us denote $F := l_n(E_n) \in P_n$ and $P^* := \text{Ult}(P_m, F)$. The diagram with the new names:

$$\begin{array}{ccc}
 P_m & \xrightarrow{j_F^{P_m}} & P^* \\
 l_m \uparrow & & \uparrow l^* \\
 M_m \restriction \alpha_m & \xrightarrow{j_{m,n+1}} & M_{n+1} \restriction \gamma
 \end{array}
 \qquad
 \begin{array}{ccc}
 P_n & & \\
 l_n \uparrow & & \\
 M_n \restriction \alpha_n & &
 \end{array}$$

17° $\alpha_{n+1} < j_{m,n+1}(\alpha_m) = \gamma$ and consequently $\alpha^* := l^*(\alpha_{n+1}) \in P^*$.

18° The following holds:

- a. $P^* \restriction \alpha^*$ is a transitive model;
- b. $l^* \restriction (M \restriction \alpha_{n+1}) : M \restriction \alpha_{n+1} \prec P^* \restriction \alpha^*$;
- d. $(\forall n' < n+1)(l^* \restriction (M_{n+1} \restriction \lambda_{n'}) = l_{n'} \restriction (M_{n'} \restriction \lambda_{n'}))$.

19° There is l^{**} such that:

- a. $P^* \restriction \alpha^*$ is a transitive model;
- b. $l^{**} : M \restriction \alpha_{n+1} \prec P^* \restriction \alpha^*$;
- c. $l^{**} \in P^*$ and $|l^{**}|^{P^*} = \omega$;
- d. $(\forall n' < n+1)(l^{**} \restriction (M_{n+1} \restriction \lambda_{n'}) = l_{n'} \restriction (M_{n'} \restriction \lambda_{n'}))$.

Proof.

1' Let $t := l^* \restriction (M_n \restriction \lambda_n)$. We will ensure $l^{**} \restriction (M_n \restriction \lambda_n) = t$, which will provide for d.

2' $\phi := l_n(\lambda_n)$ is inaccessible in P_n .

3' Since $l_n \in P_n$, we have that $t : M_n \restriction \lambda_n \prec P_n \restriction \phi$ is in P_n and $|t|^{P_n} = \omega$.

4' $t \in P_n \restriction \phi$

5' $t \in \text{Ult}(P_n, F) \restriction \phi = P^* \restriction \phi$ by proposition 3.9.

6' $M_n \restriction \alpha_n \in P_n \restriction \phi = P^* \restriction \phi$, again by proposition 3.9.

7' In addition, $|M_n \restriction \alpha_n|^{P^*} = |M_n \restriction \alpha_n|^{P^* \restriction \phi} = |M_n \restriction \alpha_n|^{P_n \restriction \phi} = \omega$.

8' Since $M_m \restriction \alpha_m \in P_m$ and $|M_m \restriction \alpha_m|^{P_m} = \omega$, we have $j_{m,n+1}(M_m \restriction \alpha_m) = M_m \restriction \alpha_m$.

9' Thus, $M_m \restriction \alpha_m \in P^*$ and $|M_m \restriction \alpha_m|^{P^*} = \omega$.

10' $M_{n+1} \restriction \gamma \in P^*$ and $|M_{n+1} \restriction \gamma|^{P^*} = \omega$. (This follows from 6', 7', and 9'. Recall that $j_{m,n+1}(\alpha_m) = \gamma$.)

11' Let us enumerate $M_{n+1} \restriction \alpha_{n+1} = \{e_i : i < \omega\}$ inside P^* . (Recall that $\alpha_{n+1} < \gamma$.)

12' Let R be the set of all partial elementary embeddings

$$r : \{e_i : i < i_0\} \subseteq M_{n+1} \restriction \alpha_{n+1} \rightarrow P^* \restriction \alpha^* \ (i_0 < \omega)$$

such that

$$(\forall i < i_0)(e_i \in \text{dom } t \implies t(e_i) = r(e_i)).$$

Set $<_R := \supseteq$.

13' $(R, <_R) \in P^*$ (Recall that $\alpha^*, t \in P^*$.)

14' $(R, <_R)$ is ill-founded, as witnessed by the sequence $(l^* \restriction \{e_i\}_{i < i_0})_{i_0 < \omega}$.

15' $P^* \models \ulcorner (R, <_R) \text{ is ill-founded} \urcorner$

16' Let $(r_n)_{n < \omega}$ witness the previous fact. Let $l^{**} := \bigcup_{n < \omega} r_n \in P^*$.

□

20° There are P_{n+1} and l_{n+1} such that:

- a. P_{n+1} is a transitive model;
- b. $l_{n+1} : M_{n+1} \restriction \alpha_{n+1} \prec P_{n+1}$;
- c. $l_{n+1} \in P_{n+1}$ and $|l_{n+1}|^{P_{n+1}} = \omega$;
- d. $(\forall n' < n+1)(l_{n+1} \restriction (M_{n+1} \restriction \lambda_{n'}) = l_{n'} \restriction (M_{n'} \restriction \lambda_{n'}))$;
- e. $P_{n+1} \in P_n$.

Proof.

1' Let $t := l^* \restriction (M_n \restriction \lambda_n)$. We will ensure $l_{n+1} \restriction (M_n \restriction \lambda_n) = t$, which will provide for d.

2' $\phi := l_n(\lambda_n)$ is inaccessible in P_n and P^* . We have $P^* \restriction (\phi + 1) = P_n \restriction (\phi + 1)$.

3' There is $H \prec P^* \restriction \alpha^*$ such that $P^* \restriction \phi \cup \{l^{**}\} \subseteq H$ and $|H|^{P^*} = \phi$.

4' Let $\pi : H \rightarrow P_{n+1}$ be the transitive collapse and let $h : P_{n+1} \prec P^* \restriction \alpha^*$ be its inverse.

5' Let $l_{n+1} := \pi(l^{**})$.

6' $l_{n+1} : M_{n+1} \restriction \alpha_{n+1} \prec P_{n+1}$

Proof.

1'' Let $F(x \dots)$ be an arbitrary formula and suppose that

$$M_{n+1} \restriction \alpha_{n+1} \models F(a \dots).$$

$$2'' P^* \restriction \alpha^* \models F(l^{**}(a) \dots)$$

$$3'' P^* \restriction \alpha^* \models F(h(l_{n+1})(h(a)) \dots)$$

$$4'' P_{n+1} \models F(l_{n+1}(a) \dots)$$

□

7' $l_{n+1} \in P_{n+1}$ and $|l_{n+1}|^{P_{n+1}} = \omega$.

Proof.

$$1'' l^{**} \in H$$

$$2'' l_{n+1} = \pi(l^{**}) \in \pi[H] = P_{n+1}$$

$$3'' |l_{n+1}|^{P_{n+1}} = |l^{**}|^H = |l^{**}|^{P^* \restriction \alpha^*} = |l^{**}|^{P^*} = \omega \text{ (The first equality is due to the elementarity of } h, \text{ while the second is due to } H \prec P^* \restriction \alpha^*.)$$

□

8' We have $h(l_{n+1}) = l^{**} \supseteq t = h(t)$ (the last equality being due to $t \in P^* \restriction \phi$). We conclude $l_{n+1} \supseteq t$ and establish 20°d.

9' By 3' and 4', P_{n+1} is recursively coded by a set in $P(\phi) \cap P^*$. By 2', that set is in P_n as well. Hence, $P_{n+1} \in P_n$.

10' We have established all the necessary properties of P_{n+1}, l_{n+1} .

□

21° The point 20° concludes the construction.

□

Theorem 6.4 (Martin, Steel). *V is $(\omega + 1)$ -iterable.*

Proof. Assume otherwise. Let $\mathcal{T}$ be an iteration on V of length ω without a well-founded cofinal branch. For $n < \omega$, let $\hat{n}$ denote the set of all cofinal branches through $\mathcal{T}$ that contain n ; the set $\hat{0}$ is exactly the set of all the cofinal branches through $\mathcal{T}$. (Some of these sets might be empty.) Notice that $m \triangleleft n$ implies that $\hat{n} \subseteq \hat{m}$. Given a branch $b \in \hat{0}$, the fact that $M_b^{\mathcal{T}}$ is ill-founded implies due to lemma 6.1 the existence of a sequence $(\alpha_n^b : n \in b)$ satisfying $j_{m,n}(\alpha_m^b) > \alpha_n^b$ for all $m \triangleleft n \in b$.

Let R be the class of all pairs (m, f) , where $m < \omega$ and $f : \hat{m} \rightarrow \text{Ord}$, and let $<_R$ be a binary relation on it defined by

$$(m, f) <_R (n, g) : \iff m \triangleright n \wedge (\forall b \in \hat{m})(f(b) < g(b)).$$

It is obvious that $<_R$ is a partial order on R ; let us verify that it is well-founded. Otherwise, there would be a sequence (m_i, f_i) ($i < \omega$) witnessing its ill-foundedness, so we would have a branch

$$b := \bigcup_{i < \omega} [0, n_i]_{\mathcal{T}} \in \bigcap_{i < \omega} \hat{n}_i$$

and consequently a sequence $f_0(b) > f_1(b) > f_2(b) > \dots$, which is a contradiction. We denote by ρ the ranking function for $(R, <_R)$.

By proposition 5.10, $\hat{n}^{M_k} = \hat{n}$ and $(\text{Ord}^{\hat{n}})^{M_k} = \text{Ord}^{\hat{n}}$ for all $n, k < \omega$. In particular, $j_{0,n}^* R = R$, the push-forward of $<_R$ by $j_{0,n}$ is again $<_R$, and $j_{0,n}^* \rho = \rho$. Let $\phi_n : \hat{n} \rightarrow \text{Ord}$ be defined by $\phi_n(b) := \alpha_n^b$. We have $(n, \phi_n) \in R$ and we can correctly define $\beta_n := \rho((n, \phi_n))$ ($n < \omega$). For $m \triangleleft n$, we want to check that $j_{m,n}(\beta_m) > \beta_n$. In order to do so, it suffices to establish $j_{m,n}((m, \phi_m)) >_R (n, \phi_n)$. Since $j_{m,n}((m, \phi_m)) = (m, j_{m,n}(\phi_m)) = (m, j_{m,n} \circ \phi_m)$ and we already know that $m \triangleleft n$, it remains to verify that $(j_{m,n} \circ \phi_m)(b) > \phi_n(b)$ for all $b \in \hat{n}$, which is just another way of saying $j_{m,n}(\alpha_m^b) > \alpha_n^b$. Thus, the tree $\mathcal{T}$ is continuously ill-founded, which contradicts lemma 6.3. □

6.2 An Additional Realizability Fact

In lemma 5.14 we showed realizability of every finite length branch of an iterations. We can now consider the case of branches of length ω . The following result shows the existence of at least one such realizable branch.

Proposition 6.5. *Let θ be large enough, let $k : M \prec V_\theta$ be a countable hull, and let $\mathcal{U}$ be an iteration on M of length ω . Then there is a cofinal branch b through $\mathcal{U}$ and an embedding σ such that the following diagram commutes.*

$$\begin{array}{ccc} & V_\theta & \\ k \uparrow & \swarrow \sigma & \\ M & \xrightarrow{j_{0,b}^\mathcal{U}} & M_b^\mathcal{U} \end{array}$$

Proof. Assume otherwise. Let $\overline{\mathcal{T}} := k_*\mathcal{U} = k(\mathcal{U})$ be the copy of the tree $\mathcal{U}$, with $(k_n : n < \omega)$ the corresponding embeddings. $\overline{\mathcal{T}}$ is an iteration on V_θ and we denote by $\mathcal{T}$ the iteration on V with the same tree ordering and the same extenders. Let us enumerate $M_n^\mathcal{U} = \{e_i^n : i < \omega\}$ ($n < \omega$). We define a partial order R in the following matter. Given $n < \omega$ and denoting $u := |[0, n]_\mathcal{U}|$, consider partial elementary embeddings

$$\sigma_m : \{e_i^m : i < u\} \rightarrow V_\theta \quad (m \leq n)$$

that satisfy

$$(\forall m' \triangleleft m'' \leq n)(\forall i', i'' < u)(e_{i''}^{m''} = j_{m', m''}^\mathcal{U}(e_{i'}^{m'}) \implies \sigma_{m''}(e_{i''}^{m''}) = \sigma_{m'}(e_{i'}^{m'})),$$

$$\sigma_0 = k \upharpoonright \{e_i^0 : i < u\};$$

let $(\sigma_m : m \leq n) \in R$. The order on R is defined as follows:

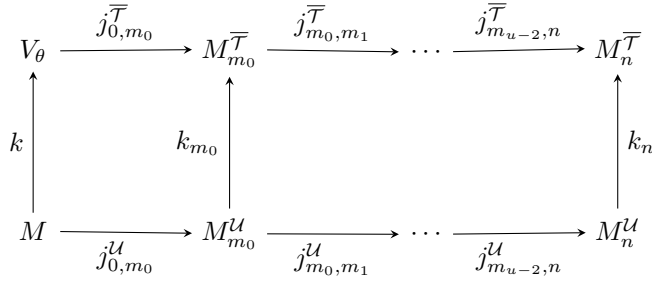
$$(\sigma_m^0 : m \leq n_0) \leq_R (\sigma_m^1 : m \leq n_1) : \iff n_0 \leq n_1 \wedge (\forall m \leq n_0)(\sigma_m^0 \subseteq \sigma_m^1).$$

The order R is a tree and a branch of length ω through it corresponds exactly to the following situation. We have the case of a branch b of length ω through $\triangleleft^\mathcal{U}$ and embeddings $\sigma_n : M_n^\mathcal{U} \prec V_\theta$ for $n \in b$ such that $\sigma_0 = k$ and the following diagram commutes for every $n' \triangleleft n'' \in b$.

$$\begin{array}{ccc} & V_\theta & \\ \sigma_{n'} \uparrow & \swarrow \sigma_{n''} & \\ M_{n'}^\mathcal{U} & \xrightarrow{j_{n', n''}^\mathcal{U}} & M_{n''}^\mathcal{U} \end{array}$$

However, this situation would provide for an embedding $\sigma : M_b^{\mathcal{U}} \prec V_\theta$ such that $\sigma \circ j_{0,b}^{\mathcal{U}} = k$, which is not the case by the assumption. Thus, $(R, >_R)$ is well-founded and we denote its ranking function by ρ .

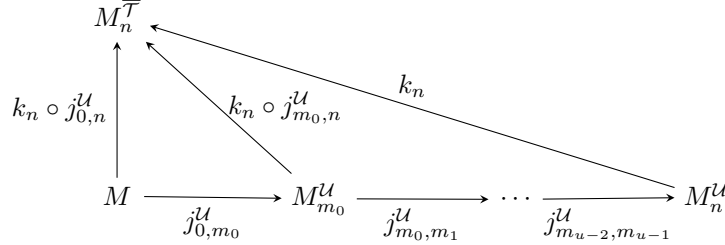
If we act by $j_{0,n}^{\mathcal{T}}$, we obtain a partial order $j_{0,n}^{\mathcal{T}}(R)$ of attempts to produce a cofinal branch through $\mathcal{U}$ which is realizable (in the sense of the previous paragraph) in $j_{0,n}^{\mathcal{T}}(V_\theta) = M_n^{\mathcal{T}} \upharpoonright \theta = M_n^{\mathcal{T}}$. The attempts of R are relative to the embedding k , so the attempts of $j_{0,n}^{\mathcal{T}}(R)$ are relative to $j_{0,n}^{\mathcal{T}}(k) = j_{0,n}^{\mathcal{T}} \circ k = j_{0,n}^{\mathcal{T}} \circ k = k_n \circ j_{0,n}^{\mathcal{U}}$. If we denote $[0, n]_{\mathcal{U}} = \{0 = m_0 \triangleleft m_1 \triangleleft \dots \triangleleft m_{u-1}\}$, we have the following situation.



This gives rise the diagram below and consequently an attempt

$$a_n := (k_n \circ j_{0,n}^{\mathcal{U}} \upharpoonright \{e_i^0 : i < u\}, k_n \circ j_{m_0,n}^{\mathcal{U}} \upharpoonright \{e_i^{m_1} : i < u\}, \dots, k_n \upharpoonright \{e_i^n : i < u\}) \in j_{0,n}^{\mathcal{T}}(R).$$

Let us define $\alpha_n := j_{0,n}^{\mathcal{T}}(\rho)(a_n)$.



We will show that the sequence $(\alpha_n : n < \omega)$ witnesses the fact that $\mathcal{T}$ is continuously ill-founded, which is a contradiction with lemma 6.3. Suppose that $n \triangleleft n'$; we want to see that $j_{n,n'}^{\mathcal{T}}(\alpha_n) > \alpha_{n'}$. Since $j_{n,n'}^{\mathcal{T}}(\alpha_n) = j_{n,n'}^{\mathcal{T}}(j_{0,n}^{\mathcal{T}}(\rho)(j_{0,n}^{\mathcal{T}}(a_n))) = j_{0,n'}^{\mathcal{T}}(\rho)(j_{n,n'}^{\mathcal{T}}(a_n))$, it suffices to verify that

$$j_{n,n'}^{\mathcal{T}}(a_n) <_{j_{0,n'}^{\mathcal{T}}(R)} a_{n'}.$$

In order to do so, first note that indices of the left tuple are contained among the indices of the right one since $n \triangleleft n'$. From here, let us consider an index $i \leq n$. Note that $j_{n,n'}^{\mathcal{T}}(a_n)(i)$ is the restriction of the embedding $j_{n,n'}^{\mathcal{T}}(k_n \circ j_{i,n}^{\mathcal{U}})$,

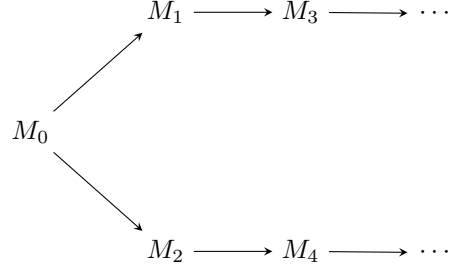
while $a_{n'}(i)$ is the restriction of the embedding $k_{n'} \circ j_{i,n'}^{\mathcal{U}}$. Now, the conclusion is obvious, having in mind the fact that

$$j_{n,n'}^{\mathcal{T}}(k_n \circ j_{i,n}^{\mathcal{U}}) = j_{n,n'}^{\mathcal{T}} \circ k_n \circ j_{i,n}^{\mathcal{U}} = j_{n,n'}^{\mathcal{T}} \circ k_n \circ j_{i,n}^{\mathcal{U}} = k_{n'} \circ j_{n,n'}^{\mathcal{U}} \circ j_{i,n}^{\mathcal{U}} = k_{n'} \circ j_{i,n'}^{\mathcal{U}}.$$

□

7 Building Iteration Trees

The goal of this section is to construct an iteration which is non-linear in a strong sense. The simplest iteration of that kind would have the following form.



We begin by introducing the necessary machinery.

Definition 7.1. A (κ, n) -type (for κ limit) is a type (in the model-theoretic sense) in the variables $x_0 \dots x_{n-1}$ over the language $\mathcal{L}_\kappa := \{\dot{\delta}, \dot{\kappa}, x : x \in V_\kappa\}$, containing the following formulas:

$$\ulcorner \dot{\kappa} < \dot{\delta} \urcorner, \quad \ulcorner \dot{\delta} \text{ is not the maximal ordinal} \urcorner, \quad \ulcorner \dot{\delta} \text{ is inaccessible} \urcorner.$$

A (κ, n) -type p can be coded as a subset of V_κ . The notion of a (κ, n) -type is a syntactical one and absolute for transitive models. The notions of a *realization* and *realizability* have their usual meaning, with some additional remarks. The models in which we are interested are rank-initial segments of inner models containing V_κ . All constant symbols except $\dot{\delta}$ have the obvious intended interpretation, which we never mention explicitly. We mention the interpretation of $\dot{\delta}$ when there is a need to do so. If κ is limit, $\delta > \kappa$ inaccessible, $\xi > \delta + 1$, and $\bar{a} \in V_\xi$, then we denote by

$$\text{tp}_\kappa^{V_\xi, \delta}(\bar{a}) = \text{tp}_\kappa(\bar{a}, V_\xi, \delta)$$

the type of the tuple $\bar{a}$ in the structure $(V_\xi, \in, \delta)$ over the language $\mathcal{L}_\kappa$. We omit δ from the notation when it is clear from the context. When relativizing this notation to an inner model M , we simply replace V_ξ by $M \upharpoonright \xi$.

Definition 7.2. Let p be a (κ, m) -type and let $\lambda \leq \kappa$ and $n \leq m$. The (λ, n) -projection of p is the (λ, n) -type defined by

$$\text{proj}_\lambda^n(p) := \{\phi(x_0 \dots x_{n-1}) \in \text{For } \mathcal{L}_\lambda : \phi(x_0 \dots x_{m-1}) \in p\}.$$

This notion covers the two usual projections between type spaces: one obtained from restricting the allowed variables and the other one from restricting the language. This notion is syntactical and absolute. We use the following abbreviations:

$$\text{proj}^n := \text{proj}_\kappa^n, \quad \text{proj}_\lambda := \text{proj}_\lambda^m.$$

Definition 7.3. A (κ, m) -type p is said to be subsumed under a (λ, n) -type q if

1. $\kappa < \lambda$;
2. $m \geq n$;
3. $q \models \ulcorner \text{there is } \xi \text{ with } x_0 \dots x_{n-1}, \delta + 1 \in V_\xi \text{ and there are } a_n \dots a_{m-1} \in V_\xi \text{ such that some permutation of } x_0 \dots x_{n-1}, a_n \dots a_{m-1} \text{ realizes } \dot{p} \text{ in } (V_\xi, \in, \dot{\delta}) \urcorner$.

The first condition provides for the fact that $\dot{p} \in \mathcal{L}_\lambda$. The third condition asserts that a certain formula belongs to q . This formula has free variables $x_0 \dots x_{n-1}$ and mentions the constants $\dot{\delta}, \dot{p}$, as well as $\dot{\kappa}$ which is implicit in our notation for an $\mathcal{L}_\kappa$ -structure. The notion introduced in the previous definition is syntactical and absolute for transitive models containing p, q . Let us consider the case when p is subsumed under q (written $p < q$) and $a_0 \dots a_{n-1}$ realize q in some V_η , relative to some δ . This means that there is $\xi < \eta$ with $a_0 \dots a_{n-1}, \delta + 1 \in V_\xi$ and some $a_n \dots a_{m-1} \in V_\xi$ such that for some permutation $\bar{a}'$ of $\bar{a}$,

$$(V_\xi, \in, \delta) \models p(\bar{a}').$$

(Here and in the next definition we allow for permutations due to technical reasons which will become apparent later.) In the other directions, if $\kappa < \lambda < \delta < \delta + 1 < \xi < \eta$, $m \geq n$, and $\bar{a} \in V_\xi$, we have

$$\text{tp}_\kappa^{V_\xi}(a_0 \dots a_{m-1}) < \text{tp}_\lambda^{V_\eta}(a_0 \dots a_{n-1}).$$

Definition 7.4. A (κ, m) -type p is said to exceed a (λ, n) -type q if

1. $\kappa > \lambda$;
2. $m \geq n$;
3. There are $\delta < \delta + 1 < \xi < \xi + 1 < \eta$ and $a_0 \dots a_m \in V_\xi$ such that

$$(V_\eta, \in, \delta) \models q(a_0 \dots a_{n-1}), \quad (V_\xi, \in, \delta) \models p(a'_0 \dots a'_{m-1}),$$

where $\bar{a}'$ is some permutation of $\bar{a}$.

This notion is conditioned on realizability in rank-initial segments and it is not absolute. One obvious way to obtain exceeding types is to start from $\lambda < \kappa < \delta < \delta + 1 < \xi < \xi + 1 < \eta$ and some $\bar{a} \in V_\xi$ and set $p := \text{tp}_\kappa^{V_\xi}(\bar{a})$, $q := \text{tp}_\lambda^{V_\eta}(\bar{a})$. A connection between the two previous notions is obtained through the notion of *elasticity* of types, which we describe below.

Definition 7.5. A (κ, n) -type p is said to be second-order-like if

1. $p \models \exists \eta (\eta = \max \text{Ord})$;
2. $p \models \bar{x}, \dot{\delta} + 1 \in V_{\max \text{Ord}}$.

The first-order part of a second-order-like (κ, n) -type p is a (κ, n) -type p^- defined by

$$p^- := \{\phi(\bar{x}) : p \models (V_{\max \text{Ord}} \models \phi(\bar{x}))\}.$$

The previous notions are purely syntactical and absolute. If p is second-order-like and $V_{\eta'} \models p(\bar{a})$, then $\eta' = \eta + 1$ for some η , with all the constants as well as $\bar{a}$ belonging to V_η . In other words, $V_{\eta'}$ can be understood as the full second order structure corresponding to $(V_\eta, \in, \delta)$, p as the second-order type of the first-order tuple $\bar{a}$, and p^- as the first-order type of the same tuple. This explains the names.

Definition 7.6. Let p be (κ, n) -type and let E be a λ -strong extender on κ . The stretch of the type p by E is defined to be the following (λ, n) -type:

$$p[E]^\lambda := \text{proj}_\lambda(j_E(p)).$$

Since p is a (κ, n) -type in V , we have that $j_E(p)$ is a $(j_E(\kappa), n)$ -type in $\text{Ult}(V, E)$, as well as that $p[E]^\lambda$ is (λ, n) -type in $\text{Ult}(V, E)$. Considering that V and $\text{Ult}(V, E)$ agree up to λ , we can conclude that $p[E]^\lambda$ is a (λ, n) -type in V .

Definition 7.7. A (κ, n) -type p is said to be elastic if

1. it is second-order-like;
2. $p \models \ulcorner$ for every limit $\lambda \in (\kappa, \dot{\delta})$, there exists a λ -strong extender $E \in V_{\dot{\delta}}$ such that

$$\text{tp}_{\kappa}^{V_\eta}(\bar{x})[E]^\lambda = \text{tp}_\lambda^{V_\eta}(\bar{x}),$$

where η is the maximal ordinal $\ulcorner$.

This is again a purely syntactical, absolute notion. If $(V_{\eta+1}, \in, \delta) \models p(\bar{a})$, then $\text{tp}_{\kappa}^{V_\eta}(\bar{a}) = p^-$ and for every $\lambda \in (\kappa, \delta)$, we have that the analogous type in the larger language $\mathcal{L}_\lambda$ is obtained by stretching p^- via an appropriate extender.

Proposition 7.8. *If a (λ, n) -type q exceeds a (κ, m) -type p and p is elastic, then there is a $(\lambda + \omega)$ -strong extender E on κ such that q is subsumed under $p[E]^{\lambda+\omega}$.*

In other words, if q exceeds an elastic type p , then p can be stretched so that it subsumes q .

Proof. Let ξ, η be ordinals with $\eta + 1 < \xi$ and let $\bar{a}, \delta \in V_\eta$ be such that

$$(V_\xi, \in, \delta) \models p(a_0 \dots a_{m-1}), \quad (V_\eta, \in, \delta) \models q(a_0 \dots a_{n-1}).$$

Since p is elastic and $\kappa < \lambda < \lambda + \omega < \delta$, we have a $(\lambda + \omega)$ -strong extender $E \in V_\delta$ satisfying

$$\text{tp}_{\lambda+\omega}^{V_{\xi-1}}(a_0 \dots a_{m-1}) = p^- \upharpoonright_E^{\lambda+\omega}.$$

However, $\eta < \xi - 1$ and

$$q = \text{tp}_\lambda^{V_\eta}(a_0 \dots a_{n-1}) < \text{tp}_{\lambda+\omega}^{V_{\xi-1}}(a_0 \dots a_{m-1}) = p^- \upharpoonright_E^{\lambda+\omega}.$$

□

Woodin cardinals imply the existence of elastic types.

Proposition 7.9. *Let δ be Woodin, $\xi > \delta + 1$, and $\bar{a} \in V_\xi$. Then the set*

$$\{\kappa < \delta : \text{tp}_\kappa^{V_{\xi+1}}(\bar{a}) \text{ is elastic}\}$$

is stationary in δ .

Proof. Let C be an arbitrary club. We set $B := \text{tp}_\delta^{V_\xi}(\bar{a})$ and $A := B \times \{0\} \cup C \times \{1\}$. There is κ witnessing the characterization of proposition 3.19. We will show that $\kappa \in C$ and that $\text{tp}_\kappa^{V_\xi}(\bar{a})$ is elastic. For $\lambda > \kappa$ limit, let E be a strong (κ, λ) -extender such that $A \cap V_\lambda = j_E(A \cap V_\kappa) \cap V_\lambda$. This implies $A \cap V_\lambda = j_E(A \cap V_\kappa) \cap V_\lambda$ and consequently $B \cap V_\lambda = j_E(B \cap V_\kappa) \cap V_\lambda$ and $C \cap V_\lambda = j_E(C \cap V_\kappa) \cap V_\lambda$.

The first equality shows that

$$\begin{aligned} \text{tp}_\lambda^{V_\xi}(\bar{a}) &= \text{tp}_\delta^{V_\xi}(\bar{a}) \cap \lambda = j_E(\text{tp}_\delta^{V_\xi}(\bar{a}) \cap V_\kappa) \cap V_\lambda \\ &= j_E(\text{tp}_\kappa^{V_\xi}(\bar{a})) \cap V_\lambda = \text{proj}_\lambda j_E(\text{tp}_\kappa^{V_\xi}(\bar{a})) = \text{tp}_\kappa^{V_\xi}(\bar{a}) \upharpoonright_E^\lambda \end{aligned}$$

(we assume the appropriate coding of the syntax). Since λ was arbitrary, we conclude elasticity. On the other hand, if we had chosen λ as a limit point of C above κ , the equality $C \cap \lambda = j_E(C \cap \kappa) \cap \lambda$ would imply that $j_E(C \cap \kappa) \cap \lambda$ is cofinal in λ . If $C \cap \kappa$ were bounded in κ , by say $\alpha < \kappa$, then $j_E(C \cap \kappa) \cap \lambda$ would be bounded by $j_E(\alpha) = \alpha < \kappa < \lambda$, which is a contradiction. Thus, $C \cap \kappa$ is cofinal in κ and since C is closed, we have $\kappa \in C$. □

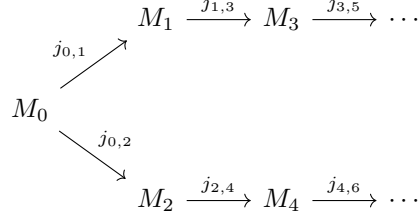
Definition 7.10. Let δ be inaccessible and let $\delta < \iota_L < \iota_H$. The pair (ι_L, ι_H) is said to be a *pair of local indiscernibles relative to δ* if

$$\text{tp}_\delta^{V_{\iota_L+\omega}}(\iota_L) = \text{tp}_\delta^{V_{\iota_H+\omega}}(\iota_H).$$

In other words ι_L, ι_H are indiscernible with respect to formulas of the form $\phi^{V_{x+\omega}}(x)$, which may use constants for elements of $V_\delta \cup \{\delta\}$. (The subscripts stand for “low” and “high”.) It is easy to establish that a pair of local indiscernibles always exists. Namely, to each $\xi \in (\delta, (2^\delta)^+)$ we may assign $\text{tp}_\delta^{V_{\xi+\omega}}(\xi)$, which can take at most 2^δ different values. The cardinality comparison now implies that two ξ ’s must have been assigned the same value; they are exactly local indiscernibles.

We are ready to construct an alternating chain iteration mentioned at the beginning of this section. In order to simplify the indexing, we allow some of the embeddings in an iteration to be trivial.

Theorem 7.11. *If there is a Woodin cardinal, then there is an iteration on V of length ω of the following form.*



Furthermore, the iteration can be constructed so that only the branch $\{0, 1, 3, 5, \dots\}$ is well-founded.

Proof. We make ad hoc adjustments to our notation in order to simplify the form of the proof. The diagram that we have in mind is the following one.

$$\begin{array}{ccccccc}
 M_0 & \xrightarrow{j_{0,1}} & M_1 & \xrightarrow{j_{1,2}} & M_2 & \xrightarrow{j_{2,3}} & \dots \\
 \parallel & & & & & & \\
 N_0 & \xrightarrow{k_{0,1}} & N_1 & \xrightarrow{k_{1,2}} & N_2 & \xrightarrow{k_{2,3}} & \dots
 \end{array}$$

In other words, we assume that the first embedding is the identity on $M_0 = V$ and that the order of the models is $M_0, N_0, M_1, N_1, M_2, \dots$. The embeddings and extenders of the M -branch are denoted by j 's and E 's respectively, while those of the N -branch are denoted by k 's and F 's. Thus, we have $j_{i,i+1} = j_{E_i}^{M_i}$ and $k_{i,i+1} = j_{F_i}^{N_i}$, while $E_i \in N_i$ and $F_i \in M_{i+1}$. The critical point of E_i , length of E_i , critical point of F_i , length of F_i are respectively denoted by $\kappa_i, \lambda_i, \mu_i, \nu_i$.

Let δ be a Woodin cardinal and let ι_L, ι_H be local indiscernibles relative to δ . We construct the iteration by recursion. We will ensure that

$$\kappa_i < \mu_i < \lambda_i < \kappa_{i+1} < \nu_i < \mu_{i+1},$$

which will provide for the required form of the tree. All the extenders will be chosen of rank $< \delta$, so δ will remain fixed throughout the iteration. We will also define types p_i, q_i in models M_i, N_i and ordinals $\alpha_i \in N_i$ satisfying

- $M_i \models (p_i = \text{tp}_{\kappa_i}^{j_{0,i}(\iota_L)+3}(j_{0,i}(\iota_L)))$ and it is elastic);
- $N_i \models (q_i = \text{tp}_{\mu_i}^{\alpha_i+1}(\alpha_i))$ and it is elastic);
- $N_i \models (q_i \text{ exceeds } p_i)$;
- $M_{i+1} \models (p_{i+1} \text{ exceeds } \text{proj}^0(q_i))$;
- $\alpha_{n+1} < k_{n,n+1}(\alpha_n)$.

The last condition ensures that the N -branch is ill-founded. Since V is $(\omega+1)$ -iterable, it will follow that M -branch is well-founded.

The steps listed below describe the base of the recursion. We are describing the following part of the diagram, together with the types p_0, q_0, p_1 and the ordinal α_0 .

$$\begin{array}{ccc}
M_0 & \longrightarrow & M_1 \\
\parallel & & \\
N_0 & &
\end{array}$$

1. $M_0 = N_0 = V$
2. Let $\kappa_0 < \delta$ be such that $p_0 := \text{tp}_{\kappa_0}^{\iota_L+3}(\iota_L)$ is elastic.
3. Let $\mu_0 \in (\kappa_0, \delta)$ be such that $q_0 := \text{tp}_{\mu_0}^{\iota_L+1}(\iota_L)$ is elastic.
4. The type q_0 exceeds p_0 . (This follows from the previous two points.)
5. $\alpha_0 := \iota_L$
6. There is an inaccessible $\lambda_0 \in (\mu_0, \delta)$ and a strong (κ_0, λ_0) -extender $E_0 \in N_0 \restriction \delta$ such that $q_0 < p_0^-[\lambda_0]_{E_0}$. (This follows from the points 4 and 2.)
7. $N_1 := \text{Ult}(M_0, E_0)$
8. $M_0 \models (V_{\iota_L+2} \models p_0^-[\iota_L])$
(This follows from point 2.)
9. $M_1 \models (V_{j_{0,1}(\iota_L)+2} \models j_{0,1}(p_0^-)[j_{0,1}(\iota_L)])$
(This follows from the previous point by elementarity.)
10. $q_0 < p_0^-[\lambda_0]_{E_0} = \text{proj}_{\lambda_0}(j_{E_0}^{N_0}(p_0^-)) = \text{proj}_{\lambda_0}(j_{E_0}^{M_0}(p_0^-)) = \text{proj}_{\lambda_0}(j_{0,1}(p_0^-))$
11. $M_1 \models ((\exists \alpha \in (j_{0,1}(\iota_L), j_{0,1}(\iota_L) + 2))(V_\alpha \models q_0[j_{0,1}(\iota_L)]))$
(This follows from the previous two points. The lower limit for α follows from the fact that the type is realized by $j_{0,1}(\iota_L)$.)
12. $M_1 \models (V_{j_{0,1}(\iota_L)+1} \models q_0[j_{0,1}(\iota_L)])$
13. $M_1 \models (V_{j_{0,1}(\iota_H)+1} \models \text{proj}^0(q_0))$
(This follows from the previous point and indiscernibility of ι 's.)
14. Inside M_1 , let $\kappa_1 \in (\lambda_0, \delta)$ be such that $p_1 := \text{tp}_{\kappa_0}^{j_{0,1}(\iota_L)+3}(j_{0,1}(\iota_L))$ is elastic.
15. $M_1 \models (p_1 \text{ exceeds } \text{proj}^0(q_0))$
(This follows from the previous two points.)

The steps below describe the inductive step. Let us assume that $N_i, q_i, \alpha_i, M_{i+1}, p_{i+1}$ have been defined. We proceed to define $N_{i+1}, q_{i+1}, \alpha_{i+1}, M_{i+2}, p_{i+2}$. The part of the diagram that we are concerned with is the following.

$$M_{i+1} \dashrightarrow M_{i+2}$$

$$N_i \dashrightarrow N_{i+1}$$

1. $M_{i+1} \models (p_{i+1} \text{ exceeds } \text{proj}^0(q_i))$
(This holds by IH.)
2. $N_i \models (q_i \text{ is elastic})$
(This holds by IH.)
3. $M_{i+1} \models (q_i \text{ is elastic})$
(This follows for the previous point, the agreement of models in an iteration, and the absoluteness of elasticity.)
4. $M_{i+1} \models (\text{proj}^0(q_i) \text{ is elastic})$
5. We work inside M_{i+1} . There is an inaccessible $\nu_i \in (\kappa_{i+1}, \delta)$ and a strong (μ_i, ν_i) -extender $F_i \in V_\delta$ such that $p_{i+1} < \text{proj}^0(q_i)^-_{[F_i]^{\nu_i}}$. (This follows from the points 1 and 4.)
6. $N_{i+1} := \text{Ult}(N_i, F_i)$
7. $(\text{proj}^0(q_i)^-_{[F_i]^{\nu_i}})^{M_{i+1}} = \text{proj}_{\nu_i}^{M_{i+1}}(j_{F_i}^{M_{i+1}}(\text{proj}^0(q_i)^-)) = \text{proj}_{\nu_i}^{N_{i+1}}(j_{F_i}^{N_{i+1}}(\text{proj}^0(q_i)^-)) = (\text{proj}_{\nu_i}^0)^{N_{i+1}}(k_{i,i+1}(q_i))^-$
8. $N_{i+1} \models (p_{i+1} < \text{proj}_{\nu_i}^0(k_{i,i+1}(q_i))^-)$
(This follows from points 5 and 7 and the absoluteness of “being subsumed”.)
9. $N_i \models (V_{\alpha_{i+1}} \models q_i[\alpha_i])$
(This holds by IH.)
10. $N_i \models (V_{\alpha_i} \models \text{proj}^0(q_i)^-)$
11. $N_{i+1} \models (V_{k_{i,i+1}(\alpha_i)} \models \text{proj}_{\nu_i}^0(k_{i,i+1}(q_i))^-)$
(This follows from the previous point by elementarity.)
12. There are $\beta < \gamma < k_{i,i+1}(\alpha_i)$ such that $N_{i+1} \models (V_\gamma \models p_{i+1}[\beta])$. (This follows from points 8 and 11.)
13. $M_{i+1} \models (V_{j_{0,i+1}(\iota_L)+3} \models p_{i+1}[j_{0,i+1}(\iota_L)])$
(This holds by IH.)
14. $p_{i+1}(x) \models \text{Ord} = x + 3$
(This follows from the previous point.)
15. $\gamma = \beta + 3$
(This follows from points 12 and 14.)

16. $\alpha_{i+1} := \beta < k_{i,i+1}(\alpha_i)$
17. $N_{i+1} \models (V_{\alpha_{i+1}+3} \models p_{i+1}[\alpha_{i+1}])$
18. Working inside N_{i+1} , let $\mu_{i+1} \in (\nu_i, \delta)$ be such that $q_{i+1} := \text{tp}_{\mu_i}^{\alpha_{i+1}+1}(\alpha_{i+1})$ is elastic.
19. $N_{i+1} \models (q_{i+1} \text{ exceeds } p_{i+1})$
(This follows from the previous two points.)
20. $M_{i+1} \models (p_{i+1} \text{ is elastic})$
(This holds by IH.)
21. $N_{i+1} \models (p_{i+1} \text{ is elastic})$
(This follows from the previous point and absoluteness of elasticity.)
22. Working inside N_{i+1} , let $\lambda_{i+1} \in (\mu_{i+1}, \delta)$ be inaccessible and let $E_{i+1} \in V_\delta$ be such that $q_{i+1} < p_{i+1}^-[\lambda_{i+1}^{E_{i+1}}]$. (These exist due to points 19 and 21.)
23. $M_{i+2} := \text{Ult}(M_{i+1}, E_{i+1})$
24. $M_{i+1} \models (V_{j_{0,i+1}(\iota_L)+2} \models p_{i+1}^-[j_{0,i+1}(\iota_L)])$
(This follows from the point 13.)
25. $M_{i+2} \models (V_{j_{0,i+2}(\iota_L)+2} \models j_{i+1,i+2}(p_{i+1})^-[j_{0,n+2}(\iota_L)])$
(This follows from the previous point by elementarity.)
26. $(p_{i+1}^-[\lambda_{i+1}^{E_{i+1}}])^{N_{i+1}} = \text{proj}_{\lambda_{i+1}}^{N_{i+1}}(j_{E_{i+1}}^{N_{i+1}}(p_{i+1}^-)) = \text{proj}_{\lambda_{i+1}}^{M_{i+2}}(j_{E_{i+1}}^{M_{i+1}}(p_{i+1}^-)) = \text{proj}_{\lambda_{i+1}}^{M_{i+2}}(j_{i+1,i+2}(p_{i+1}^-))$
27. $M_{i+2} \models (q_{i+1} < \text{proj}_{\lambda_{i+1}}(j_{i+1,i+2}(p_{i+1}^-)))$
(This follows from points 22 and 26.)
28. $M_{i+2} \models (V_{j_{0,n+2}(\iota_L)+1} \models q_{i+1}[j_{0,n+2}(\iota_L)])$
(This follows from points 25 and 27.)
29. $M_{i+2} \models (V_{j_{0,n+2}(\iota_H)+1} \models \text{proj}^0(q_{i+1}))$
(This follows from the previous point by indiscernibility of ι 's.)
30. Working inside M_{i+2} , let $\kappa_{i+2} \in (\lambda_{i+1}, \delta)$ be such that
$$p_{i+2} := \text{tp}_{\kappa_{i+2}}^{j_{0,n+2}(\iota_L)+3}(j_{0,n+2}(\iota_L))$$
is elastic.
31. $M_{i+2} \models (p_{i+2} \text{ exceeds } \text{proj}^0(q_{i+1})^-)$
(This follows from the previous two points.)

□

8 Homogeneously Suslin Sets

In this section we introduce notions of a *homogeneous tree* and a *homogeneously Suslin set*. We will show that homogeneously Suslin sets are determined, hence reducing the problem of projective determinacy to showing that projective sets are homogeneously Suslin.

Definition 8.1. A *tower of measures with support Z* is a sequence $(\mu_n : n < \omega)$ such that each μ_n is a measure (i.e. ω_1 -complete ultrafilter) on Z^n and they are coherent:

$$(\forall m < n < \omega)(\forall A \subseteq Z^m)(\mu_m(A) = \mu_n(\{s \in Z^n : s \restriction m \in A\})).$$

If $f : Z^m \rightarrow V$ and $n > m$, we can define $f^n : Z^n \rightarrow V$ by simply putting $f^n(s) := f(s \restriction m)$ for $s \in Z^n$. For $m < n < \omega$, we define $j_{m,n} : \text{Ult}(V, \mu_m) \rightarrow \text{Ult}(V, \mu_n)$ by setting $j_{m,n}([f]_{\mu_m}) := [f^n]_{\mu_n}$. This mapping is correctly defined and elementary. To see this, consider a formula $\phi(x \dots)$ and sets $[f] \dots \in \text{Ult}(V, \mu_m)$ such that

$$\text{Ult}(V, \mu_m) \models \phi([f] \dots).$$

Łoś's theorem implies that $\mu_m(\{s \in Z^m : \phi(f(s) \dots)\}) = 1$, so we conclude by coherence that

$$\begin{aligned} \mu_n(\{t \in Z^n : \phi(f^n(t) \dots)\}) &= \mu_n(\{t \in Z^n : \phi(f(t \restriction m) \dots)\}) = \\ \mu_m(\{s \in Z^m : \phi(f(s) \dots)\}) &= 1. \end{aligned}$$

The reverse direction Łoś's theorem now yields $\text{Ult}(V, \mu_n) \models \phi([f^n] \dots)$, i.e.

$$\text{Ult}(V, \mu_n) \models \phi(j_{m,n}([f]) \dots).$$

It is a matter of routine to verify that $j_{m,p} = j_{n,p} \circ j_{m,n}$ for $m < n < p$, implying that

$$(\text{Ult}(V, \mu_m), j_{m,n} : m < n < \omega)$$

is a directed system. We denote by $\text{Ult}(V, (\mu_n : n < \omega))$ its direct limit. This class is a model of ZFC, but it need not be well-founded. When it is well-founded, we also say that the tower of measures is well-founded. Let us characterize the case when that occurs.

Definition 8.2. A tower μ of measures is said to be *complete* if for every sequence of measure-one sets $A_n \subseteq Z^n$ ($n < \omega$), there is a function $f : \omega \rightarrow Z$ such that $(\forall n < \omega)(f \restriction n \in A_n)$.

Proposition 8.3. A tower μ of measures is well-founded if and only if it is complete.

Proof. Let us assume that μ is ill-founded. Then there is an increasing sequence $(n_i)_{i < \omega}$ of naturals and a sequence $(\alpha_i)_{i < \omega}$ of ordinals such that $j_{n_i, n_{i+1}}(\alpha_i) > \alpha_{i+1}$ for every $i < \omega$. We let $f_i : Z^{n_i} \rightarrow \text{Ord}$ be such that $\alpha_i = [f_i]$. Then

$$\mu_{n_{i+1}}(\{s \in Z^{n_{i+1}} : f_i(s \restriction n_i) > f_{i+1}(s)\}) = 1.$$

Let us set

$$A_j := \begin{cases} \{s \in Z^{n_{i+1}} : f_i(s \restriction n_i) > f_{i+1}(s)\}, & \exists i(j = n_{i+1}) \\ Z^j & \text{otherwise.} \end{cases}$$

The sequence $(A_j : j < \omega)$ witnesses the incompleteness of μ . For otherwise, there would be a thread $x : \omega \rightarrow Z$ such that $x \restriction j \in A_j$. In particular, $x \restriction n_{i+1} \in A_{n_{i+1}}$, i.e. $f_i(x \restriction n_i) > f_{i+1}(x \restriction n_{i+1})$. The decreasing sequence of ordinals $(x \restriction n_i)_{i < \omega}$ provides for a contradiction.

On the other hand, let us assume that μ is incomplete, i.e. there is a sequence $(A_i)_{i < \omega}$ of measure-one sets that cannot be threaded. We define by induction the sets $B_0 := A_0$, $B_{i+1} = \{s \in A_{i+1} : s \restriction i \in B_i\}$ ($i < \omega$). Due to the coherence condition, these are also measure-one sets. Let the set $T := \bigcup_{i < \omega} B_i$ be ordered by inclusion. This poset is a tree. Since A_i 's cannot be threaded, neither can B_i 's, so this tree is well-founded. Let ρ be its ranking function (i.e. the ranking function for $(T, \supseteq)$) and set

$$f_i(s) := \begin{cases} \rho(s), & s \in B_i \\ 0, & \text{otherwise} \end{cases}$$

for $i < \omega$ and $s \in Z^i$. If $s \in B_{i+1}$, then $s \restriction i \in B_i$ and $f_i(s \restriction i) = \rho(s \restriction i) > \rho(s) = f_{i+1}$. In other words, $j_{i,i+1}([f_i]) > [f_{i+1}]$ ($i < \omega$), implying that $\text{Ult}(V, \mu)$ is ill-founded. $\square$

Definition 8.4. A *homogeneity system* over X with support Z is a family $(\mu_s : s \in X^{<\omega})$ such that every μ_s is a measure on $Z^{|s|}$ and

$$(\forall s \triangleleft t \in X^{<\omega})(\forall A \subseteq Z^{|s|})(\mu_s(A) = \mu_t(\{u \in Z^{|t|} : u \restriction |s| \in A_s\})).$$

A κ -*homogeneity system* is one all of whose measures are κ -complete.

In other words, a family of measures $(\mu_s : s \in X^{<\omega})$ is a homogeneity system exactly in the case when for every $x \in X^\omega$, the family $(\mu_{x \restriction i} : i < \omega)$ is a tower of measures. Before proceeding further, let us recall some of the terminology related to trees. If T is a tree on $X \times Y$, we understand it as a subset of $X^{<\omega} \times Y^{<\omega}$ whose coordinates have the same length. (A similar note holds for trees on products of more sets.) The fiber of T corresponding to $s \in X^{<\omega}$ is defined by

$$T_s := \{t \in Y^{<\omega} : (s, t) \in T\},$$

while the fiber corresponding to $x \in X^\omega$ is defined by

$$T_x := \bigcup_{n < \omega} T_{x \restriction n} = \{t \in Y^{<\omega} : (\exists n < \omega)((x \restriction n, t) \in T)\}.$$

We denote by $p[T]$ the projection of the set $[T] \subseteq X^\omega \times Y^\omega$ of branches of T to the first coordinate; thus,

$$\begin{aligned} x \in p[T] &\iff \exists y((x, y) \in [T]) \iff \exists y \forall n((x \restriction n, y \restriction n) \in T) \\ &\iff \exists y \forall n(y \restriction n \in T_{x \restriction n}) \iff [T_x] \neq \emptyset. \end{aligned}$$

Definition 8.5. A tree T on $X \times Z$ is said to be *homogeneous* if there is a $|X|^+$ -homogeneity system μ over X with support Z satisfying $(\forall s \in X^{<\omega}) \mu_s(T_s) = 1$ and for every $x \in p[T]$, the model $M_x^\mu := \text{Ult}(V, (\mu_{x \upharpoonright n} : n < \omega))$ is well-founded. The tree T is κ -homogeneous if μ can be chosen to be a κ -homogeneity system.

The definition implies that if M_x^μ is well-founded for some $x \in X^\omega$, then $x \in p[T]$. To see this, note that $\mu_{x \upharpoonright n}(T_{x \upharpoonright n}) = 1$ ($n < \omega$). The well-foundedness, i.e. the completeness, of the tower $(\mu_{x \upharpoonright n})_{n < \omega}$ implies the existence of a fiber $z \in Z^\omega$ satisfying $z \upharpoonright n \in T_{x \upharpoonright n}$ for all $n < \omega$. This exactly means that $x \in p[T]$. Thus,

$$p[T] = \{x \in X^\omega : M_x^\mu \text{ is well-founded}\}.$$

Definition 8.6. A set $A \subseteq X^\omega$ is said to be *homogeneously Suslin* if there is a homogeneous tree T such that $A = p[T]$. The set A is κ -homogeneously Suslin if a tree T can be chosen to be κ -homogeneous.

If $(\mu_s : s \in X^{<\omega})$ witnesses the homogeneity of an appropriate T , we have that

$$A = \{x \in X^\omega : \text{Ult}(V, (\mu_{x \upharpoonright n})_{n < \omega}) \text{ is well-founded}\}.$$

The purpose of the notion of being homogeneously Suslin is given by the following theorem.

Theorem 8.7 (Martin). *If $A \subseteq X^\omega$ is homogeneously Suslin, then it is determined.*

Proof. The game G defined by A is played as follows:

$$\begin{array}{c|ccc} \text{I} & x_0 & x_2 & \cdots \\ \hline \text{II} & & x_1 & x_3 \quad \cdots \end{array}$$

where $x_i \in X$ ($i < \omega$) and I wins if and only if $x = (x_i)_{i < \omega} \in A$. We are going to define a closed auxiliary game G^* . Let T be a homogeneous tree on $X \times Z$ witnessing the fact that A is homogeneously Suslin. Let $\mu = (\mu_s : s \in X^{<\omega})$ be a homogeneity system on Z witnessing the fact that T is homogeneous. The game G^* is played as follows:

$$\begin{array}{c|cccc} \text{I} & x_0, z_0 & z_1, x_2, z_2 & z_3, x_4, z_4 & \cdots \\ \hline \text{II} & & x_1 & x_3 & \cdots \end{array}$$

where $x_i \in X$ and $z_i \in Z$ ($i < \omega$), while I must observe the rule that $(x \upharpoonright i, z \upharpoonright i) \in T$ for every $i < \omega$. I wins if he is able to make ω moves, otherwise II wins. In this game I tries to produce a witness z to the fact that $x \in A = p[T]$. The game is obviously closed (for I), so it is determined.

If I has a winning strategy in G^* , then the same strategy wins G after omitting z_i 's. If II has a winning strategy σ^* in G^* , then he also has a winning strategy σ in G , which we describe in below. These two facts conclude the proof that G is determined. Let us assume that the first $2k+1$ moves in G have been made and let us describe the move x_{2k+1} .

$$\begin{array}{c|ccc} \text{I} & x_0 & \cdots & x_{2k} \\ \hline \text{II} & x_1 & \cdots & x_{2k+1}? \end{array}$$

Denoting $s := x \upharpoonright (2k+1)$, we note that for every $t \in T_s$, there is a move $\sigma^*(s*t) \in X$ in G^* according to σ^* after the position $s*t := (x_0, t_0 \dots x_{2k}, t_{2k})$. Since μ_s is $|X|^+$ -complete and $\mu_s(T_s) = 1$, we have a unique move $\sigma(s) \in X$ such that $\sigma(s) = \sigma^*(s*t)$ for μ_s -almost every $t \in T_s$, i.e. for every $t \in U_s$, where $U_s \subseteq T_s$ is of measure one. The strategy σ is winning for II in G . Otherwise, there would be a play $x \in A$ in G according to σ . Since $x \in p[T]$, the tower $(\mu_{x \upharpoonright i} : i < \omega)$ is complete, so there is a fiber z through $(U_{x \upharpoonright i} : i < \omega)$. Now, the play

$$\begin{array}{c|cccc} \text{I} & x_0, z_0 & z_1, x_2, z_2 & z_3, x_4, z_4 & \cdots \\ \hline \text{II} & x_1 & & x_3 & \cdots \end{array}$$

of G^* is played according to σ^* , so II wins it. This means that I broke the rule at some point, i.e. $z \upharpoonright i \notin T_{x \upharpoonright i} \supseteq U_{x \upharpoonright i}$ for some $i < \omega$, which is a contradiction. $\square$

To witness the homogeneity of a tree, we may use families of inner models and elementary embeddings instead of families of measures.

Proposition 8.8. *A tree T on $X \times X$ is homogeneous if and only if there is a family*

$$(M_s, j_{s,t}, w_s : s \triangleleft t \in X^{<\omega})$$

satisfying

1. M_s 's are inner models for ZFC, with $M_\emptyset = V$;
2. $j_{s,t} : M_s \prec M_t$ for $s \triangleleft t$, with $j_{s,s} = \text{id}$;
3. $j_{t,u} \circ j_{s,t} = j_{s,u}$ for $s \triangleleft t \triangleleft u$;
4. $\text{crit } j_{\emptyset,s} > |X|$;
5. $w_s \in j(T_s)$ and $j_{s,t}(w_s) \triangleleft w_t$ for $s \triangleleft t$;
6. if $x \in p[T]$, then the direct limit of the system

$$(M_{x \upharpoonright m}, j_{x \upharpoonright m, x \upharpoonright n} : m < n < \omega)$$

is well-founded.

Proof. For the implication $(\Rightarrow)$, let μ witness the homogeneity of T and set $M_s := \text{Ult}(V, \mu_s)$ and $j_{s,t}([f]_{\mu_s}) := [f \upharpoonright^t]_{\mu_t}$ for $s \triangleleft t$. These are the notions already seen above and we have already established all the points except 4 and 5. For 4, we simply note that all the measures μ_s are $|X|^+$ -complete. For 5, we set $w_s := [\text{id}_{Z^{|s|}}]_{\mu_s}$, while the verification is a matter of routine ($w_s \in j_{\emptyset,s}(T_s)$ follows from the fact that $\mu_s(T_s) = 1$).

For the implication $(\Leftarrow)$, we need to define an appropriate tower of measures. We set

$$A \in \mu_s : \iff w_s \in j_{\emptyset,s}(A) \quad (A \subseteq Z^{|s|})$$

for $s \in X^{<\omega}$. It is readily verifiable that these are indeed $|X|^+$ -complete measures (the completeness follows from the point 4). Let us check the coherence. Given $s \triangleleft t$, we reason as follows (for $A \subseteq Z^{|s|}$):

$$\begin{aligned}
\mu_s(A) = 1 &\iff w_s \in j_{\emptyset,s}(A) \\
&\iff j_{s,t}(w_s) \in j_{s,t}(j_{\emptyset,s}(A)) \\
&\iff j_{s,t}(w_s) \in j_{\emptyset,t}(A) \\
&\iff w_t \in j_{\emptyset,t}(A \times Z^{|t|-|s|}) \\
&\iff \mu_t(A \times Z^{|t|-|s|}) = 1
\end{aligned}$$

where the first and last equivalences are by definition, the second is due to point 2, the third is due to point 3, and the fourth is due to point 5. We also easily verify that $\mu_s(T_s) = 1$, since this is equivalent to $w_s \in j_{\emptyset,s}(T_s)$.

It remains to check that $(\mu_{x \upharpoonright n} : n < \omega)$ is complete for $x \in p[T]$. Let $A_n \subseteq Z^n$ ($n < \omega$) be a family of measure-one sets. This means that $w_{x \upharpoonright n} \in j_{\emptyset,x \upharpoonright n}(A_n)$ for every $n < \omega$. Denoting by M_x the transitive direct limit of the system

$$(M_s, j_{s,t} : s \triangleleft t \triangleleft x)$$

and by $j_{s,x}$ the appropriate embeddings, we conclude that

$$j_{s,x}(w_s) \in j_{\emptyset,x}(A_{|s|})$$

and $j_{s,x}(w_s) \triangleleft j_{t,x}(w_t)$ for $s \triangleleft t$ (this follows from the point 5). The tree

$$\bigcup_{n < \omega} j_{\emptyset,x}(A_n) = j_{\emptyset,x} \left(\bigcup_{n < \omega} A_n \right) \in M_x$$

is ill-founded in V , as witnessed by $(j_{x \upharpoonright n, x}(w_{x \upharpoonright n}) : n < \omega)$. This implies that it is also ill-founded in M_x , so by elementarity, the tree $\bigcup_{n < \omega} A_n$ is ill-founded in V , i.e. there is a fiber through A_n 's. $\square$

In order to characterize τ -homogeneity, it suffices to replace the condition 4 by the condition “ $\text{crit } j_{\emptyset,s} \geq \max\{|X|^+, \tau\}$ ”.

9 Proof of the Projective Determinacy

9.1 Structure of the Proof

In this section, we prove that if X is a set and if there are n Woodin cardinals above it and a measurable above them, then any Π_{n+1}^1 subset of X^ω is homogeneously Suslin and hence determined. The proof is realized by induction on n . For the base, we have the following theorem.

Theorem 9.1 (Martin). *If X is a set and κ a measurable cardinal above it, then every co-analytic subset of X^ω is κ -homogeneously Suslin.*

This theorem is proved in the second subsection. For the inductive step, we prove the following theorem, which is also designated as a *transfer theorem*. Most of the theory developed in this thesis is for the purpose of proving this theorem.

Theorem 9.2 (Martin, Steel). *Let X be a set and δ a Woodin cardinal above it. Let $A \subseteq X^\omega \times \omega^\omega$ be δ^+ -homogeneously Suslin. Then $p[A]^\mathbb{C}$ is τ -homogeneously Suslin for every $\tau < \delta$.*

In the third subsection, we study some technical notions necessary for proving this theorem. In the fourth subsection, we construct a τ -homogeneously Suslin tree that projects onto $p[A]^\mathbb{C}$. Once we have proven the two theorems from above, we can easily conclude the projective determinacy.

Theorem 9.3 (Martin, Steel). *If X is a set and if there are n Woodin cardinals above it and a measurable cardinal above them, then every Π_{n+1}^1 subset of X^ω is determined.*

Proof. Let $\delta_n < \delta_{n-1} < \dots < \delta_1$ be Woodin cardinals and $\delta_0 > \delta_1$ a measurable. We show by induction on n that for every $X \in V_{\delta_n}$, every Π_{n+1}^1 set $A \subseteq X^\omega$ is τ -homogeneously Suslin for every $\tau < \delta_n$. This suffices for the conclusion. The base $n = 0$ is simply the theorem 9.1. Let us assume that the statement holds for n and prove it for $n + 1$. We have that $A = p[B]^\mathbb{C}$ for some Π_{n+1}^1 set $B \subseteq X^\omega \times \omega^\omega$. By applying the IH to $X \times \omega \in V_{\delta_n}$ and the cardinal $\delta_{n+1}^+ < \delta_n$, we conclude that B is δ_{n+1}^+ -homogeneously Suslin. We can now apply the Main Theorem and obtain that $A = p[B]^\mathbb{C}$ is τ -homogeneously Suslin for every $\tau < \delta_{n+1}$. $\square$

9.2 Co-analytic Determinacy

Definition 9.4. The Kleene-Brouwer order on $\omega^{<\omega}$ is given by

$$s >_{\text{KB}} t : \iff s \triangleleft t \vee (|s| = |t| > 0 \wedge s(|s| - 1) > t(|t| - 1)).$$

Proposition 9.5. $<_{\text{KB}}$ is a linear order extending $\triangleright$. $<_{\text{KB}}$ is well-founded on a tree T on ω if and only if T is well-founded.

Proof. The first part is readily verifiable. Since $\triangleright \upharpoonright T \subseteq <_{\text{KB}} \upharpoonright T$, we conclude the direction $(\Rightarrow)$ of the second part. Let us assume that $<_{\text{KB}} \upharpoonright T$ is ill-founded. We want to conclude that $\triangleright \upharpoonright T$ is also ill-founded. We have a sequence

$$s_0 >_{\text{KB}} s_1 >_{\text{KB}} s_2 >_{\text{KB}} \dots$$

in T . For any $i < \omega$ there is $j > i$ such that $s_i \triangleleft s_j$. Otherwise, for every $j > i$, we have $|s_j| = |s_i| =: n$ and $s_j(n-1) < s_i(n-1)$. In particular, $\{s_j(n-1) : i < j < \omega\}$ is finite. However, we also have $s_{j_1}(n-1) < s_{j_2}(n-1)$ for $i < j_1 < j_2$, so $\{s_j(n-1) : i < j < \omega\}$ is infinite. This is obviously a contradiction. Thus, there is a sequence

$$s_{i_0} \triangleright s_{i_1} \triangleright s_{i_2} \triangleright \dots$$

in T , witnessing its ill-foundedness. $\square$

Let X be a set and let A be a Π_1^1 subset of X^ω . There is a tree S on $X \times \omega$ such that $A = p[S]^\mathbb{G}$. Fix such a tree. Let us also fix a recursive bijection $\mathbf{t} : \omega \rightarrow \omega^{<\omega}$ satisfying

$$\mathbf{t}_i \triangleleft \mathbf{t}_j \implies i < j$$

and let $\langle \cdot \rangle : \omega^{<\omega} \rightarrow \omega$ be its inverse.

Definition 9.6. For $x \in X^{<\omega}$, we define a relation $<_x$ on $|x|$ in the following manner:

$$m >_x n \iff \begin{cases} m > n, & \mathbf{t}_m, \mathbf{t}_n \notin S_x \\ \top, & \mathbf{t}_m \in S_x, \mathbf{t}_n \notin S_x \\ \perp, & \mathbf{t}_m \notin S_x, \mathbf{t}_n \in S_x \\ \mathbf{t}_m >_{\text{KB}} \mathbf{t}_n, & \mathbf{t}_m, \mathbf{t}_n \in S_x. \end{cases}$$

Proposition 9.7.

1. For $x \in X^{<\omega}$, $<_x$ is a linear order on $|x|$.
2. For $s \triangleleft t \in X^{<\omega}$, $<_s \subseteq <_t$.
3. For $x \in X^\omega$, $<_x = \bigcup_{i < \omega} <_{x \upharpoonright i}$.
4. For $x \in X^\omega$, $<_x$ is well-founded if and only if $x \in A$.

Proof. The first three points are easily established. Let us consider the fourth one. If $x \notin A$, then $x \in p[S]$, so there is branch $y \in [S_x]$. By setting $n_i := \langle y \upharpoonright i \rangle$ ($i < \omega$), we get an infinite $<_x$ -descending chain. On the other hand, if $<_x$ is ill-founded, as witnessed by a sequence

$$n_0 >_x n_1 >_x n_2 >_x \cdots,$$

it is not the case that some $\mathbf{t}_i \notin S_x$ since that would yield a decreasing sequence $(n_j : j > i)$ of naturals. Thus, we have a $<_{\text{KB}}$ -decreasing sequence $(\mathbf{t}_{n_i} : i < \omega)$ of nodes in S_x . This implies that S_x is ill-founded and consequently $x \in p[S] = A^\mathbb{G}$. $\square$

Let κ be an uncountable cardinal above X . A tree T on $X \times \kappa$ is defined by setting

$$(s, f) \in T : \iff s \in X^{<\omega} \wedge f : (|s|, <_s) \hookrightarrow (\kappa, <).$$

Proposition 9.8. $A = p[T]$

Proof. Branches through T exactly correspond to pairs (x, F) , where $x \in X^\omega$ and $F : (\omega, <_\omega) \hookrightarrow (\kappa, <)$. Hence,

$$\begin{aligned} x \in A & \iff <_x \text{ is well-founded} \\ & \iff \exists F : (\omega, <_x) \hookrightarrow (\kappa, <) \\ & \iff \exists F ((x, F) \in [T]) \\ & \iff x \in p[T]. \end{aligned}$$

$\square$

If κ is in addition measurable, we will show that T is κ -homogeneously Suslin, which will conclude the argument. Let μ be a normal measure on κ .

Definition 9.9. For $s \in X^{<\omega}$ and $C \subseteq \kappa$, we define

$$C^s := \{f : (|s|, <_s) \hookrightarrow (C, <)\} \subseteq \kappa^{|s|}.$$

We also set

$$\mu_s := \{D : (\exists C \in \mu)(C^s \subseteq D)\} \subseteq P(\kappa^{|s|}).$$

Proposition 9.10. $(\mu_s : s \in X^{<\omega})$ witnesses κ -homogeneity for T .

Proof. It is easily seen that every μ_s is a κ -complete filter. To see that it is in fact an ultrafilter, let us consider arbitrary $D \subseteq \kappa^{|s|}$. For $u \in [\kappa]^{|s|}$, there is a unique $f_u : (|s|, <_s) \hookrightarrow (\kappa, <)$ with $\text{im } f_u = u$. Let us define a coloring $c : [\kappa]^{|s|} \rightarrow 2$ by setting $c(u) := \chi_D(f_u)$. By Rowbottom's theorem, there is $C \in \mu$ homogeneous for c . If $c \restriction [C]^{|s|} \equiv 0$, then $C^s \subseteq D^c$ and $D^c \in \mu_s$; if $c \restriction [C]^{|s|} \equiv 1$, then $C^s \subseteq D$ and $D \in \mu_s$. To verify the coherence, let $s \triangleleft t \in X^{<\omega}$ and let $D \subseteq \kappa^{|s|}$. If $C^s \subseteq D$ for some $C \in \mu$, then $C^t \subseteq D \times \kappa^{|t|-|s|}$ and $D \times \kappa^{|t|-|s|} \in \mu_t$. On the other hand, if $C^t \subseteq D \times \kappa^{|t|-|s|}$ for some $C \in \mu$, then $C^s \subseteq D$ and $D \in \mu_s$. Hence, $\mu_s(D) = \mu_t(D \times \kappa^{|t|-|s|})$.

We have established that the given family of measures is a κ -homogeneity system. Let us verify that $\mu_s(T_s) = 1$ for $s \in X^{<\omega}$ and that $(\mu_{x \restriction n} : n < \omega)$ is well-founded for $x \in X^\omega$. For the first part, we simply notice that $T_s = \kappa^s \in \mu$. For the second part, let $D_n \in \mu_{x \restriction n}$ ($n < \omega$) be an arbitrary family. Fix $C_n \in \mu$ such that $C_n^{x \restriction n} \subseteq D_n$, for every $n < \omega$. Since $C := \bigcap_{n < \omega} C_n \in \mu$, there is an $f : (\omega, <_x) \hookrightarrow (C, <)$. Note that

$$f[n] \in C^{x \restriction n} \subseteq C_n^{x \restriction n} \subseteq D_n$$

for every $n < \omega$. □

This concludes the proof of the theorem 9.1.

9.3 Nice Types

We now work towards a proof of the transfer theorem. Let X be a set and δ a Woodin cardinal above it. Let $A \subseteq X^\omega \times \omega^\omega$ be a δ^+ -homogeneously Suslin set, as witnessed by a tree S on $X \times \omega \times Z$ and a homogeneity system $\mu = (\mu_{s,t} : (s,t) \in (X \times \omega)^{<\omega})$.

Definition 9.11. An (s,t) -nice $(\kappa, n+2)$ -type p is determined by

- a $(\kappa, n+2)$ -type $p(S, \bar{u}, \alpha)$
- together with a label $(s,t) \in X^n \times \omega^n$

satisfying the following properties:

- $\kappa \in (\text{rank}(X), \delta)$;

- $p \models (\alpha > \max\{\text{rank}(S), \dot{\delta}\})$;
- the set

$$Z_p := \{u \in Z^n : \exists \xi \exists \alpha (V_\xi \models p[S, u, \alpha])\}$$

is a $\mu_{s,t}$ -measure-one set.

In this case, a function $h_p : Z_p \rightarrow \text{Ord}$ is defined by setting

$$h_p(u) := \min\{\xi : \exists \alpha (V_\xi \models p[S, u, \alpha])\}.$$

Remark 9.12. Note that every nice type $p(S, \bar{u}, \alpha)$ is realized, so

$$p \models ((u_0 \dots u_{n-1}) \in S_{s,i}).$$

If $n = 0$, the $(\emptyset, \emptyset)$ -niceness is equivalent to the fact of set $Z_p \subseteq Z^0 = \{\emptyset\}$ being non-empty, i.e. to the existence of ordinals $\xi > \alpha$ such that $V_\xi \models p[S, \alpha]$, i.e. $p = \text{tp}_\kappa^\xi(S, \alpha)$.

Remark 9.13. Consider the case $q(S, \bar{v}, \alpha) < p(S, \bar{u}, \alpha)$. This does not mean that the part $\bar{u}$ of a realization of p is an initial segment of the part $\bar{v}$ of the appropriate realization of q . The reason for this is the fact that we have allowed for permutations of variables when we defined subsumed types. To correct this issue, in the definition above, we can work with types realized by $S, (0, u(0)) \dots (n-1, u(n-1)), \alpha$ instead of those realized by $S, u(0) \dots u(n-1), \alpha$. In this manner, first coordinates ensure the order of realizations. We sacrifice this precision for the sake for readability.

Definition 9.14.

- Let p be an (s, t) -nice $(\kappa, n+2)$ -type,
- let p' is an (s', t') -nice $(\kappa', n'+2)$ -type.

The relation $p \succ p'$ holds if

- $(s, t) \trianglelefteq (s', t')$,
- $h_p(u' \upharpoonright n) > h_{p'}(u')$ for $\mu_{s', t'}$ -almost every $u' \in Z^{n'}$.

One easily verifies that this relation is transitive. This is a third way to compare types, in addition to the notions of being subsumed and exceeding. In the case that $\kappa < \kappa'$, the relation $p \succ p'$ implies that p' exceeds p . This is follows by taking arbitrary $u' \in Z_{p'}$ and noting that

$$V_{h_p(u' \upharpoonright n)} \models p[S, u' \upharpoonright n, \alpha], \quad V_{h_{p'}(u')} \models p'[S, u', \alpha], \quad h_p(u' \upharpoonright n) > h_{p'}(u').$$

A relationship of the new notion with the notion of being subsumed is described in the following proposition. Before stating it, we introduce a notation: the *decap* of a $(\kappa, n+2)$ -type p is the $(\kappa, n+1)$ -type $\text{dcp}(p) := \text{proj}^{n+1}(p)$, obtained by omitting the last variable.

Proposition 9.15. *If*

- p is an (s, t) -nice $(\kappa, n + 2)$ -type,
- p' be a $(\kappa', n + 2)$ -type,
- $p' < \text{dcp}(p)$,
- $p' \models (\alpha > \max\{\text{rank}(S), \delta\})$,
- $\kappa' > \text{rank}(X)$,

then

- p' is an (s, t) -nice type,
- $p' \prec p$.

Proof. Let $u \in Z_p$ be arbitrary. Setting $\xi := h_p(u)$, we have $\alpha > \max\{\text{rank}(S), \delta\}$ such that $V_\xi \models p[S, u, \alpha]$. Consequently, $V_\xi \models \text{dcp}(p)[S, u]$, so there are some $\alpha' < \xi' < \xi$ such that $V_{\xi'} \models p'[S, u, \alpha']$. We have $u \in Z_{p'}$ and $h_{p'}(u) \leq \xi' < \xi = h_p(u)$. Thus, $Z_{p'}$ contains a measure-one set Z_p , so it is a measure-one set as well. The latter inequality holds on a measure-one set Z_p , ensuring that $p' \prec p$. $\square$

Let us describe a way to ensure that some $x \in X^\omega$ belongs to $p[A]^\mathbb{G}$. First notice that

$$x \in p[A]^\mathbb{G} \iff (\forall y \in \omega^\omega)((x, y) \notin A) \quad (17)$$

$$\iff (\forall y \in \omega^\omega)((\mu_{x \upharpoonright n, y \upharpoonright n} : n < \omega) \text{ is ill-founded,}) \quad (18)$$

so we only need to ensure that every branch of the homogeneity system $\mu^x := (\mu_{x \upharpoonright |t|, t} : t \in \omega^{<\omega})$ is ill-founded.

Proposition 9.16. *Let $(p_t : t \in \omega^{<\omega})$ satisfy*

- p_t is an $(x \upharpoonright |t|, t)$ -nice type $(t \in \omega^{<\omega})$;
- $p_t \succ p_{t'}$ for $t \triangleleft t' \in \omega^{<\omega}$.

Then every branch of μ^x is ill-founded and consequently $x \in p[A]^\mathbb{G}$.

Proof. Let us denote by M_s and $j_{s,t}$ ($s \triangleleft t \in \omega^\omega$) the models and embeddings of μ^x . The function h_{p_s} is defined on a measure-one subset of $Z^{|s|}$ with values in Ord, thus describing an ordinal $\alpha_s := [h_{p_s}]_{\mu_s^x} \in M_s$. Since $h_{p_s}(u \upharpoonright |s|) > h_{p_t}(u)$ for almost every $u \in Z^{|t|}$ and $s \triangleleft t$, we have $j_{s,t}(\alpha_s) > \alpha_t$ for every $s \triangleleft t$. This last observation implies that all the branches of μ^x are ill-founded, i.e. $x \in p[A]^\mathbb{G}$. $\square$

9.4 Proof of the Transfer Theorem

In this section we show that the set $B := p[A]^\mathbb{C}$ is τ -homogeneously Suslin, for a fixed $\tau < \delta$. To that end, we will first describe a tree T that projects to B and then show that it is a τ -homogeneously Suslin tree. This later fact will be established by invoking proposition 8.8, where the required models will come from a family of iteration trees. These iteration trees are constructed recursively, and the following lemma will be used to make a step in the construction.

Lemma 9.17.

- Let q be an (s_q, t_q) -nice $(\phi, n+2)$ -type,
- satisfying $q \models (\alpha + 2 \in \text{Ord})$;
- let $(s, t) \supseteq (s_q, t_q)$;
- let $\psi < \delta$.

There is a type p and a cardinal κ satisfying

- p is an (s, t) -nice $(\kappa, m+2)$ -type,
- $p \prec q$,
- $\kappa > \psi$,
- $\text{dcp}(p)$ is elastic,
- $p \models (\alpha = \max \text{Ord})$.

(Notice that the above conditions imply by proposition 9.15 that p exceeds q .)

Proof. Let $u \in Z^m$ be such that $u \restriction n \in Z_q$. Denoting $\eta := h_q(u \restriction n)$, there is an $\alpha \in (\max\{\text{rank}(S), \delta\}, \eta)$ such that

$$V_\eta \models q[S, u \restriction n, \alpha].$$

There is $\kappa \in (\max\{\phi, \psi\}, \delta)$ such that the type $\text{tp}_\kappa^{\alpha+1}(S, u)$ is elastic. Set $p' := \text{tp}_\kappa^{\alpha+1}(S, u, \alpha)$. It follows that $u \in Z_{p'}$ and $h_{p'}(u) \leq \alpha + 1 < \xi = h_q(u \restriction n)$.

Note that $p' = p'(u)$ depends on a u from the measure-one set $\{u \in Z^m : u \restriction n \in Z_q\}$ and takes the values in the set V_δ of cardinality δ . Since all the measures are δ^+ -complete, we have that this function is constant almost everywhere, with some value p . Note that the set Z_p contains a measure-one set $\{u \in Z^m : u \restriction n \in Z_q \wedge p'(u) = p\}$, so it is itself measure-one; in other words, p is nice. Also, $h_p(u) < h_q(u \restriction n)$ for every u in the same measure-one set, implying $p \prec q$. Other properties of p are immediate from the choices made above. $\square$

Let us fix a recursive bijection $\mathbf{t} : \omega \rightarrow \omega^{<\omega}$ satisfying

$$\mathbf{t}_i \triangleleft \mathbf{t}_j \implies i < j.$$

This implies in particular that $\mathbf{t}_0 = 0$ and $l_i := |\mathbf{t}_i| \leq i$. The inverse of this bijection is denoted by $\langle \cdot \rangle$. Hence, we have

$$\mathbf{t}_{\langle s \rangle} = s, \quad \langle \mathbf{t}_i \rangle = i, \quad |\mathbf{t}_i| = l_i.$$

For $i > 0$, we denote by i^* the index of the predecessor of $\mathbf{t}_i$, i.e.

$$\mathbf{t}_i \upharpoonright (l_i - 1) = \mathbf{t}_{i^*}, \quad l_{i^*} = l_i - 1.$$

Let us also fix a pair (ι_L, ι_H) of local indiscernibles relative to $\max\{\text{rank}(S), \delta\}$. We describe a partial function Γ whose arguments are of the form (s, q) , where

- $s \in X^n$ for some $n < \omega$,
- $q = (q_i : i < n)$,
- q_i is an $(s \upharpoonright l_i, \mathbf{t}_i)$ -nice type,
- $q_i \models (\alpha + 2 = \max \text{Ord})$.

Function Γ is computed by induction on n .

- If $n = 0$, we set $\Gamma(\emptyset, \emptyset) \downarrow$ and let its value be some $p_0 := \text{tp}_{\kappa_0}^{\iota_L+1}(S, \iota_L)$, where $\kappa_0 \in (\max\{|X|, \tau\}, \delta)$ is chosen such that $\text{dcp}(p_0)$ is elastic.
- If $n > 0$ and $\Gamma(s \upharpoonright (n-1), q \upharpoonright (n-1)) \uparrow$, we set $\Gamma(s, q) \uparrow$.
- If $n > 0$ and $\Gamma(s \upharpoonright (n-1), q \upharpoonright (n-1)) \downarrow = p_{n-1}$ is some (κ_{n-1}, l_{n-1}) -type, the computation goes as follows.

– If $q_{n-1} \prec p_{n-1}$, we set $\Gamma(s, q) \downarrow$ and for its value we choose a type p_n satisfying

- * p_n is an $(s \upharpoonright l_n, \mathbf{t}_n)$ -nice (κ_n, l_n) -type,
- * $p_n \prec q_{n^*}$,
- * $\kappa_n > \kappa_{n-1}$,
- * $\text{dcp}(p_n)$ is elastic,
- * $p_n \models (\alpha = \max \text{Ord})$,

which is provided by the previous lemma.

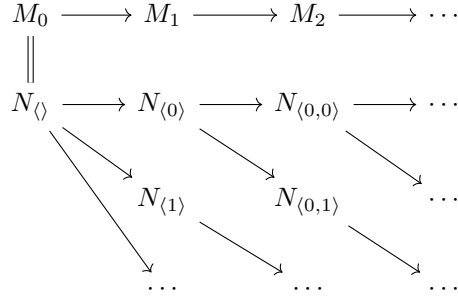
– If $p_{n-1} \not\prec q_{n-1}$, then $\Gamma(s, q) \uparrow$.

We define $T := \text{dom } \Gamma$, which is clearly a tree on $X \times V_\delta$.

Lemma 9.18. $p[T] \subseteq B$

Proof. Let $(x, q) \in [T]$ and let us show that $x \in B = p[A]^{\mathbb{C}}$. We set $p_n := \Gamma(x \upharpoonright n, q \upharpoonright n)$ for $n < \omega$. Notice that $q_n \prec p_n$ since $\Gamma(x \upharpoonright (n+1), q \upharpoonright (n+1)) \downarrow$, and $p_n \prec q_{n^*}$ by definition of Γ . This implies $q_n \prec q_{n^*}$ for $n > 0$, so the family $(q_{\langle t \rangle} : t \in \omega^{<\omega})$ satisfies the hypotheses of proposition 9.16. This proposition then yields $x \in p[A]^{\mathbb{C}}$. $\square$

Before establishing the other inclusion, we work towards a system of proposition 8.8. Given a parameter $x \in \omega^\omega$, we describe an iteration tree $\mathcal{T}^x$ of the form



dependent on that parameter. The order of the models is $M_0, N_0, M_1, N_1 \dots$, while the N -models form a tree isomorphic to $\omega^{<\omega}$. The embeddings between M -models are denoted by $j_{m,n}$ ($m < n < \omega$), while those between N -models are denoted by $k_{\langle s \rangle, \langle t \rangle}$ ($s \triangleleft t \in \omega^{<\omega}$). The notation for the extenders is as follows: E_n is a (κ_n, λ_n) -extender in N_n and $M_{n+1} = \text{Ult}(M_n, E_n)$; F_n is a (ϕ_n^*, χ_n) -extender in M_n and $N_n = \text{Ult}(N_{n^*}, F_n)$ (for $n > 0$).

The iteration is described by induction. For the base, we describe models M_0, N_0, M_1 ; for the step, we describe models N_n, M_{n+1} . When defining model M_n , we also define a type p_n ; when defining model N_n , we also define a type q_n and a tuple u_n . We require the following conditions:

- $N_n \models \ulcorner q_n \text{ is an } (x \upharpoonright l_n, \mathbf{t}_n)\text{-nice } (\phi_n, l_n + 2)\text{-type} \urcorner$;
- $N_n \models (q_n = \text{tp}_{\phi_n}^{k_{0,n}(\iota_L)+3}(k_{0,n}(S), u_n, k_{0,n}(\iota_L)))$;
- $N_n \models \ulcorner q_n \text{ is elastic} \urcorner$;
- $M_n \models (j_{0,n}(\Gamma)(x \upharpoonright n, (q_i : i < n)) = p_n)$;
- $M_n \models \ulcorner \mathcal{L}_{\kappa_n} \text{ is the language of } p_n \urcorner$.

Every one of the four cardinal parameters $\kappa_n, \lambda_n, \phi_n, \chi_n$, when chosen, will be larger than all of those defined before that point. This will ensure the desired structure of the tree. These parameters will also be less than δ , which means that δ will remain fixed throughout the iteration.

The base of the induction.

1. $M_0 := V$
2. $p_0 := \Gamma(\emptyset, \emptyset)$
3. Let $\kappa_0 \in (\max\{|X|, \tau\}, \delta)$ be such that $p_0 = \text{tp}_{\kappa_0}^{\iota_L+1}(S, \iota_L)$.
4. $N_0 := M_0 = V$
5. Let $\phi_0 \in (\kappa_0, \delta)$ be such that $q_0 := \text{tp}_{\phi_0}^{\iota_L+3}(S, \iota_L)$ is elastic.
6. $u_0 := \emptyset$
7. q_0 exceeds $\text{tp}_{\kappa_0}^{\iota_H+1}(S) = \text{dcp}(p_0)$.
(This follows from point 3 and the indiscernibility.)
8. There is an inaccessible $\lambda_0 \in (\phi_0, \delta)$ and a (κ_0, λ_0) -extender E_0 such that $q_0 < \text{dcp}(p_0)^{-\lambda_0}_{E_0}$.
9. $M_1 := \text{Ult}(M_0, E_0)$
10. $(\text{dcp}(p_0)^{-\lambda_0}_{E_0})^{N_0} = (\text{proj}_{\lambda_0}(\text{dcp}(j_{0,1}(p_0))^-))^{M_1}$
11. $M_1 \models (q_0 < \text{dcp}(j_{0,1}(p_0)))$
(This follows from points 8 and 10.)
12. $M_1 \models (j_{0,1}(\Gamma)(\emptyset, \emptyset) = j_{0,1}(p_0))$
(This follows from the point 2.)
13. $M_1 \models ((x_0, q_0) \in \text{dom } j_{0,1}(\Gamma))$
(This follows from the previous two points.)
14. Inside M_1 , let $p_1 := j_{0,1}(\Gamma)((x_0), (q_0))$ and let κ_1 correspond to its language.
15. $\kappa_1 > j_{0,1}(\kappa_0) \geq \lambda_0$
(The first inequality follows from the previous point and the fact that $j_{0,1}(\kappa_0)$ exactly corresponds to the language of $j_{0,1}(p_0)$. The second inequality follows from the fact that $j_{0,1}$ comes from a (κ_0, λ_0) -extender.)

The inductive step.

1. $M_n \models (j_{0,n}(\Gamma)(x \upharpoonright n, (q_i : i < n)) = p_n)$
(This follows by the IH.)
2. $M_n \models (p_n \prec q_{n^*})$
(This follows from the previous point by definition of Γ .)
3. $M_n \models (p_n \text{ exceeds } q_{n^*})$
(This follows from the previous point and the fact that $\kappa_n > \phi_{n^*}$.)
4. $N_{n^*} \models (q_{n^*} \text{ is elastic})$
(This follows by the IH.)

5. $M_n \models (q_{n^*} \text{ is elastic})$
6. Inside M_n , choose an inaccessible $\chi_n \in (\kappa_n, \delta)$ and a (ϕ_{n^*}, χ_n) -extender F_n such that $p_n < q_{n^*}^- [_{F_n}^{\chi_n}]$.
7. $N_n := \text{Ult}(N_{n^*}, F_n)$
8. $(q_{n^*}^- [_{F_n}^{\chi_n}])^{M_n} = \text{proj}_{\chi_n}(k_{n^*,n}(q_{n^*})^-)$
9. $N_n \models (p_n < \text{proj}_{\chi_n}(k_{n^*,n}(q_{n^*})^-))$
(This follows from the points 6 and 8.)
10. $N_{n^*} \models (q_{n^*} = \text{tp}_{\phi_{n^*}}^{k_{0,n^*}(\iota_L)+3}(k_{0,n^*}(S), u_{n^*}, k_{0,n^*}(\iota_L)))$
(This follows by the IH.)
11. $N_n \models (\text{proj}_{\chi_n}(k_{n^*,n}(q_{n^*})) = \text{tp}_{\chi_n}^{k_{0,n}(\iota_L)+3}(k_{0,n}(S), k_{n^*,n}(u_{n^*}), k_{0,n}(\iota_L)))$
12. Inside N_n , there is a $u_n \triangleright k_{n^*,n}(u_{n^*})$ such that

$$p_n = \text{tp}_{\kappa_n}^{k_{0,n}(\iota_L)+1}(k_{0,n}(S), u_n, k_{0,n}(\iota_L)).$$

(This follows from points 9 and 11.)

13. Inside N_n , chose a $\phi_n \in (\chi_n, \delta)$ such that

$$q_n := \text{tp}_{\phi_n}^{k_{0,n}(\iota_L)+3}(k_{0,n}(S), u_n, k_{0,n}(\iota_L))$$

is elastic.

14. $N_n \models (q_n \text{ exceeds } \text{tp}_{\phi_n}^{k_{0,n}(\iota_H)+1}(k_{0,n}(S), u_n) = \text{dcp}(p_n))$
(This follows from point 12 and the indiscernibility.)
15. $M_n \models \ulcorner \text{dcp}(p_n) \text{ is elastic} \urcorner$
(This follows from point 1.)
16. $N_n \models \ulcorner \text{dcp}(p_n) \text{ is elastic} \urcorner$
17. Inside N_n , there is an inaccessible $\lambda_n \in (\phi_n, \delta)$ and a (κ_n, λ_n) -extender E_n such that $q_n < \text{dcp}(p_n)^- [_{E_n}^{\lambda_n}]$.
(This follows from points 14 and 16.)
18. $M_{n+1} := \text{Ult}(M_n, E_n)$
19. $(\text{dcp}(p_n)^- [_{E_n}^{\lambda_n}])^{N_n} = \text{proj}_{\lambda_n}(\text{dcp}(j_{n,n+1}(p_n)^-))$
20. $M_{n+1} \models (q_n < \text{dcp}(j_{n,n+1}(p_n)))$
(This follows from points 17 and 19.)

21. $M_{n+1} \models (j_{n,n+1}(p_n) = j_{0,n+1}(\Gamma)(x \upharpoonright n, (q_i : i < n)))$
(This follows from point 1 by elementarity.)
22. $M_{n+1} \models ((x \upharpoonright (n+1), (q_i : i \leq n)) \in \text{dom } j_{0,n+1}(\Gamma))$
(This follows by the previous two points.)
23. Inside M_{n+1} , let $p_{n+1} := j_{0,n+1}(\Gamma)(x \upharpoonright (n+1), (q_i : i \leq n))$ and let κ_{n+1} correspond to its language.
24. $\kappa_{n+1} > j_{n,n+1}(\kappa_n) \geq \lambda_n$

Proposition 9.19. $x \in B$ if and only if the M -branch of $\mathcal{T}^x$ is well-founded.

Proof. Let us assume that the M -branch is well-founded. We denote its direct limit by M_ω . By the construction, we have that $(x \upharpoonright n, (q_i : i < n)) \in \text{dom } j_{0,n}(\Gamma) = j_{0,n}(T)$. This implies $(q_i : i < n) \in j_{0,\omega}(T)_x$ for every $n < \omega$. Hence, the tree $j_{0,\omega}(T)_x$ is ill-founded in V , so by absoluteness, it is ill-founded in M_ω . This means that

$$M_\omega \models x \in p[j_{0,\omega}(T)],$$

so the elementarity implies $x \in p[T] \subseteq B$.

Let us now assume that the M -branch is ill-founded. Since V is $(\omega + 1)$ -iterable, there is a well-founded branch b ; this branch is not the M -branch, so it is made entirely of N -models. Let us denote its limit by N_b . For $n \in b$, we have inside N_n that q_n is an $(x \upharpoonright l_n, \mathbf{t}_n)$ -nice type and that $u_n \models q_n$. If we set $y := \bigcup_{n \in b} \mathbf{t}_n \in \omega^\omega$, we see that

$$u_n \in k_{0,n}(S)_{x \upharpoonright l_n, \mathbf{t}_n} = k_{0,n}(S)_{x \upharpoonright l_n, y \upharpoonright l_n}$$

and consequently

$$k_{n,b}(u_n) \in k_{0,b}(S)_{x,y}.$$

Since $k_{n^*,n}(u_{n^*}) \triangleleft u_n$ for $n > 0$, we see that $(k_{n,b}(u_n) : n \in b)$ is a $\triangleleft$ -increasing sequence witnessing that $k_{0,b}(S)_{x,y}$ is ill-founded (in V). This means that $k_{0,b}(S)_{x,y}$ is ill-founded in N_b , so

$$N_b \models (x, y) \in p[k_{0,b}(S)].$$

By the elementarity, we conclude that $(x, y) \in p[S] = A$. This yields that $x \in p[A] = B^{\mathfrak{C}}$. $\square$

Notice that the construction from above can be executed for $s \in X^{<\omega}$ (instead of x) up to model $M_{|s|}$; we denote such a tree by $\mathcal{T}^s$. Hence, length of $\mathcal{T}^s$ is $2|s| + 1$ and $\mathcal{T}^s \triangleleft \mathcal{T}^x$ for $s \triangleleft x$. Let us define

$$W_s := M_{|s|}^s, \quad \pi_{s,t} := j_{s,t}^t, \quad w_s := (q_i^s : i < |s|) \quad (s \triangleleft t \in X^\omega).$$

(A superscript s means that the object corresponds to the one in construction $\mathcal{T}^s$.) The family just described witnesses the fact that T is τ -homogeneous,

in the sense of proposition 8.8. All the points of that proposition are obvious, except that

$$x \in p[T] \implies (W_s, \pi_{s,t} : s \triangleleft t \triangleleft x) \text{ is well-founded.}$$

To see this, it suffices to note that the ill-foundedness of the family in the consequent simply means the ill-foundedness of the M -branch of tree $\mathcal{T}^x$. By the previous proposition, this yields $x \in B^{\mathcal{C}} \subseteq p[T]^{\mathcal{C}}$. We are now ready to conclude the argument.

Proposition 9.20. *B is τ -homogeneously Suslin.*

Proof. We have shown that T is τ -homogeneous and that $p[T] \subseteq B$. We are done if we show that $B \subseteq p[T]$. To see this, if $x \in B$, then the M -branch of $\mathcal{T}^x$ is well-founded. This means that the family

$$(W_s, \pi_{s,t} : s \triangleleft t \triangleleft x)$$

is well-founded. Since T is homogeneous, this implies that $x \in p[T]$. $\square$

Remark 9.21. The family $(\mathcal{T}^s : s \in X^{<\omega})$ of trees constructed above can be chosen to be injective. We have the following manner. Suppose that $\mathcal{T}^s$ has been constructed for $s \in X^n$. The next step is to construct the models $N_n^{s \frown i}$ of $\mathcal{T}^{s \frown i}$ for $i < \omega$. These models might or might not be different; regardless, we ensure injectivity in the next step, when choosing extenders $E_n^{s \frown i}$ for the construction of $M_{n+1}^{s \frown i}$. To accomplish this, the points 8. of the base and 17. of the step should read: “Inside $N_n^{s \frown i}$, there is an inaccessible

$$\lambda_n^{s \frown i} \in (\max\{\phi_n^{s \frown i}, \lambda_n^{s \frown j} : j < i\}, \delta)$$

and ...” This provides for the extenders $E_n^{s \frown i}$ of different lengths (and thus different among themselves). Note that this construction is done recursively with respect to (1) $|s|$ and (2) $s(|s| - 1)$, ordered lexicographically.

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